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Back to the 1980s or Not? The Drivers of Inflation and Real Risks in Treasury Bonds

--Manuscript Draft--

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Abstract:	This paper shows that supply shock uncertainty interacts with the monetary policy rule to drive bond risks in a New Keynesian asset pricing model. In my model, positive nominal bond-stock betas emerge as the result of volatile supply shocks but only if the monetary policy rule features a high inflation weight. Habit formation preferences generate endogenously time-varying risk premia, explaining the volatility and predictability of bond and stock excess returns in the data, and implying that bond-stock betas price the expected equilibrium mix of shocks rather than realized shocks. The model explains the change from positive nominal and real bond-stock betas in the 1980s to negative nominal and real bond-stock betas in the 2000s with a shift from dominant supply shocks and an inflation-focused monetary policy rule, to demand shocks in the 2000s. Post-pandemic nominal and real bond-stock betas are explained with dominant supply shocks and a late increase in the monetary policy inflation coefficient.
Additional Information:	
Question	**Response**
To complete your submission you must select a statement which best reflects the availability of your research data/code. IMPORTANT: this statement will be published alongside your article. If you have selected "Other", the explanation text will be published verbatim in your article (online and in the PDF). (If you have not shared data/code and wish to do so, you can still return to Attach Files. Sharing or referencing research data and code helps other researchers to evaluate your findings, and increases trust	The authors will follow the journal's code and data policy.

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To complete your submission you must select a statement which best reflects the availability of your research data/code. IMPORTANT: this statement will be published alongside your article. If you have selected "Other", the explanation text will be published verbatim in your article (online and in the PDF). (If you have not shared data/code and wish to do so, you can still return to Attach Files. Sharing or referencing research data and code helps other researchers to evaluate your findings, and increases trust in your article. Find a list of supported data repositories in Author Resources, including the free-to-use multidisciplinary open Mendeley Data Repository.)	The authors will follow the journal’s code and data policy.

Dear Professor Whited,

Thank you very much for the opportunity to submit a revised version of my manuscript, and for the clear guidance. I am thrilled to resubmit a revised manuscript. Detailed responses are included below, but first I would like to provide an overview of the main changes.

- The **revised introduction states clearly the key contribution that sets this paper apart** from my own prior work and the broader literature on changing bond-stock comovements. In short, macroeconomic dynamics have changed so much over the past 30 years that changing one set of parameters is insufficient to return bonds to their risky state of the 1980s. Instead, a "perfect storm" consisting of a change towards more volatile supply shocks and a more hawkish monetary policy rule is needed to generate such reversion in bond risks.

- The revised paper provides additional discussion and results supporting the calibration for the first subperiod. Following the AE's suggestion, the paper includes the **new Figure 8** showing that a **high volatility of supply shocks is needed to replicate the bond-stock betas of the first subperiod starting from the second subperiod**. The revised paper also includes new examples and references for supply shocks during the 1980s and 1990s, such as the sudden drop in oil prices in 1986. I also added a newspaper count exercise in the **new Appendix Figure A1** showing that supply shocks were a substantial concern during the 1980s and 1990s.

- The revised paper provides an **updated discussion of recent events**. The **new Section 5** emphasizes the difference between an initially weak Fed inflation response during the early post-pandemic and a later much stronger Fed inflation response during the late post-pandemic period. I thereby address the referee's and AE's observation that the Fed eventually did respond strongly to the recent spike in inflation.

Thank you again for the opportunity to submit a revised version of this manuscript to the *JFE*. My apologies for taking slightly longer for this revision. I had a baby in September 2024, and I believe this timeline enabled me to respond fully to the thoughtful comments by the AE and the referees.

Sincerely,

Carolin

Responses to Associate Editor

I would like to thank you for the opportunity to revise and resubmit this manuscript, and especially for your constructive and highly detailed feedback. These comments have enabled me to produce a revised version of the manuscript that I believe makes a much clearer contribution. My apologies for taking slightly longer for this revision. I had a baby in September 2024, and I believe this timeline enabled me to respond fully and carefully to your and the referees' thoughtful comments.

It's useful to start with the common threads from both reports. Both referees want to see:

1. Better positioning of the manuscript in the literature, both in relation to the author's earlier publications and other relevant contributions (a few scattered comments across the footnotes won't suffice).

   Reply: I substantially re-wrote the introduction. The revised introduction now contains an updated discussion of how the manuscript relates to the literature, both in terms of my own prior work and other authors' contributions. See the **third paragraph on p.1** and the broader literature discussion starting at the bottom of **p.4** (**"This paper contains distinct contributions over the literature (…)"**). For a more detailed response, see my reply to your comment 1. below.

2. A more extensive discussion/defense of the calibration of the monetary policy side of the economy across the two sub-periods, especially for the first portion of the sample, which the author uses as an example of a time dominated by supply-side shocks.

   Reply: Following your suggestion, I added the **new Figure 8** showing that a **high volatility of supply shocks is needed to replicate the bond-stock betas of the first subperiod starting from the second subperiod**. The revised paper also includes new examples and references for supply shocks during the 1980s and 1990s, such as the sudden drop in oil prices in 1986. I also added a newspaper count exercise in the **new Appendix Figure A1** showing that supply shocks were a substantial concern during the 1980s and 1990s. For a more detailed response, see my reply to your comment 3. below.

If you decide to invite a resubmission, it should be made clear that the author must convincingly address these two points to have a chance at eventual publication in this Journal. The more negative referee raises several additional relevant points that the author should carefully address:

1. The manuscript builds heavily on some of the author's earlier contributions. While I certainly agree on this point, I do not concur with the conclusion that this diminishes the relevance of the current work to the point of not being worthy of publication in a top finance journal. The paper analyzes an important empirical question: why didn't the stock-risk exposures of nominal and real bonds change during the recent spike in inflation, despite perceived similarities with the 1970s and 1980s? The author answers this by estimating an equilibrium model across two subsamples, showing that the real and nominal sides of the economy have changed so much in the last 20 years that altering one set of parameters at a time (to match those of the earlier years) isn't enough to revert the risk exposures. It takes a "perfect storm" (as the author puts it) to make nominal and real bonds as risky as they used to be. This is a potentially important point in the asset pricing literature, one, in my opinion, not automatically inferred from the author's earlier papers. However, it's paramount that the author

provides a compelling argument about why the previous papers aren't enough to fully examine this empirical fact in both the response and the next draft.

Reply: Thank you for these fantastic suggestions. I substantially re-wrote the introduction to clarify the contribution of this paper relative to my own prior work and the broader literature on bond-stock comovements. The main changes can be found in the following locations:

- I revised the **first paragraph of the introduction** to state the new empirical puzzle up-front, following your suggestion. Here, I say

> "Maybe surprisingly, Figure 1 newly shows that nominal bond-stock betas remained negative for much of the inflationary post-pandemic period. This raises the question why -- different from the 1980s -- nominal bond-stock betas did not rise during the recent spike in inflation, despite perceived macroeconomic similarities."

- The paper's main conclusion that a "perfect storm" is needed is now stated up-front at the beginning of the introduction (**see p.1, paragraph starting "Calibrating a New Keynesian equilibrium model (…)"**).

- On **p.1** of the introduction, I now state how the model in this paper is different from the literature, and in particular my own prior work (**see p.1, paragraph starting "Despite the substantial literature studying historical bond-stock comovements (…)"**).

- A more detailed discussion of how this manuscript contributes over the existing literature follows on **pp.4-6** of the re-written introduction (**starting on p.4 "This paper contains distinct contributions (…)**).

In summary, while this paper builds on the previous literature, its model contains several changes giving it a unique ability to conduct counterfactual exercises with respect to the composition of macroeconomic shocks and the Taylor rule coefficients on bond-stock betas. One important predecessor paper, Campbell, Pflueger, and Viceira (2020, CPV), contains the same Euler equation, and prices bonds and stocks using the same habit-formation preferences with countercyclical risk premia. However, CPV is not as microfounded as the current manuscript, limiting its ability to conduct counterfactual exercises. While CPV contains reduced-form equations describing inflation and interest rate dynamics that bear some resemblance to a Phillips curve and Taylor rule, in CPV these are equilibrium dynamics and not structural equations. Concretely, the coefficients in CPV's reduced-form inflation and interest rate dynamics do not even have the signs that one would expect in structural equations and their shocks are correlated, so any change in these reduced-form dynamics in CPV must be interpreted as a combination of changes to monetary policy and the macroeconomic environment. By contrast, the current manuscript requires that the Phillips curve (20) and Taylor rule (24) have coefficients in line with the prior structural literature, and their shocks are uncorrelated, as is standard for structural shocks. The model in CPV is therefore not sufficient to understand how counterfactual changes in macroeconomic volatilities and Taylor rule coefficients impact bond-stock betas.

The model in Pflueger and Rinaldi (2022, PR) is more similar to the current one, containing a structural Phillips curve and a Taylor-type monetary policy rule. However, PR do not study the implications for bond-stock betas and the PR model is insufficient to reach the central conclusion of this manuscript. The current model contains some important changes compared to PR, in particular both demand and supply shocks (PR has neither), and adaptive inflation expectations. The ability to perform counterfactual exercises to assess the impact of changes in the macroeconomic shock volatilities and Taylor rule coefficients on bond-stock betas therefore sets this paper apart from CPV and PR.

Regarding the broader literature on changing bond-stock comovements, there are several features that set the current manuscript apart. As discussed by referee 2, various channels have been proposed in the literature to explain the changes in bond-stock betas over the past four decades, including changing inflation risks (Piazzesi and Schneider (2006), David and Veronesi (2013), Song (2017), Bekaert and Engstrom (2017), Campbell, Pflueger, and Viceira (2020), Gourio and Ngo (2020), and Fang, Liu, and Roussanov (2022)); liquidity, flight-to-safety, and time-varying risk aversion (Baele, Bekaert, and Inghelbrecht (2010), Kozak (2022), and Laarits (2022)); permanent-transitory compositional changes of consumption growth (Chernov, Lochstoer, and Song (2021)); and changes in the monetary-fiscal mix regimes (Li, Zha, Zhang, and Zhou (2022)). The current paper incorporates these channels in a relatively parsimonious setup, where inflation risk and the permanent-transitory decomposition arise endogenously from the nature of macroeconomic shocks and monetary policy, and habit formation preferences give rise to time-varying risk aversion akin to endogenous "flight-to-safety".

One particularly closely related paper is Song (2017), which however uses a more reduced-form setup featuring an endowment economy, limiting its ability to speak to the role of macroeconomic shocks vs. monetary policy for bond-stock betas.

The model in this paper is also distinct from prior work that models the impact of changes to the volatilities of macroeconomic shocks or to monetary policy separately, whereas it is central to the conclusion in this paper to model these changes jointly. Different from the current paper, the production-based models of Fang, Liu, and Roussanov (2022), and Kozak (2022) do not incorporate changes in monetary policy. Conversely, Gourio and Ngo (2020) and Li et al. (2022) model changes in monetary and/or fiscal policy but do not incorporate changes in the composition of macroeconomic shocks. Those latter two models also acknowledge their limitations as models of asset prices, with basic moments such as the equity premium and equity return volatility either not reported or deviating substantially from the data. Finally, Gourio and Ngo (2022) consider a model where bond-stock comovements are closely linked to the level of inflation, assuming a constant volatility of technology shocks and no other shocks. Because of the close link between the level of inflation and bond-stock comovements in Gourio and Ngo (2022), their model is not suited to understanding the key empirical finding in this paper, namely that during 2021 bond-stock betas remained negative even as inflation rose to levels not seen in 40 years. Hence, different from the current manuscript, none of these previous papers are sufficient to reach the conclusion in this paper, namely that supply shocks and an inflation-centric monetary policy rule interact and are both necessary to make nominal bonds risky.

2. This referee doesn't like some of the modeling assumptions. I'm not convinced by the "Taylor rule" comment (the author does include a shock in her MP rule that we could interpret as the discretionary component) and would like to see richer wage-price inflation dynamics. The author can address the latter (e.g., by introducing an additional shock to break the connection between the two inflation processes) and acknowledge some limitations in the modeling approaches.

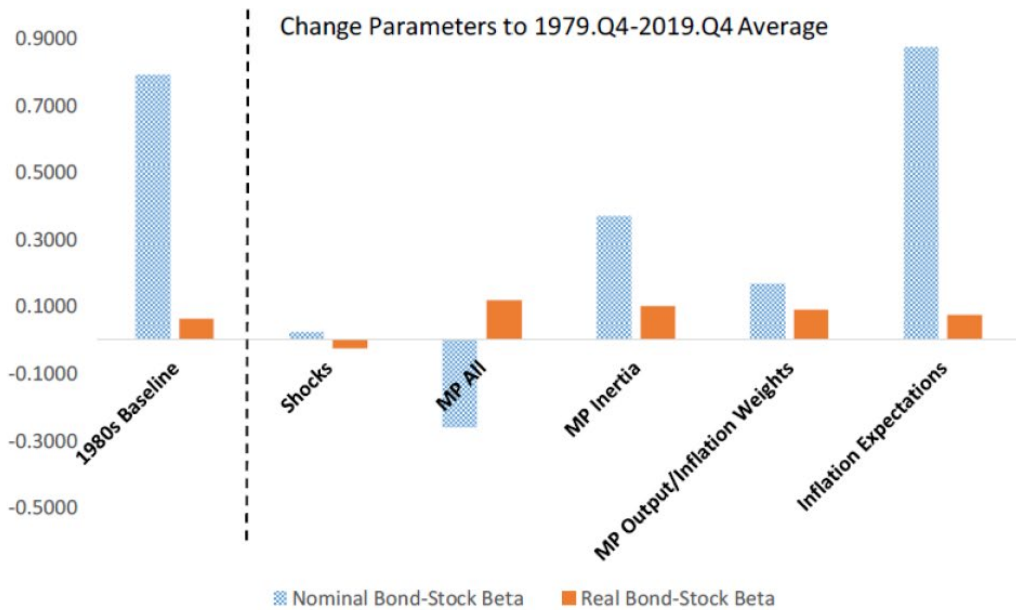
Reply: The revised manuscript now clearly acknowledges the limitations of the Taylor rule as a model of monetary policy. While a monetary policy rule is a standard feature of many models of monetary policy, in practice monetary policy clearly also contains a discretionary component as captured by the monetary policy shock in equation (24) of the revised manuscript. This is now discussed at the **top of p.13 in the paragraph starting "While there is an active debate to what extent monetary policy can be described by rules versus discretion (…)"**.

To address the issue of richer wage-price inflation dynamics, the **new Appendix Section G (online Appendix pp. 44-48)** shows results for a model extension with richer price-wage dynamics. This new appendix section first solves the model with richer price-wage dynamics and then re-produces the paper's main results analogous to Table 2 and Figure 4. The version of Figure 4 with a richer price-wage dynamics is reproduced below for your convenience **(Appendix Figure A7)**.

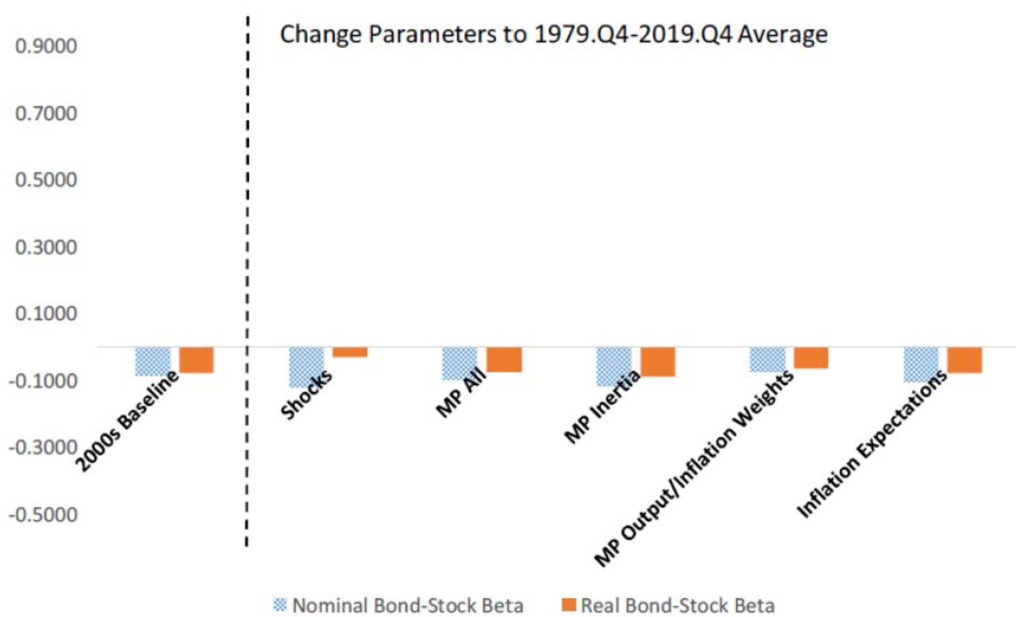
The new **Appendix Figure A7** shows that the asset pricing moments and key counterfactuals are robust and indeed quantitatively indistinguishable even when the volatility of the price-wage inflation shock is set to a plausible upper bound ($\sigma_p = 0.55$). To keep the model exposition as simple as possible, I therefore abstract from richer price-wage inflation dynamics in the main paper. Following your suggestion, the revised manuscript newly acknowledges the limitations of the simple wage-price inflation dynamics and notes that the results are robust to relaxing the tight link between wage and price inflation dynamics. For this discussion, see the **new footnote 9 on p. 12** in the paper.

Figure A7: Counterfactual Bond-Stock Betas with Richer Price-Wage Dynamics

Panel A: Starting from 1979.Q4-2001.Q1 Calibration



Panel B: Starting from 2001.Q2-2019.Q4 Calibration



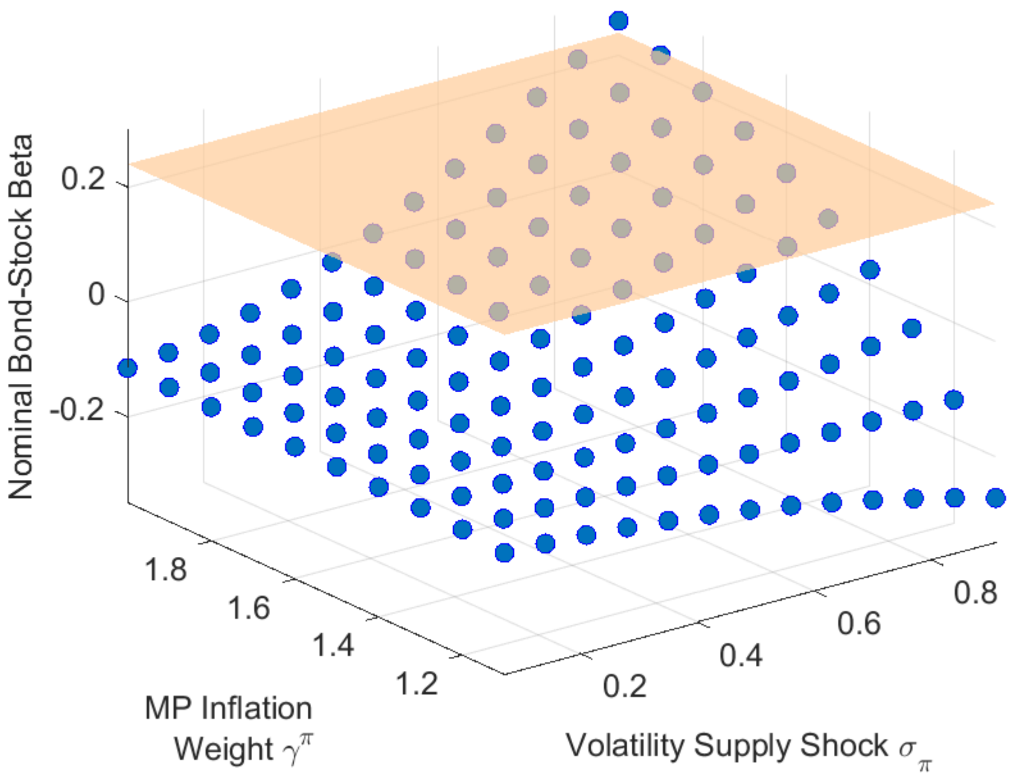
3. This referee is concerned that viewing the 1980s as a prototypical example of supply-side shocks is incorrect. I'm sympathetic to this, as the 1970s are typically the more natural benchmark (although, in fairness, the JEP article by Goodfriend that this referee mentions in footnote 1 does say at the bottom of page 51 that "In January and February 1980 [. . .], the spike in underlying inflation in connection with the ongoing oil price shock, the run-up in the dollar price of gold to $850 per ounce [. . .]", which one could argue as manifestations of supply shocks). Two potential avenues to address this issue are:

- Extend the estimation of the first sample period to include the 1970s, though finding inflation expectations data may be challenging (I am not sure that Blue Chip forecasts go back that far in time). The author could use parametric forecasts of inflation in the spirit of Stock and Watson instead.
- Calculate the necessary changes in parameters to replicate the stock betas from the first half of the sample in the second half. For example, is there a degree of MP inertia that would make the betas as big as they were in the first half of the sample? How much bigger should the volatilities of the shocks be to engineer the same betas?

Reply: I have addressed the issue twofold. First, I followed your suggestion to calculate the necessary changes in parameters to explain the nominal bond-stock betas during the 1980s from the second subperiod calibration (see the **new Figure 8**). This new figure clearly shows that volatile supply shocks are required to explain risky nominal Treasury bonds observed during the 1980s. Panel A of that figure is reproduced below for your convenience. The nominal bond-stock beta during the first subperiod is depicted with an orange plane, while blue circles depict model bond-stock betas. All parameters not explicitly mentioned are set to their values in the 2000s calibration listed in Table 1. The new Figure 8 shows that a high volatility of supply shocks is necessary to reverse-engineer the bond-stock betas observed during the first subperiod, further supporting a high supply shock volatility for the first calibration period. These new results are described in the revised introduction (**p.4, "The third counterfactual (…)"**). For a detailed discussion, see the **new Section 4.4 (pp.31-33)** in the revised manuscript.

I refrained from calibrating the model to the 1970s period. While I agree that this was plausibly a period of substantial supply shock volatility, monetary policy during the 1970s was more complicated, which would complicate the analysis. In particular, it has been argued that during this period monetary policy opened up the economy to self-fulfilling sunspots or "inflation scares" (Clarida, Gali, and Gertler (1999), Goodfriend (2007)). While important, these issues are beyond the scope of this paper. The 1980s hence provide an ideal calibration period for this paper, as this period featured substantial supply shock uncertainty and a sufficiently inflation-responsive monetary policy rule, thereby sidestepping the issue of sunspots and self-fulfilling inflation expectations. I added a brief discussion to the introduction explaining why I calibrate the model to the 1980s rather than the 1970s (**p3, paragraph starting "The calibrated changes…"**).

Figure 8: Model-Implied Nominal Bond-Stock Beta vs. 1980s Bond-Stock Beta
Panel A: MP Inertia $\rho^i = 0.54$



Second, the revised manuscript includes new anecdotes and references for examples of supply shocks during the 1980s and 1990s. It is important to keep in mind that in my model bond-stock betas depend on the volatility of supply shocks, but the sign of these shocks can be either positive or negative. While the 1970s were a period of dramatic adverse supply shocks, the 1980s and 1990s featured substantial supply shocks that tended to be positive, such as in 1986 when Saudi Arabia more than doubled its own oil output leading to a dramatic drop in oil prices (Gately, 1986). These shocks were viewed as highly relevant for monetary policy during 1980s and 1990s, as illustrated by the following quote from Greg Mankiw in a Fortune article from December 1997:

> "SUPPLY SHOCKS: Sometimes outside events affect the prices at which firms supply their goods and services. The classic inflationary supply shocks were the OPEC oil price increases of the 1970s. As the cost of all oil-related products rose, the Fed had to cope with a less favorable tradeoff between inflation and unemployment. Greenspan has been more fortunate. Shortly before he was appointed Fed chairman in 1987, world oil prices plummeted, improving the inflation-

> unemployment tradeoff. Over the past two years, the foreign-exchange market has provided a similar benefit. The rise in the exchange rate from 90 to 120 yen per dollar has put downward pressure on the prices of imports and import-competing goods. And because wages respond to consumer prices, the strong dollar has also kept down growth in labor costs."
>
> (N. Gregory Mankiw, "Alan Greenspan's Tradeoff", Fortune, December 8, 1997)

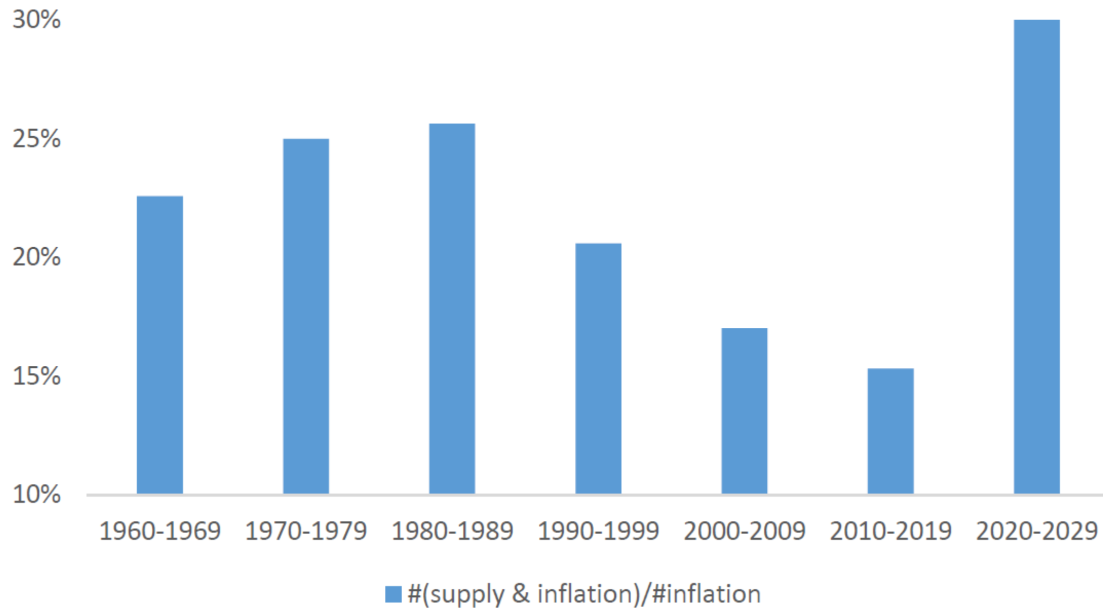
Supply shocks arising from new technology and the "New Economy" were also highly salient to monetary policy makers during my first subperiod, and in particular during the 1990s. The puzzle for economists and policy makers during the 1990s was how the US economy was able to perform so well while inflation remained low. It is immediate that these empirical observations would contribute to the negative empirical inflation-output gap relationship displayed in the top-left panel of Figure 2, and hence to the identification of volatile supply shocks during my first calibration period. Fed Chair Alan Greenspan himself famously attributed the performance of the US economy with low inflation during this period to technology shocks and the "New Economy" in statements such as the following:

> "As we have discussed before, powerful forces have evolved in the past few years to help contain inflationary tendencies. An ever-increasing share of our nation's workforce uses the tools of new technologies. Microchips embodied in physical capital make it work more efficiently, and sophisticated software adds to intellectual capital."
>
> (Chairman Alan Greenspan, Semiannual monetary policy report, U.S. Senate July 18, 1996)

Not all supply shocks during the 1980s and 1990s were positive, and as you mention Goodfriend (2007) refers to the 1980 and 1990/1991 (first gulf war) oil price increases as examples of adverse supply shocks. Also consistent with my break date, Goodfriend (2007) argues that the late 1990s represented a significant break in monetary policy towards a "new consensus". I therefore view his insightful account of US monetary policy as supportive of the break date used in this manuscript.

To quantify this anecdotal evidence, I added a simple quantification for the perceived relevance of supply shocks across the decades using newspaper counts to the revised manuscript. The results are shown in the **new Appendix Figure A1**, reproduced below for your convenience. This figure shows for each decade the number of Wall Street Journal articles mentioning both "supply" and "inflation", scaled by the number of articles mentioning "inflation" from Proquest Historical Newspapers. This new analysis shows that the share of supply mentions peaked in the 1980s. It even slightly increased from the 1970s to the 1980s. This seems plausible because the dramatic supply shocks during the 1970s arguably struck as a surprise, whereas by the 1980s and 1990s economists, the public, and investors had learned to worry about supply shocks and these worries were plausibly reflected in more frequent news mentions. The figure also shows a spike in the concern about supply-driven inflation during the post-pandemic period starting in 2020, similar to the level last observed during the 1980s. Overall, this analysis based on newspaper counts further supports the notion that the public and investors were highly attuned to macroeconomic supply shocks during the 1980s and again during the post-pandemic period after 2020.

Figure A1: Relative Frequency of Supply and Inflation in WSJ Articles



The revised manuscript discusses prominent examples of supply shocks during the first subperiod and the news count evidence in the following locations of the revised manuscript:

- **p.3, paragraph starting "The calibrated changes in the monetary policy rule (…)**
- **new footnote 4 on p.3**
- **p.18, paragraph starting "A simple newspaper count exercise (…)"**
- **new footnote 14 on p.18**
- **new footnote 27 on p.37**

4. The referee provides an alternative explanation for recent events, which the author should consider. It would also be useful to examine the likelihood of observing four 75bps rate increases in a year without changing some of the Taylor rule's parameters. I suspect that one way to examine the story proposed by the referee at the top of page 3 would consist in changing some of the parameters of the model to levels that are similar to the first part of the sample. There is a sense in which this aligns with what the author does in her counterfactuals.

I think the reports contain relevant and well-crafted comments. Similarly, I think the paper has the potential to make an important contribution to the literature. In my opinion, it would be appropriate to offer the author an opportunity to address these comments.

Reply: Thank you for these excellent suggestions. The **new Section 5** contains an updated discussion of recent events, dividing the analysis into an early post-pandemic (until 2023.Q2) and a late post-pandemic period (2023.Q3-Q4)**.** I find that the model can explain nominal and real bond-stock betas during the early post-pandemic with a low inflation coefficient. However, the model requires a high inflation coefficient during the later post-pandemic. I hence find that in the late post-pandemic period, bond-stock betas indeed require changing the monetary policy rule to parameters that are more similar to the first part of the sample, just as you suggested.

Following your suggestion, the new Section 5 puts the model-implied monetary policy rule in the context of the recent inflation spike and the subsequent energetic policy rate hikes. The timeline of inflation and policy rates during the post-pandemic period are depicted in the **new Figure 9**. The key takeaway from this figure is that inflation rose already in 2021, well before the Fed started lifting interest rates. The policy rate initially remained constant, lifted off from zero in March 2022, and reached its new plateau in late 2023. This timeline suggests that supply shocks became salient in 2021, but a change in the monetary policy stance only occurred with a substantial delay. Importantly for the analysis here, this change in the monetary policy stance was plausibly understood and priced only gradually as it unfolded.

To support this timeline of events, the revised paper also references my recent complementary paper Bauer, Pflueger, and Sunderam, (2024), which estimates the monetary policy inflation coefficient as perceived by professional forecasters from rich survey data, and links the change in the perceived monetary policy rule to policy rate surprises on FOMC dates. Table 2 of Bauer, Pflueger and Sunderam (2024) suggests that the perceived monetary policy inflation coefficient peaked during the second half of 2023. Even though the data and methodology in this complementary work are completely separate, it is interesting that the timing lines up closely with the significant increases in rolling nominal, real and breakeven betas in Figure 10 in the present manuscript.

The revised paper discusses the timeline of recent inflation and monetary policy on **p.4** of the introduction and in more detail on **pp.34-35**. Bauer, Pflueger, and Sunderam, (2024a,b) do not speak to bond-stock betas, so this paper has a unique ability to explain bond-stock betas. I clarify the unique contribution of this paper compared to Bauer, Pflueger, and Sunderam, (2024a,b) in the literature review on **p.6** ("**Complementary to this paper, Bauer, Pflueger, and Sunderam (2024a,b) (…)**").

Figure 10 in Section 5 depicts rolling nominal, real, and breakeven betas from daily data during the post-pandemic period. Panel A shows that while nominal bond-stock betas show some higher frequency movements, they overall remained negative during the early post-pandemic and only turned decisively positive at the very end of sample in the second half of 2023. Panel B depicts rolling breakeven betas (or the difference between nominal and real bond-stock betas) with confidence intervals. This shows that until mid-2023 nominal bond-stock betas were either significantly lower than real bond-stock betas, or at least not statistically significantly different, indicating that the inflationary dynamics priced in bond-stock betas were still procyclical (since breakeven returns are inversely related to inflation expectations). However, in the second half of 2023, breakeven betas did turn significantly positive, suggesting that the priced cyclicality of inflation expectations had changed. I interpret these patterns as providing out-of-sample confirmation of the key model predictions, if supply shocks were a significant concern throughout the post-pandemic period, but the priced monetary policy rule changed from a low inflation coefficient in the early post-pandemic to a high inflation coefficient in the later post-pandemic. The manuscript contains a revised discussion of Figure 10 on **pp.35-36**.

The **revised Table 3** provides a simple reverse-engineering exercise, showing that observed nominal and real bond-stock betas in the early and late post-pandemic periods can be matched with 1980s-style macroeconomic shocks, and a low monetary policy inflation coefficient during the early post-pandemic but a higher monetary policy inflation coefficient during the later post-pandemic. The results in Table 3 hence indicate that the monetary policy rule implied by bond-stock betas was similar to the 2000s during 2021 and 2022, but became more similar to the 1980s monetary policy rule during 2023 after the series of energetic rate hikes. The revised description of Table 3 can be found on **pp. 37-38** of the paper.

<u>Responses to Referee 1</u>

Thank you for your careful reading of the manuscript and your insightful comments. I addressed them following the AE's guidance, and am thrilled to resubmit a clearer and much improved manuscript. I include detailed point-by-point responses to your comments below.

1. **Lack of methodological novelty.** This paper is too close to CPV. The model is mostly the same: in fact, most of the equations in Section 2 are the same as in CPV. The difference is the Phillips curve and the monetary policy rule described in Section 2.4 and 2.5. Methodologically, this is a relatively minor extension. Moreover, the structural additions are either not really structural or not very convincing. Specifically:

Reply: I substantially re-wrote the introduction to clarify this paper's contribution following the detailed suggestions by the AE and referee 2. The unique contribution of this paper is now stated up-front in **the second paragraph on p.1** as follows:

> "Calibrating a New Keynesian equilibrium model for asset prices separately to the 1980s and the 2000s, I find that over the past 30 years monetary policy and the real side of the economy have changed sufficiently that reverting either set of parameters is insufficient to revert bond risks to their levels in the 1980s. Instead, a "perfect storm" consisting of a change towards more volatile supply shocks and a more hawkish monetary policy rule is needed to generate a reversion back to the risky bond markets of the 1980s."

The next paragraph (**p.1, starting "Despite the substantial literature (…)"**) clarifies how the model in this paper differs from CPV and other key papers. Finally, the revised paper discusses how this paper contributes to the broader literature in a new literature review (**pp. 4- 6, starting "This paper contains distinct contributions (…)**)

In summary, while this paper builds on the previous literature, its model contains several changes giving it a unique ability to conduct counterfactual exercises with respect to the composition of macroeconomic shocks and the Taylor rule coefficients on bond-stock betas. One important predecessor paper, Campbell, Pflueger, and Viceira (2020, CPV), contains the same Euler equation, and prices bonds and stocks using the same habit-formation preferences with countercyclical risk premia. However, CPV is not as microfounded as the current manuscript, limiting its ability to conduct counterfactual exercises. While CPV contains reduced-form equations describing inflation and interest rate dynamics that bear some resemblance to a Phillips curve and Taylor rule, in CPV these are equilibrium dynamics and not structural equations. Concretely, the coefficients in CPV's reduced-form inflation and interest rate dynamics do not even have the signs that one would expect in structural equations and their shocks are correlated, so any change in these reduced-form dynamics in CPV must be interpreted as a combination of changes to monetary policy and the macroeconomic environment. By contrast, the current manuscript requires that the Phillips curve (20) and Taylor rule (24) have coefficients in line with the prior structural literature, and their shocks are uncorrelated, as is standard for structural shocks. The model in CPV is therefore not sufficient to understand how counterfactual changes in macroeconomic volatilities and Taylor rule coefficients impact bond-stock betas.

The model in Pflueger and Rinaldi (2022, PR) is more similar to the current one, containing a structural Phillips curve and a Taylor-type monetary policy rule. However, PR do not study the implications for bond-stock betas and the PR model is insufficient to reach the central conclusion of this manuscript. The current model contains some important changes compared to PR, in particular both demand and supply shocks (PR has neither), and adaptive inflation expectations. The ability to perform counterfactual exercises to assess the impact of changes in the macroeconomic shock volatilities and Taylor rule coefficients on bond-stock betas therefore sets this paper apart from CPV and PR.

Regarding the broader literature on changing bond-stock comovements, there are several features that set the current manuscript apart. As discussed by referee 2, various channels have been proposed in the literature to explain the changes in bond-stock betas over the past four decades, including changing inflation risks (Piazzesi and Schneider (2006), David and Veronesi (2013), Song (2017), Bekaert and Engstrom (2017), Campbell, Pflueger, and Viceira (2020), Gourio and Ngo (2020), and Fang, Liu, and Roussanov (2022)); liquidity, flight-to-safety, and time-varying risk aversion (Baele, Bekaert, and Inghelbrecht (2010), Kozak (2022), and Laarits (2022)); permanent-transitory compositional changes of consumption growth (Chernov, Lochstoer, and Song (2021)); and changes in the monetary-fiscal mix regimes (Li, Zha, Zhang, and Zhou (2022)). The current paper incorporates these channels in a relatively parsimonious setup, where inflation risk and the permanent-transitory decomposition arise endogenously from the nature of macroeconomic shocks and monetary policy, and habit formation preferences give rise to time-varying risk aversion akin to endogenous "flight-to-safety".

One particularly closely related paper is Song (2017), which however uses a more reduced-form setup featuring an endowment economy, limiting its ability to speak to the role of macroeconomic shocks vs. monetary policy for bond-stock betas.

The model in this paper is also distinct from prior work that models the impact of changes to the volatilities of macroeconomic shocks or to monetary policy separately, whereas it is central to the conclusion in this paper to model these changes jointly. Different from the current paper, the production-based models of Fang, Liu, and Roussanov (2022), and Kozak (2022) do not incorporate changes in monetary policy. Conversely, Gourio and Ngo (2020) and Li et al. (2022) model changes in monetary and/or fiscal policy but do not incorporate changes in the composition of macroeconomic shocks. Those latter two models also acknowledge their limitations as models of asset prices, with basic moments such as the equity premium and equity return volatility either not reported or deviating substantially from the data. Finally, Gourio and Ngo (2022) consider a model where bond-stock comovements are closely linked to the level of inflation, assuming a constant volatility of technology shocks and no other shocks. Because of the close link between the level of inflation and bond-stock comovements in Gourio and Ngo (2022), their model is not suited to understanding the key empirical finding in this paper, namely that during 2021 bond-stock betas remained negative even as inflation rose to levels not seen in 40 years. Hence, different from the current manuscript, none of these previous papers are sufficient to reach the conclusion in this paper, namely that supply shocks and an inflation-centric monetary policy rule interact and are both necessary to make nominal bonds risky.

- The Taylor rules are not structural relations. In practice, central banks set policy **with discretion** to achieve their mandates; they do not follow Taylor rules (although they might use them as benchmark). These rules are best interpreted as reduced form relations that approximate central bank behavior.

I followed the AE's suggestion, and added a discussion of the limitations of the Taylor rule as a model of monetary policy. While a monetary policy rule is a standard feature of many models of monetary policy, in practice monetary policy clearly also contains a discretionary component as captured by the monetary policy shock in equation (24) of the revised manuscript. This new discussion can be found at the **top of p.13** in the paragraph starting **"While there is an active debate to what extent monetary policy can be described by rules versus discretion (…)"**.

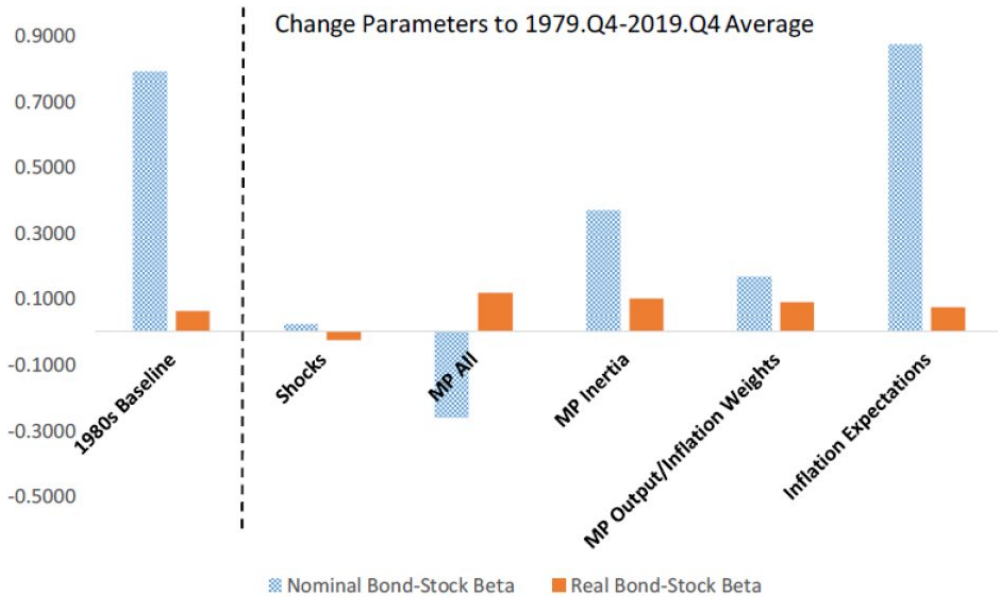
- The Phillips curve is generally the weakest part of New Keynesian models since there is much we don't understand about inflation (we have a *qualitative* understanding). This paper's Phillips curve is no exception. The paper assumes prices are flexible and wages are sticky, and derives a wage Phillips curve with backward and forward looking elements. This approach will miss a lot of richness that matters in practice, e.g., the Covid-19 shock showed that price and wage inflation can have very different dynamics.

I followed the AE's guidance and considered a model extension featuring an additional shock to price vs. wage inflation, and showed that the model asset pricing moments and counterfactuals are robust. To address the issue of richer wage-price inflation dynamics, the **new Appendix Section G (online Appendix pp. 44-48)** shows results for a model extension with richer price-wage dynamics. This new appendix section first solves the model with richer price-wage dynamics and then re-produces the paper's main results analogous to Table 2 and Figure 4. The version of Figure 4 with a richer price-wage dynamics is reproduced below for your convenience **(Appendix Figure A7)**.

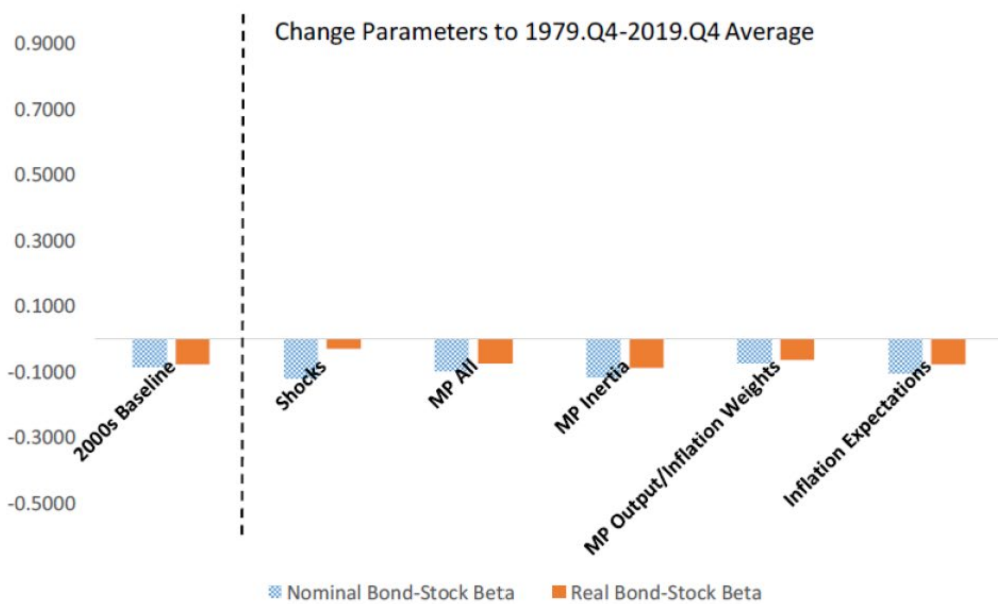
The new **Appendix Figure A7** shows that the asset pricing moments and key counterfactuals are robust and indeed quantitatively indistinguishable even when the volatility of the price-wage inflation shock is set to a plausible upper bound ($\sigma_p = 0.55$). To keep the model exposition as simple as possible, I therefore abstract from richer price-wage inflation dynamics in the main paper. Following your suggestion, the revised manuscript newly acknowledges the limitations of the simple wage-price inflation dynamics and notes that the results are robust to relaxing the tight link between wage and price inflation dynamics. For this discussion, see the **new footnote 9 on p. 12** in the paper.

Figure A7: Counterfactual Bond-Stock Betas with Richer Price-Wage Dynamics

Panel A: Starting from 1979.Q4-2001.Q1 Calibration



Panel B: Starting from 2001.Q2-2019.Q4 Calibration



The paper does not persuasively explain any new fact (beyond what we can tell based on CPV). Given the lack of methodological novelty, the paper should be evaluated in terms of how well it explains the motivating facts—specifically, the changing bond-stock betas. The paper claims that betas were positive in the earlier decades because "supply shocks" were more relevant and the Fed was more inflation focused and more alert (less inertial). This explanation is problematic for a number of reasons:

Reply: I revised the **first paragraph of the introduction** to state the new empirical puzzle up-front, following the AE's suggestion. Here, I say

> "Maybe surprisingly, Figure 1 newly shows that nominal bond-stock betas remained negative for much of the inflationary post-pandemic period. This raises the question why -- different from the 1980s -- nominal bond-stock betas did not rise during the recent spike in inflation, despite perceived macroeconomic similarities."

I view post-pandemic betas as a particularly informative moment about the macroeconomic drivers of bond-stock betas because this recent period featured a spike in inflation not experienced in the US since the 1980s. CPV cannot speak to this empirical puzzle because its sample ends before the pandemic.

As discussed in detail in response to your previous comment, I also re-wrote the introduction to clarify how this paper's conclusion and model are different from CPV and the broader literature on bond-stock betas.

- A qualitative reading of history suggests that there were no major "supply shocks" in the 1980s (either empirically or "in investors' minds"). Supply shocks were relevant in the 1970s—think oil/food price shocks and associated recessions. In the 1980s, the main issue was elevated inflation and its persistence (e.g., due to inflation expectations or experience affecting price-setting behavior). Chair Volcker hiked rates and generated a recession to reduce inflation. Afterward, there were several "inflation scares"—the markets were testing the Fed's resolve to see whether they are serious about controlling inflation or whether inflation would flare up again. Over time, the Fed managed to keep inflation at its implicit target (2%). The Fed eventually gained credibility and inflation expectations became anchored. This credibility is why (among other things) bond-stock betas eventually turned negative.This [1]

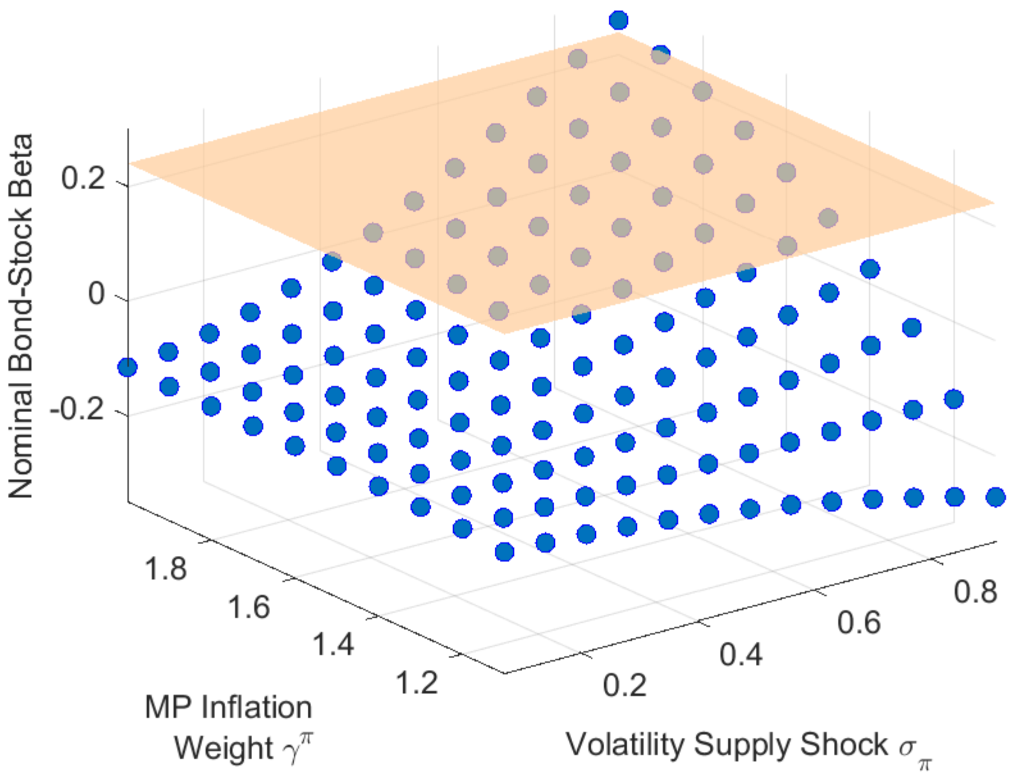
  Given this history, what should we make of this paper's explanation that this period featured dominant "supply shocks" with a fast and inflation-focused monetary policy? We can *perhaps* interpret the supply shocks as an "as-if" modeling device to capture inflation scares driven by weak credibility. But we could do a similar interpretation for CPV. This paper's ambition appears to go deeper than CPV and provide a "structural" interpretation rather than an as-if interpretation. But the structural interpretation does not seem correct.

[1] This paper has a nice review of this period https://www.aeaweb.org/articles?id=10.1257/jep.21.4.47

Reply: I have addressed the issue twofold. First, I followed the AE's suggestion to calculate the necessary changes in parameters to explain the nominal bond-stock betas during the 1980s from the second subperiod calibration (see the **new Figure 8**). This new figure clearly shows that volatile supply shocks are required to explain risky nominal Treasury bonds observed during the 1980s. Panel A of that figure is reproduced below for your convenience. The nominal bond-stock beta during the first subperiod is depicted with an orange plane, while blue circles depict model bond-stock betas. All parameters not explicitly mentioned are set to their values in the 2000s calibration listed in Table 1. The new Figure 8 shows that a high volatility of supply shocks is necessary to reverse-engineer the bond-stock betas observed during the first subperiod, further supporting a high supply shock volatility for the first calibration period. For a detailed discussion of this figure see the **new Section 4.4 (pp.31-32)** in the manuscript.

I refrained from calibrating the model to the 1970s period. While I agree that this was plausibly a period of substantial supply shock volatility, monetary policy during the 1970s was more complicated, which would complicate the analysis. In particular, it has been argued that during this period monetary policy opened up the economy to self-fulfilling sunspots or "inflation scares" (Clarida, Gali, and Gertler (1999), Goodfriend (2007)). While important, these issues are beyond the scope of this paper. The 1980s hence provide an ideal calibration period for this paper, as this period featured substantial supply shock uncertainty and a sufficiently inflation-responsive monetary policy rule, thereby sidestepping the issue of sunspots and self-fulfilling inflation expectations. I added a brief discussion to the introduction explaining why I calibrate the model to the 1980s rather than the 1970s (**p3, paragraph starting "The calibrated changes…"**).

Figure 8: Model-Implied Nominal Bond-Stock Beta vs. 1980s Bond-Stock Beta
Panel A: MP Inertia $\rho^i = 0.54$



Second, the revised manuscript includes new anecdotes and references for examples of supply shocks during the 1980s and 1990s. It is important to keep in mind that in my model bond-stock betas depend on the volatility of supply shocks, but the sign of these shocks can be either positive or negative. While the 1970s were a period of dramatic adverse supply shocks, the 1980s and 1990s featured substantial supply shocks that tended to be positive, such as in 1986 when Saudi Arabia more than doubled its own oil output leading to a dramatic drop in oil prices (Gately, 1986). These shocks were viewed as highly relevant for monetary policy during 1980s and 1990s, as illustrated by the following quote from Greg Mankiw in a Fortune article from December 1997:

> "SUPPLY SHOCKS: Sometimes outside events affect the prices at which firms supply their goods and services. The classic inflationary supply shocks were the OPEC oil price increases of the 1970s. As the cost of all oil-related products rose, the Fed had to cope with a less favorable tradeoff between inflation and unemployment. Greenspan has been more fortunate. Shortly before he was appointed Fed chairman in 1987, world oil prices plummeted, improving the inflation-unemployment tradeoff. Over the past two years, the foreign-exchange market has provided a similar benefit. The rise in the exchange rate from 90 to 120 yen per dollar has put downward

> pressure on the prices of imports and import-competing goods. And because wages respond to consumer prices, the strong dollar has also kept down growth in labor costs."
> (N. Gregory Mankiw, "Alan Greenspan's Tradeoff", Fortune, December 8, 1997)

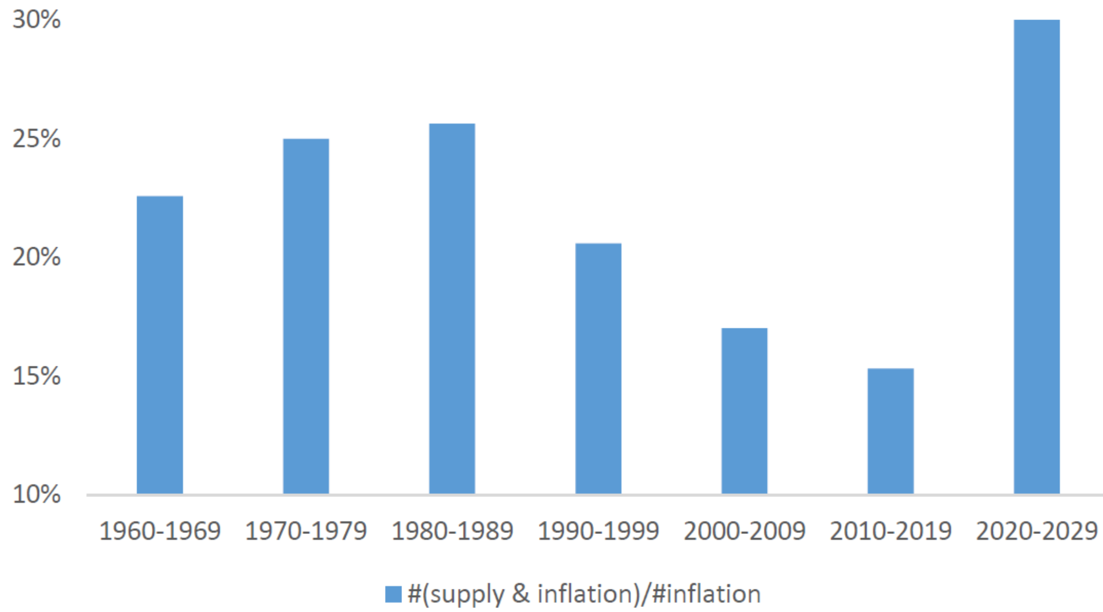
Supply shocks arising from new technology and the "New Economy" were also highly salient to monetary policy makers during my first subperiod, and in particular during the 1990s. The puzzle for economists and policy makers during the 1990s was how the US economy was able to perform so well while inflation remained low. It is immediate that these empirical observations would contribute to the negative empirical inflation-output gap relationship displayed in the top-left panel of Figure 2, and hence to the identification of volatile supply shocks during my first calibration period. Fed Chair Alan Greenspan himself famously attributed the performance of the US economy with low inflation during this period to technology shocks and the "New Economy" in statements such as the following:

> "As we have discussed before, powerful forces have evolved in the past few years to help contain inflationary tendencies. An ever-increasing share of our nation's workforce uses the tools of new technologies. Microchips embodied in physical capital make it work more efficiently, and sophisticated software adds to intellectual capital."
> (Chairman Alan Greenspan, Semiannual monetary policy report, U.S. Senate July 18, 1996)

Not all supply shocks during the 1980s and 1990s were positive. As mentioned by the AE, Goodfriend (2007) refers to the 1980 and 1990/1991 (first gulf war) oil price increases as examples of adverse supply shocks. Consistent with my break date, Goodfriend (2007) argues that the late 1990s represented a significant break in monetary policy towards a "new consensus". I therefore view Goodfriend (2007)'s insightful account of US monetary policy as supportive of the break date used in this manuscript.

To quantify this anecdotal evidence, I added a simple quantification for the perceived relevance of supply shocks across the decades using newspaper counts to the revised manuscript. The results are shown in the **new Appendix Figure A1**, reproduced below for your convenience. This figure shows for each decade the number of Wall Street Journal articles mentioning both "supply" and "inflation", scaled by the number of articles mentioning "inflation" from Proquest Historical Newspapers. This new analysis shows that the share of supply mentions peaked in the 1980s. It even slightly increased from the 1970s to the 1980s. This seems plausible because the dramatic supply shocks during the 1970s arguably struck as a surprise, whereas by the 1980s and 1990s economists, the public, and investors had learned to worry about supply shocks and these worries were plausibly reflected in more frequent news mentions. The figure also shows a spike in the concern about supply-driven inflation during the post-pandemic period starting in 2020, similar to the level last observed during the 1980s. Overall, this analysis based on newspaper counts further supports the notion that the public and investors were highly attuned to macroeconomic supply shocks during the 1980s and again during the post-pandemic period after 2020.

Figure A1: Relative Frequency of Supply and Inflation in WSJ Articles



The revised manuscript discusses this news count evidence and prominent examples of supply shocks during my first subperiod in the following locations of the revised manuscript:

- **p.3, paragraph starting "The calibrated changes in the monetary policy rule (…)**
- **new footnote 4 on p.3**
- **p.18, paragraph starting "A simple newspaper count exercise (…)"**
- **new footnote 14 on p.18**
- **new footnote 27 on p.37**

- A similar qualitative look at the recent decades suggests the Fed is not less focused on inflation and it is not more inertial, compared to the 1980s. As I already mentioned, the Fed managed to keep inflation around 2% for decades until the Covid-19 shock. Inflation flared up recently, but that is because Covid-19 was a large and unusual supply shock. The Fed made a mistake in thinking that this shock was more transitory than it was (the markets made the same mistake throughout 2021). There is no evidence that the Fed was less inflation focused or less inertial. In fact, once the Fed realized its mistake, it quickly reversed course and started a very aggressive hiking campaign: raising interest rates by 500 basis points in one year. This seems like *the opposite* of an inertial central bank.

In terms of the behavior of inflation expectations and stock-bond betas in the Covid-19 cycle, there is a simpler explanation than what this paper proposes. The Fed has gained credibility so inflation expectations have not changed very much (beyond a few years). At the same time, because of large macroeconomic shocks (both supply and demand), the Fed had to move interest rates much more than in the past to shift financial conditions and control inflation. The prevalence of macroeconomic shocks increased the interest rate volatility and the bond-stock betas starting from a negative level.[2] This story does not require a change in the Fed's inertia or inflation focus, and it does not require a complicated model with habit formation.

Reply: I completely re-wrote the account of recent events in the **new Section 5**. Following the AE's guidance, the clarified explanation of recent events emphasizes the difference between the early post-pandemic – characterized by large supply shocks but little response by the Fed – and the later post-pandemic – characterized by supply shocks and a much aggressive Fed response. I believe this model-implied account of recent events lines up closely with the narrative you suggest, and thereby addresses your comment.

The timeline of inflation and policy rates during the post-pandemic period is depicted at the beginning of Section 5 in the **new Figure 9**. The key takeaway from this figure is that inflation rose already in 2021, well before the Fed started lifting interest rates. The policy rate initially remained constant, lifted off from zero in March 2022, and reached its new plateau in late 2023. This timeline suggests that supply shocks became salient in 2021, but a change in the monetary policy stance only occurred with a substantial delay. The revised paper discusses the model implications in relation to the timeline of recent inflation and monetary policy on **p.4** of the introduction and in more detail on **pp.34-35**.

The **revised Table 3** provides a simple reverse-engineering exercise, showing that observed nominal and real bond-stock betas in the early and late post-pandemic periods can be matched with 1980s-style macroeconomic shocks, and a low monetary policy inflation coefficient during the early post-pandemic but a higher monetary policy inflation coefficient during the later post-pandemic. The results in Table 3 hence indicate that the monetary policy rule priced in bond-stock betas remained similar to the 2000s during 2021 and 2022, but became more similar to the 1980s monetary policy rule during 2023 after the series of energetic rate hikes. The revised description of Table 3 can be found on **pp. 37-38** of the paper.

To support this timeline of events, the revised paper also references my recent complementary paper Bauer, Pflueger, and Sunderam, (2024), which estimates the monetary policy inflation coefficient as perceived by professional forecasters from rich survey data, and links the change in the perceived monetary policy rule to policy rate surprises on FOMC dates. Note that by drawing on rich cross-sectional survey data, Bauer, Pflueger, and Sunderam (2024) can distinguish between transitory inflation expectations and a perceived low monetary policy inflation coefficient, and still finds a substantial role for an initially low monetary policy inflation coefficient, followed by an increase in the perceived monetary policy inflation coefficient, peaking in the second half of 2023. Even though the data and methodology in this complementary work are completely separate, it is interesting that the timing lines up closely with the significant increases in rolling nominal, real and breakeven betas in Figure 10 in the present manuscript.

[2] The baseline negative bond-stock correlations are likely driven by risk premium or sentiment shocks (or any financial shock that affects stock prices more than bond prices). These shocks are common and they affect stock prices without changing potential output by much. The Fed naturally responds to these shocks in a stabilizing direction, e.g., they cut rates after a stock price decline (the Fed put). This response (and its anticipation) creates a negative correlation between stocks and bonds.

The revised paper discusses the timeline of recent inflation and monetary policy on **p.4** of the introduction and in more detail on **pp.34-35**. Bauer, Pflueger, and Sunderam, (2024a,b) do not speak to bond-stock betas, so this paper has a unique ability to explain bond-stock betas. I clarify the unique contribution of this paper compared to Bauer, Pflueger, and Sunderam, (2024a,b) in the literature review on **p.6** ("**Complementary to this paper, Bauer, Pflueger, and Sunderam (2024a,b) (…)**").

More broadly, the general equilibrium framework in this paper unifies various channels priced in bond-stock betas in a relatively parsimonious manner, including "sentiment" or risk premium shocks, the rationality of inflation expectations, and the monetary policy response to macroeconomic shocks. The revised manuscript states this contribution clearly in the introduction (**p.5, "The model in this paper incorporates these channels in a relatively parsimonious setup (…)"**). The revised Section 4 makes clear how "sentiment" shocks are accounted for in the model (**p. 30, "One might wonder whether "sentiment" shocks could turn bond-stock betas negative (…)"**). To address your comment about the anchoring of inflation expectations during the recent period, the revised manuscript says clearly that I assume rational inflation expectations when interpreting recent events (**top of p.37, "To account for strongly anchored long-term inflation expectations during the recent period (…)"**).

Responses to Referee 2

Thank you for your suggestions, and especially your lucid discussion of the literature. Following your and the AE's suggestions, I re-wrote the introduction to delineate more clearly how the current manuscript stands out from the existing literature. The revised paper also contains additional discussion and results supporting the calibration for the first subperiod. Detailed point-by-point responses to your comments are included below.

In this sense, the paper's setup builds on the author's previous two published papers, Campbell, Pflueger, and Viceira (2020) and Pflueger and Rinaldi (2022), as they both include (P1) and (P2). While the model (P1) in Campbell, Pflueger, and Viceira (2020) is not as micro-founded as the one in the current paper, (P1) in Pflueger and Rinaldi (2022) is very similar, but with some important changes such as enriching structural shocks to include both demand and supply, and considering adaptive subjective inflation expectations.

The calibration of the model for two distinct subperiods sets the current paper apart from Pflueger and Rinaldi (2022), while its unique ability to perform counterfactual exercises assessing the impact of Taylor-rule coefficients and macroeconomic shock volatilities on bond-stock betas amplifies its significance.

In the literature, various channels explain bond-stock comovement, including the inflation-risk narrative proposed by David and Veronesi (2013), Song (2017), Campbell, Pflueger, and Viceira (2020), Gourio and Ngo (2020), and Fang, Liu, and Roussanov (2022); the liquidity, flight-to-safety, and time-varying risk aversion argument presented by Baele, Bekaert, and Inghelbrecht (2010), Kozak (2022), and Laarits (2022); examination of permanent-transitory compositional changes discussed by Chernov, Lochstoer, and Song (2021); and the examination of monetary-fiscal mix regimes explored by Li, Zha, Zhang, and Zhou (2022), among others.

What sets the current paper apart is its remarkable ability to integrate these popular narratives within a relatively simple unified framework. It adeptly incorporates elements such as the inflation risk narrative through the differentiation of demand versus supply shocks, time-varying risk aversion via habit-formation preferences, adjustments in the persistence of state variables, achieved by calibrating the model for two distinct subperiods, and modifications in the monetary policy rule, all contributing to a comprehensive understanding of bond-stock comovement.

Overall, I believe the paper merits publication in the *Journal of Financial Economics*. While I find the current paper already quite clear, I will only raise two relatively minor points.

Firstly, considering that the paper's primary contribution focuses on explaining changing bond-stock comovement, I recommend that the author provide a more precise explanation of her contribution to the literature. Although the author cites most of the papers mentioned above in the paper, many of them are relegated to footnotes. For example, among the more pertinent papers, the author's paper undoubtedly presents advancements beyond Song (2017), which also discussed changes in the monetary policy rule within an endowment economy, and it extends Campbell, Pflueger, and Viceira (2020), where the monetary policy rule isn't specified. None of the production-based papers mentioned above, such as Gourio and Ngo (2020), Fang, Liu, and Roussanov (2022), and Kozak (2022), incorporate changes in the monetary policy rule. However, the work of Li, Zha, Zhang, and Zhou (2022) is somewhat unclear, as it also explores changes in policy rules, encompassing both monetary and fiscal aspects. The distinction may lie in the fact that Li, Zha, Zhang, and Zhou (2022) do not consider changes in volatility related to structural shocks. Clarifying both the precise contribution of this paper and that of others would be beneficial for readers to comprehend its position in the literature.

> **Reply:** I liked your summary a lot and found it very helpful to clarify the current manuscript's relationship to the literature. The revised manuscript now highlights the papers' key difference relative to the key papers you mentioned at the very beginning of the introduction **(see p.1, paragraph starting "Despite the substantial literature studying bond-stock comovements (…)").**
>
> A new detailed discussion of how this paper contributes over the broader literature on bond-stock betas is included at the end of the introduction (**starting at the bottom of p.4 "This paper contains distinct contributions over the literature studying changing bond-stock comovements. (…)"**) All papers mentioned in your report are now described in the text of the introduction, and previous footnotes containing these references have been moved to the main text.

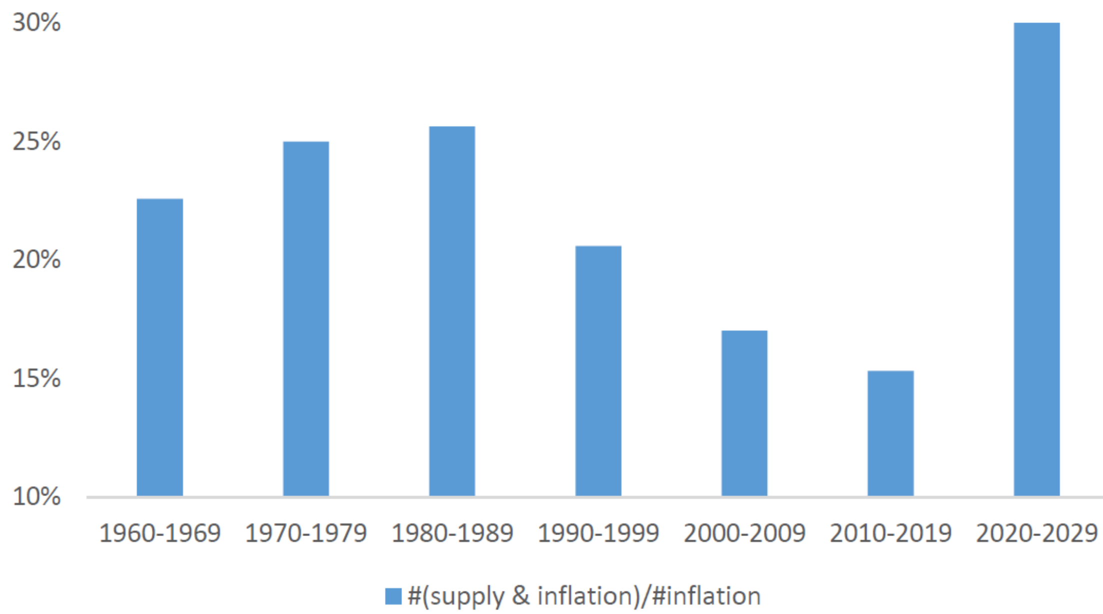
Secondly, while I generally agree with the author's calibration across two periods, denoted as R1 and R2 in my report, I recommend that the author provide more discussion regarding the existing macro literature on calibration. The author has touched upon this to some extent in footnote 4, where she discussed the shift to a more inertial monetary policy rule with greater emphasis on output, aligning with anecdotal evidence. However, the part concerning dominant supply shocks in the 80s versus demand shocks in the 00s may warrant further discussion. In particular, Table 1 highlights a substantial difference in the magnitudes of volatilities σ_j for $j \in \{x, \pi, i\}$ across the two periods. Two notable references, Liu, Waggoner, and Zha (2011) and Bianchi and Ilut (2017), offer diverse volatility estimates within regime-switching New Keynesian models, both of which are highly cited. However, given the intricacies of their respective frameworks, which incorporate a broader array of structural shocks, direct comparisons may be challenging. In light of this complexity, it becomes imperative for the reader to discern whether the current paper's calibration, particularly on (P1), aligns broadly with the trends observed in the macro DSGE literature. Should disparities arise, elucidating the rationale behind such deviations becomes essential. This clarity is paramount, especially considering that subsequent counterfactual analyses pivot critically on the calibration values delineated in Table 1.

> **Reply:** Thank you for this helpful comment. I added a detailed discussion of the shock volatility calibrations and their relationship to the DSGE literature with regime switches to the revised manuscript.
>
> Liu, Waggoner and Zha (2011) and Bianchi and Ilut (2017) both rely on samples that start significantly earlier and end earlier than the one in this paper, and allow only for two different volatility regimes. As a result, they naturally detect one switch in shock volatilities around or shortly after 1980, characterized by a roughly proportional decline in the volatilities of all shocks. My sample starts roughly at the time of their switch in volatilities. I therefore cannot detect the shift from high volatility to low volatility that they focus on. I use a break around 2000 and estimate a shift in the relative volatilities of supply-type v. demand-type shocks. This break date is supported by the more reduced-form approach to regime switches of Sims and Zha (2006) and the monetary policy model with term premia of Bianchi, Lettau, and Ludvigson (2022), who also detect switches around 2000.
>
> In addition, a common shift in all volatilities would leave regression coefficients, such as the target moments depicted in Figure 2, unchanged, so my calibration procedure naturally is more sensitive to

relative shifts in the volatilities of shocks. The overall volatility of shocks is identified in my calibration from the macroeconomic volatilities in the bottom panel of Table 2, but is less central to the question studied here than in the macroeconomics literature on regime switches. The revised manuscript clarifies the relationship to Liu, Waggoner, and Zha (2011) and Bianchi and Ilut (2017) in the introduction (**p.6 "The paper is also complementary to a long-standing macroeconomics literature on DSGE models with regime switches (…)"**) and in the calibration section (**p.21, paragraph starting "The change in the shock volatilities in Table 1 is consistent (…)"**).

To further support the high supply shock volatility in my 1980s calibration, I added a discussion of examples of supply shocks during this period, and a simple newspaper count exercise to the revised manuscript. The newspaper count exercise is shown in the **new Appendix Figure A1**, reproduced below for your convenience. The examples are discussed in the introduction (**p.3 "The calibrated changes in the monetary policy rule and the nature of macroeconomic shocks are consistent with anecdotal evidence…"**) and the calibration section (**last paragraph on p.18, and new footnote 14).**

Figure A1: Relative Frequency of Supply and Inflation in WSJ Articles



Following the suggestion of the AE, I also added a new reverse-engineering exercise showing that a high volatility of supply shocks is needed to explain the bond-stock betas of the 1980s starting from the 2000s calibration. These results are shown in the **new Figure 8** in the paper and discussed in the **new Section 4.4 starting on p.31**.

Dear Professor Whited,

Thank you very much for the opportunity to submit a revised version of my manuscript, and for the clear guidance. I am thrilled to resubmit a revised manuscript. Detailed responses are included below, but first I would like to provide an overview of the main changes.

- The **revised introduction states clearly the key contribution that sets this paper apart** from my own prior work and the broader literature on changing bond-stock comovements. In short, macroeconomic dynamics have changed so much over the past 30 years that changing one set of parameters is insufficient to return bonds to their risky state of the 1980s. Instead, a "perfect storm" consisting of a change towards more volatile supply shocks and a more hawkish monetary policy rule is needed to generate such reversion in bond risks.

- The revised paper provides additional discussion and results supporting the calibration for the first subperiod. Following the AE's suggestion, the paper includes the **new Figure 8** showing that a **high volatility of supply shocks is needed to replicate the bond-stock betas of the first subperiod starting from the second subperiod**. The revised paper also includes new examples and references for supply shocks during the 1980s and 1990s, such as the sudden drop in oil prices in 1986. I also added a newspaper count exercise in the **new Appendix Figure A1** showing that supply shocks were a substantial concern during the 1980s and 1990s.

- The revised paper provides an **updated discussion of recent events**. The **new Section 5** emphasizes the difference between an initially weak Fed inflation response during the early post-pandemic and a later much stronger Fed inflation response during the late post-pandemic period. I thereby address the referee's and AE's observation that the Fed eventually did respond strongly to the recent spike in inflation.

Thank you again for the opportunity to submit a revised version of this manuscript to the *JFE*. My apologies for taking slightly longer for this revision. I had a baby in September 2024, and I believe this timeline enabled me to respond fully to the thoughtful comments by the AE and the referees.

Sincerely,

Carolin

Title Page with Author Information

Back to the 1980s or Not? The Drivers of Inflation and Real Risks in Treasury Bonds*

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December 20, 2024

Abstract

This paper shows that supply shock uncertainty interacts with the monetary policy rule to drive bond risks in a New Keynesian asset pricing model. In my model, positive nominal bond-stock betas emerge as the result of volatile supply shocks but only if the monetary policy rule features a high inflation weight. Habit formation preferences generate endogenously time-varying risk premia, explaining the volatility and predictability of bond and stock excess returns in the data, and implying that bond-stock betas price the expected equilibrium mix of shocks rather than realized shocks. The model explains the change from positive nominal and real bond-stock betas in the 1980s to negative nominal and real bond-stock betas in the 2000s with a shift from dominant supply shocks and an inflation-focused monetary policy rule, to demand shocks in the 2000s. Post-pandemic nominal and real bond-stock betas are explained with dominant supply shocks and a late increase in the monetary policy inflation coefficient.

Keywords: Bond betas, stagflation, soft landing, supply shocks, demand shocks, monetary policy, New Keynesian, time-varying risk premia
JEL Classifications: E43, E52, E58

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1 Introduction

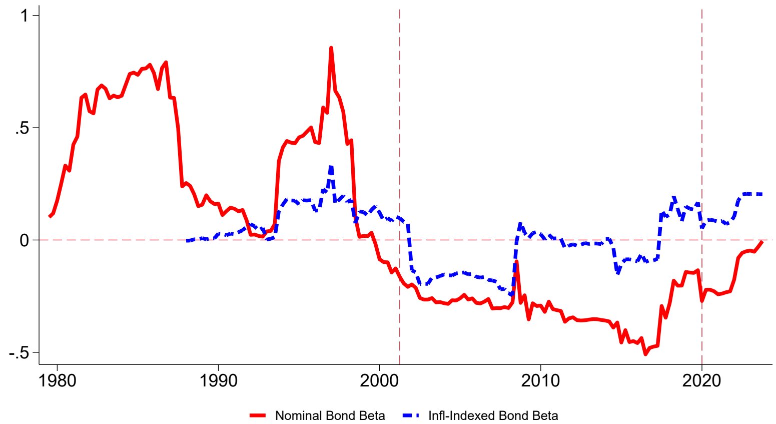
What do nominal and real Treasury bond risks tell us about supply shocks and stagflation or, conversely, the Fed's ability to engineer a "soft landing"? Nominal and inflation-indexed bonds have different inflation exposures, so nominal bond-stock betas are intuitively informative about inflation dynamics, whereas inflation-indexed bond-stock betas are informative about the dynamics of real rates. Figure 1 shows that during the 1980s nominal bond betas were positive and significantly larger than inflation-indexed bond betas, as one would expect if inflation expectations rise during recessions and investors worry about stagflation.[1] Nominal bond betas changed sign and the gap between nominal and inflation-indexed bond betas narrowed during the 2000s, as one would expect if inflation is less volatile and tends to rise in expansions. Maybe surprisingly, Figure 1 newly shows that nominal bond-stock betas remained negative for much of the post-pandemic period. This raises the question why – different from the 1980s – nominal bond-stock betas did not rise during the recent spike in inflation, despite perceived macroeconomic similarities.

Calibrating a New Keynesian equilibrium model for asset prices separately to the 1980s and the 2000s, I find that over the past 30 years monetary policy and the real side of the economy have changed sufficiently that reverting either set of parameters is insufficient to revert bond risks to their levels in the 1980s. Instead, a "perfect storm" consisting of a change towards more volatile supply shocks and a more hawkish monetary policy rule is needed to generate a reversion back to the risky bond markets of the 1980s. The model hence explains the different bond-stock betas during the recent period with substantial supply shock uncertainty and an initially low but increasing monetary policy inflation coefficient.

Despite the substantial literature studying historical bond-stock comovements (e.g. David and Veronesi (2013), Song (2017), Campbell, Sunderam, and Viceira (2017), Campbell, Pflueger, and Viceira (2020, CPV)), the link to monetary policy and macroeconomic shocks within a general equilibrium asset pricing framework has still been elusive. The habit formation preferences in this paper build on CPV and Pflueger and Rinaldi (2022, PR), giving rise to countercyclical risk premia, volatile and predictable stock returns, and a log-linear macroeconomic Euler equation linking output, consumption, and the monetary policy rate. However, the model in this paper contains several innovations over CPV and PR, in particular demand and supply shocks and adaptive inflation expectations. These innovations give it a unique ability to conduct counterfactual exercises to assess the impact of changes in the

[1] Figure 1 depicts five-year rolling nominal and inflation-indexed bond-stock return betas from 1979.Q4 through 2022. I use UK inflation-linked bond yields prior to 1999 and yields on US Treasury Inflation-Protected Securities (TIPS) after 1999, when TIPS data becomes available. Campbell, Shiller, and Viceira (2009) find similar changes in US and UK nominal and inflation-indexed bond-stock betas.

Figure 1: Rolling Nominal and Real Bond-Stock Betas



Note: This figure shows betas from regressing quarterly ten-year Treasury bond excess returns onto quarterly US equity excess returns over five-year rolling windows for the period 1979.Q4-2023.Q4. Quarterly excess returns are in excess of three-month T-bills. Prior to 1999, I replace US Treasury Inflation-Protected (TIPS) returns with UK ten-year linker returns. Bond excess returns are computed from changes in yields. Zero-coupon yield curves are from Gürkaynak, Sack, and Wright (2007, 2010) and the Bank of England. Vertical lines indicate 2001.Q2 and the start of the pandemic 2020.Q1.

macroeconomic shock volatilities and Taylor rule coefficients on bond-stock betas. Different from CPV and PR, a bond preference shock gives rise to a negative demand shock, lowering real consumption at a given policy rate, similar to the safety shock that has been increasingly successful in international finance.[2] The supply side of the model features adaptive wage-setter inflation expectations and sticky wages in the manner of Rotemberg (1982). Wage markup shocks give rise to supply shocks to the wage Phillips curve. Monetary policy is modeled as a Taylor (1993)-type rule, whereby the nominal monetary policy rate increases with inflation and the output gap and has an inertial weight on the lagged policy rate. Stocks in the model represent a levered claim to firm profits, and investors are assumed to have rational inflation expectations. Demand, supply, and monetary policy shocks are assumed to be uncorrelated, as is standard for structural shocks.

I calibrate the model separately to macroeconomic data from the 1980s and 2000s, thereby using the well-understood changes in the macroeconomy over these decades as a laboratory for my model of bond-stock betas. The model matches the changing bond-stock betas from the 1980s to the 2000s with a change from a supply-shock driven economy in the 1980s to a demand-shock driven one in the 2000s, and a change from a quick-acting and inflation-focused monetary policy rule in the 1980s to an inertial and more output-focused

[2] E.g. Bianchi and Lorenzoni (2021), Kekre and Lenel (2021), Jiang, Krishnamurthy, and Lustig (2021).

monetary policy rule in the 2000s. The volatilities of shocks and monetary policy parameters for each subperiod target inflation-output gap, fed funds rate-output gap, and inflation-fed funds rate relationships, as well as the volatilities of consumption growth, long-term inflation expectations, and the fed funds rate. I set the adaptiveness of wage-setters' inflation expectations to match the well-known predictability of bond excess returns of Campbell and Shiller (1991). I use a break date of 2001.Q2, when the correlation between inflation and the output gap turned from negative (i.e. stagflations) to positive (Campbell et al. (2020)), and Goodfriend (2007) argues monetary policy changed to a "new consensus".

The calibrated changes in the monetary policy rule and the nature of macroeconomic shocks are consistent with anecdotal evidence. For example, in recent decades central bankers have tended to move in incremental policy steps, and have shown substantial concern for output.[3] Since it has been argued that monetary policy opened up the economy to self-fulfilling sunspots or "inflation scares" during the 1970s (Clarida, Gali, and Gertler (2000), Goodfriend (2007)), I refrain from calibrating my model for the 1970s and instead focus on the 1980s and 1990s as an example of a supply-shock dominated economy. Examples of supply-type shocks during my first calibration period include the sudden drop in oil prices in 1986 (Gately, 1986) and the "New Economy" of the 1990s.[4]

I next use the calibrated model for a series of counterfactual analyses. The first counterfactual shows that combining 1980s-style shocks with a 2000s-style monetary policy rule implies negative nominal bond betas but positive real bond-stock betas, thereby explaining the empirical pattern of nominal and real bond-stock betas during 2021 and 2022. Intuitively, when the systematic component of monetary policy allows the real rate to fall in response to an inflationary supply shock, the recession is mitigated and a "soft landing" ensues. Nominal bond prices fall with higher inflation expectations, while stocks do not fall due to the accommodative monetary policy response, implying a slightly negative nominal bond-stock beta. The drop in the real rate boosts real bond prices, as well as the economy and stock prices, and a positive real bond-stock beta ensues. Nominal bond-stock betas in the model do not depend strongly on the adaptiveness of wage-setters' inflation expectations, but the predictability of nominal bond excess returns does.

[3]See Cieslak and Vissing-Jorgensen (2021), Bauer and Swanson (2023), and Bauer, Pflueger, and Sunderam (2024a) for direct empirical evidence of the Fed's output concern after the mid-1990s.

[4]In my model the volatility of shocks matters for bond-stock betas, but not the sign of the realized shocks. Different from the 1970s, the supply shocks of the 1980s and 1990s were often positive rather than negative. Greg Mankiw noted these supply shocks during the 1980s and 1990s in a *Fortune* article arguing: "Greenspan has been more fortunate. Shortly before he was appointed Fed chairman in 1987, world oil prices plummeted, improving the inflation-unemployment tradeoff. Over the past two years, the foreign-exchange market has provided a similar benefit (...)" N. Gregory Mankiw, "Alan Greenspan's Tradeoff", Fortune, December 8, 1997.

The second counterfactual shows that prevalent shocks – or the expected mix of shocks going forward – are priced in bond-stock betas, whereas realized shocks matter little. I show that model nominal bond-stock betas remain positive as long as the 1980s calibration is priced in equilibrium, even if the realized shocks are drawn from the 2000s distribution. The mechanism relies on habit formation preferences, which generate endogenous time-varying risk premia. Intuitively, in the 1980s equilibrium nominal bonds are priced as risky assets because they are expected to pay out in low marginal utility states. An increase in risk aversion—whether caused by realizations of demand, supply, or monetary policy shocks—causes investors to have a lower willingness to pay for risky nominal bonds and stocks, driving down nominal bond and stock prices simultaneously. Bond betas in the model should hence be interpreted as forward-looking indicators.

The third counterfactual computes the changes in parameters necessary to replicate 1980s bond-stock betas starting from the 2000s calibration. I find that a high volatility of supply shocks is necessary to reverse-engineer the bond-stock betas observed during the first subperiod. Consistent with the notion that a "perfect storm" is needed to make bonds risky again, a high supply shock volatility is necessary but not sufficient to replicate the high nominal bond-stock betas of the 1980s. A high monetary policy inflation coefficient and low monetary policy inertia are also needed. This counterfactual hence further supports a high supply shock volatility for the first calibration period.

Finally, I show that the model gives an intuitive account of the recent period, which was characterized by high inflation not seen since the 1980s. For this, I turn to bond-stock betas from daily returns to measure the risks of Treasury bonds at higher frequency. The model can explain negative nominal bond-stock betas and slightly positive real bond-stock betas from daily returns during the early post-pandemic (until 2023.Q2), and positive nominal, real, and breakeven (i.e. nominal-minus-real) bond betas from daily returns during the later post-pandemic (2023.Q3-Q4). It does so by selecting a low monetary policy inflation coefficient for the early post-pandemic and a high monetary policy inflation coefficient for the later post-pandemic, as well as 1980s-style shocks for the entire post-pandemic period. The late model-implied change towards a more inflation-focused monetary policy rule is consistent with the timeline of events, as inflation rose already in 2021, well before the Fed started lifting interest rates. The delayed change in the model-implied monetary policy rule is also consistent with evidence from rich survey data (Bauer, Pflueger, and Sunderam (2024b)). Taken together, the framework in this paper is a useful tool for understanding the monetary policy stance and macroeconomic shocks priced by nominal and real bond-stock betas during the recent inflationary spike.

This paper contains distinct contributions over the literature studying changing bond-stock comovements. Various channels have been proposed in the literature to explain the changes in bond-stock betas over the past four decades, including changing inflation risks (Piazzesi and Schneider (2006), David and Veronesi (2013), Song (2017), Campbell, Sunderam, and Viceira (2017), Campbell, Sunderam, and Viceira (2017), Gourio and Ngo (2020), and Fang, Liu, and Roussanov (2022)); liquidity, flight-to-safety, and time-varying risk aversion (Baele, Bekaert, and Inghelbrecht (2010), Kozak and Santosh (2020), and Laarits (2022)); permanent-transitory compositional changes of consumption growth (Chernov, Lochstoer, and Song (2021)); and changes in the monetary-fiscal mix regimes (Li, Zha, Zhang, and Zhou (2022)). The model in this paper incorporates these channels in a relatively parsimonious setup, where inflation risk and the permanent-transitory decomposition arise endogenously from the nature of macroeconomic shocks and monetary policy, and habit formation preferences give rise to time-varying risk aversion and hence endogenous "flight-to-safety".

Compared to CPV, PR, and Song (2017), the ability to perform counterfactual exercises assessing how changes in Taylor rule coefficients *and* macroeconomic volatilities impact bond-stock betas within a general equilibrium model sets this paper apart. In contrast to this paper, in CPV the monetary policy rule is not specified. CPV's reduced-form equation for interest rate dynamics cannot be interpreted as a structural monetary policy rule because its innovations are correlated with inflation innovations, and it has a negative coefficient on lagged inflation. The model in Pflueger and Rinaldi (2022) is more similar to the one in this paper, containing a structural Phillips curve and Taylor-type monetary policy rule. The calibration of the model for two distinct subperiods sets the current paper apart from PR. The model in this paper also contains innovations that are crucial to reach its conclusion, such as demand and supply shocks (PR has neither), and adaptive inflation expectations. Another closely related paper is Song (2017), which however considers an endowment economy.

The model in this paper is also distinct from prior work that models the impact of changes to the volatilities of macroeconomic shocks or to monetary policy separately, whereas it is central to the conclusion in this paper to model these changes jointly. Different from the current paper, the production-based models of Fang, Liu, and Roussanov (2022), and Kozak (2022) do not incorporate changes in monetary policy. Conversely, Gourio and Ngo (2020) and Li, Zha, Zhang, and Zhou (2022) model changes in monetary policy but do not incorporate changes in the composition of macroeconomic shocks. Finally, Gourio and Ngo (2022) consider a model where the only macroeconomic shocks, technology shocks, have constant volatility and bond-stock comovements are closely linked to the level of inflation. Because of the close link between the level of inflation and bond-stock comovements, their model is not suited to understanding the key empirical finding in this paper that bond-stock betas remained negative even as inflation surged during 2021-2022. Hence, none of those previous papers are sufficient to reach the key conclusion that a "perfect storm" is needed to revert bond risks to the levels of the 1980s.

The paper is also complementary to a long-standing macroeconomics literature on DSGE models with regime switches in policy and/or the volatility of shocks, such as Liu, Waggoner, and Zha (2011) and Bianchi and Ilut (2017). Using samples that start and end substantially earlier than mine, these two papers detect a change from a high-volatility to a low-volatility regime around 1980, i.e. the start of my sample period. The more reduced form approach to regime switches of Sims and Zha (2006) and the macro-asset pricing approach of Bianchi, Lettau, and Ludvigson (2022a) also detect breaks around 2000, consistent with the start of my second subperiod. Consistent with the recent macroeconomics and inflation literature, I find that bond-stock betas support a combination supply shocks with initially easy monetary policy during the post-pandemic period (e.g. Gagliardone and Gertler (2023), Rubbo (2022), Hazell and Hobler (2024), Bocola et al. (2024), Cieslak et al. (2024)).

Finally, this paper more broadly contributes to the literatures jointly modeling term premia within New Keynesian models (Kung (2015), Rudebusch and Swanson (2012), Kekre, Lenel, and Mainardi (2024)), stock and credit risk premia within New Keynesian models (Kekre and Lenel (2022), Swanson (2021), Bianchi, Lettau, and Ludvigson (2022a), Bianchi, Ludvigson, and Ma (2022c)), and optimal monetary policy with time-varying risk premia (e.g. Gilchrist and Leahy (2002), Caballero and Simsek (2022)). Complementary to this paper, Bauer, Pflueger, and Sunderam (2024a,b) cannot speak to the role of supply shocks and contain no general equilibrium asset pricing model, so they are insufficient to reach the key conclusion of the present paper that a "perfect storm" is needed to turn bonds risky. Bond-stock betas studied here are informative about monetary policy and economic shocks over the lifetime of the assets, and matter separately, for example for portfolio allocation and Treasury debt issuance.

2 Model

I use lower-case letters to denote logs throughout, π_t to denote log price inflation, and π_t^w to denote log wage inflation. I refer to price inflation and inflation interchangeably.

2.1 Preferences

A representative agent derives utility from real consumption C_t relative to a slowly moving habit level H_t:

$$U_t = \frac{(C_t - H_t)^{1-\gamma} - 1}{1-\gamma}. \tag{1}$$

Habits are external, meaning that they are shaped by aggregate consumption and households do not internalize how habits might respond to their personal consumption choices. The parameter γ is a curvature parameter. Relative risk aversion equals $-U_{CC}C/U_C = \gamma/S_t$, where surplus consumption is the share of consumption available to generate utility:

$$S_t = \frac{C_t - H_t}{C_t}. \tag{2}$$

Risk aversion therefore increases when consumption has fallen close to habit. As equation (2) makes clear, a model for market habit implies a model for surplus consumption and vice versa. As in Campbell, Pflueger, and Viceira (2020), I model market consumption habit implicitly by assuming that log surplus consumption, s_t, satisfies:

$$\begin{aligned} s_{t+1} &= (1-\theta_0)\bar{s} + \theta_0 s_t + \theta_1 x_t + \theta_2 x_{t-1} + \lambda(s_t)\varepsilon_{c,t+1}, &(3)\\ \varepsilon_{c,t+1} &= c_{t+1} - \mathrm{E}_t c_{t+1}. &(4)\end{aligned}$$

Here, x_t equals the log output gap, defined as log real output minus log potential real output at perfectly flexible prices and wages. In Section 2.4, potential depends on a moving average of past output and there is no real investment, yielding a useful expression linking the output gap and consumption (up to a constant):

$$x_t = c_t - (1-\phi)\sum_{j=0}^{\infty} \phi^j c_{t-1-j}. \tag{5}$$

Here, ϕ is a smoothing parameter. Expression (5) implies that log consumption growth is stationary – as is standard in asset pricing – but the output gap is stationary in levels – as is standard in macroeconomics. It is also consistent with the smoothing embedded in typical empirical proxies of potential. Provided that the parameter ϕ is close to one, asset price dynamics are relatively insensitive to its precise value.

The sensitivity function $\lambda(s_t)$ takes the form as in Campbell and Cochrane (1999)

$$\begin{aligned} \lambda(s_t) &= \begin{cases} \frac{1}{\bar{S}}\sqrt{1-2(s_t-\bar{s})} - 1 & s_t \leq s_{max} \\ 0 & s_t > s_{max} \end{cases}, &(6)\\ \bar{S} &= \sigma_c\sqrt{\frac{\gamma}{1-\theta_0}},\ \bar{s} = \log(\bar{S}),\ s_{max} = \bar{s} + 0.5(1-\bar{S}^2). &(7)\end{aligned}$$

This function is decreasing in log surplus consumption, so marginal utility becomes more sensitive to consumption surprises when surplus consumption is already low, as would be the case after a sequence of bad shocks. Here, σ_c denotes the standard deviation of the consumption surprise $\varepsilon_{c,t+1}$, and $\bar{s}$ is the steady-state value for log surplus consumption. Both consumption and the output gap are equilibrium objects that depend on fundamental shocks, and in equilibrium they are conditionally homoskedastic and lognormal. As shown in Campbell, Pflueger, and Viceira (2020), implied log habit follows approximately a weighted average of lagged consumption and lagged consumption expectations. The asset pricing habit formation preferences used here are known to capture a wide range of asset pricing moments, including the high volatility of stock returns, their high Sharpe ratio, the predictability of stock excess returns (Campbell and Cochrane (1999)), and the risk premium effect of high-frequency monetary policy surprises (Bernanke and Kuttner (2005), Pflueger and Rinaldi (2022)). However, the mechanism more broadly relies on countercyclical risk premia, whether they are generated from the price of risk as here, the quantity of risk as in Jurado, Ludvigson, and Ng (2015), or heterogeneous agents with different risk aversion (Chan and Kogan (2002), Kekre and Lenel (2022), Caballero and Simsek (2020), Drechsler, Savov, and Schnabl (2018)).

2.2 Asset Pricing Equations and Bond Preference Shock

The stochastic discount factor (SDF) M_{t+1} is derived from (1):

$$M_{t+1} = \beta \frac{\frac{\partial U_{t+1}}{\partial C}}{\frac{\partial U_t}{\partial C}} = \beta \exp\left(-\gamma(\Delta s_{t+1} + \Delta c_{t+1})\right). \tag{8}$$

I model stocks as a levered claim on consumption or equivalently firm profits, while preserving the cointegration of consumption and dividends. The asset pricing recursion for a claim paying consumption at time $t+n$ and zero otherwise takes the following form

$$\frac{P^c_{n,t}}{C_t} = E_t\left[M_{t+1}\frac{C_{t+1}}{C_t}\frac{P^c_{n-1,t+1}}{C_{t+1}}\right]. \tag{9}$$

The price-consumption ratio for a claim to all future consumption then equals

$$\frac{P^c_t}{C_t} = \sum_{n=1}^{\infty}\frac{P_{n,t}}{C_t}. \tag{10}$$

At time t the aggregate levered firm buys P^c_t and sells equity worth δP^c_t, with the remainder of the firm's position financed by one-period risk-free debt worth $(1-\delta)P^c_t$, so the price of the levered equity claim equals $P^\delta_t = \delta P^c_t$. Log zero coupon yields are defined by $y_{n,t} = -\log P_{n,t}$ and $y^\$_{n,t} = -\log P^\$_{n,t}$, and log excess returns in excess of the corresponding 1-quarter rate from t to $t+h$ are denoted by $xr^\$_{n,t\to t+h}$ and $xr_{n,t\to t+h}$.

The bond asset pricing recursions are subject to a bond preference shock ξ_t. The Euler equation for the one-period risk-free rate is given by:

$$1 = E_t \left[M_{t+1} exp \left(r_t - \xi_t \right) \right], \tag{11}$$

and one-period real and nominal interest rates are linked via the Fisher equation

$$i_t = E_t \pi_{t+1} + r_t. \tag{12}$$

Equation (12) is an approximation, effectively assuming that the inflation risk premium in one-period nominal bonds is zero. Longer-term bond prices do not use this approximation and are given by the recursions:

$$P^{\$}_{1,t} = exp(-i_t), \; P_{1,t} = exp(-r_t), \tag{13}$$
$$P^{\$}_{n,t} = exp(-\xi_t) E_t \left[M_{t+1} exp(-\pi_{t+1}) P^{\$}_{n-1,t+1} \right], \; P_{n,t} = exp(-\xi_t) E_t \left[M_{t+1} P_{n-1,t+1} \right], \tag{14}$$

where all expectations are rational. Because all bonds are priced with the preference shock ξ_t, the expectations hypothesis holds when investors are risk-neutral.

Fundamentally, the shock ξ_t corresponds to a disconnect between the bond and the stock markets, as for example documented empirically during March 2020 by He, Nagel, and Song (2022). A positive shock ξ_t acts like a decline in Treasury bond convenience, analogous to the time-varying intermediation capacity that has been successful at reconciling several empirical puzzles in international finance, and may reflect that households do not have direct access to the government bond market.[5] Alternatively, the shock ξ_t can be microfounded as an optimism or growth shock, similar to expectations-based demand shocks in Beaudry and Portier (2006), Bordalo, Gennaioli, LaPorta, and Shleifer (2022) and Caballero and Simsek (2022)'s "traditional financial forces" shock. For microfoundations see Appendix D.

2.3 Macroeconomic Euler Equation from Preferences

I next show that the bond preference shock enters equilibrium macroeconomic dynamics just like a demand shock in the Euler equation. Starting from the asset pricing equation for a one-period risk-free bond (11), and substituting for the SDF and surplus consumption

[5] See e.g. Krishnamurthy and Vissing-Jorgensen (2012), Du, Im, and Schreger (2018), Bernanke and Gertler (2001), Bianchi and Lorenzoni (2021), Gabaix and Maggiori (2015), Jiang, Krishnamurthy, and Lustig (2021), Itskhoki and Mukhin (2021). Pflueger, Siriwardane, and Sunderam (2020) provide US evidence that preference for safety not immediately driven by aggregate risk aversion can forecast business cycle variables.

dynamics gives (up to a constant):

$$r_t = \gamma \mathrm{E}_t \Delta c_{t+1} + \gamma \mathrm{E}_t \Delta s_{t+1} - \frac{\gamma^2}{2} (1 + \lambda(s_t))^2 \sigma_c^2 + \xi_t, \tag{15}$$

$$= \gamma \mathrm{E}_t \Delta c_{t+1} + \gamma \theta_1 x_t + \gamma \theta_2 x_{t-1} + \underbrace{\gamma(\theta_0 - 1)s_t - \frac{\gamma^2}{2} (1 + \lambda(s_t))^2 \sigma_c^2}_{=0} + \xi_t. \tag{16}$$

The sensitivity function (6) through (7) has the advantageous property that the two bracketed terms drop out, and the real risk-free rate has the familiar log-linear form, and much lower volatility than the stock market. Substituting (5) then gives the exactly log-linear macroeconomic **Euler equation**:

$$x_t = f^x E_t x_{t+1} + \rho^x x_{t-1} - \psi r_t + v_{x,t}. \tag{17}$$

Imposing the restriction that the forward- and backward-looking terms in the Euler equation add up to one, the Euler equation parameters are given by

$$\rho^x = \frac{\theta_2}{\phi - \theta_1}, f^x = \frac{1}{\phi - \theta_1}, \psi = \frac{1}{\gamma(\phi - \theta_1)}, \theta_2 = \phi - 1 - \theta_1. \tag{18}$$

Non-zero values for the habit parameters, θ_1 and θ_2, are therefore needed to generate the standard New Keynesian block with forward- and backward-looking coefficients. The demand shock in the Euler equation equals

$$v_{x,t} = \psi \xi_t. \tag{19}$$

Because consumption is endogenous, a decrease in the preference for government bonds ($\xi_t \uparrow$) tends to raise both consumption and the real rate through (17). Higher consumption, in turn, raises dividends and lowers risk aversion. The preference shock ξ_t hence raises stock prices, even though it does not appear directly in the Euler equation for consumption claims. The demand shock $v_{x,t}$ is conditionally homoskedastic, serially uncorrelated, and uncorrelated with supply and monetary policy shocks because ξ_t is. Its standard deviation is denoted by σ_x.[6]

[6]While a shock to the discount factor shared by bonds and stocks (Albuquerque, Eichenbaum, Luo, and Rebelo (2016)) also generates a demand shock in the Euler equation, a joint discount rate shock tends to move bonds and stocks in the same direction unless the endogenous cash flow effect is very strong. Different from a bond preference shock, a joint discount rate shock hence cannot explain the negative real bond-stock beta observed during the pre-pandemic 2000s.

2.4 Supply Side

I keep the supply side as simple as possible to generate a standard log-linearized Phillips curve, and the link between consumption and the output gap. Details are relegated to the Appendix. There is no real investment, and the aggregate resource constraint simply states that aggregate consumption equals aggregate output, i.e. $C_t = Y_t$. Following Lucas (1988) I assume that productivity depends on past economic activity. Potential output is defined as the level of real output that would obtain with flexible prices and wages taking current productivity as given. The log output gap is the difference between log real output and log potential real output and in equilibrium satisfies (5).

Wage unions charge sticky wages but firms' product prices are flexible. Specifically, I assume that wage-setters face a quadratic cost as in Rotemberg (1982) if they raise wages faster than past inflation. The indexing to past inflation is analogous to the indexing assumption in Smets and Wouters (2007) and Christiano, Eichenbaum, and Evans (2005). I assume that households experience disutility of working outside the home due to the opportunity cost of home production as in Greenwood, Hercowitz, and Huffman (1988), with external home production habit defined so that home production drops out of the intertemporal consumption decision and the asset pricing stochastic discount factor. Log-linearizing the intratemporal first-order condition of wage-setting unions gives the **Phillips curve**:

$$\pi_t^w \quad = \quad f^\pi E_t \pi_{t+1}^w + \rho^\pi \pi_{t-1}^w + \kappa x_t + v_{\pi,t}, \tag{20}$$

for constants ρ^π, f^π, and κ. The parameter κ is a wage-flexibility parameter. The supply or Phillips curve shock $v_{\pi,t}$ is assumed to be conditionally homoskedastic with standard deviation $\sigma_{\pi,t}$, serially uncorrelated, and uncorrelated with other shocks. This supply shock can arise from a variety of sources, such as variation in optimal wage markups charged by unions or shocks to the marginal utility of leisure.[7]

Following Fuhrer (1997) wage-setters have partially adaptive subjective inflation expectations

$$\tilde{E}_t \pi_{t+1}^w \quad = \quad (1-\zeta) E_t \pi_{t+1}^w + \zeta \pi_{t-1}^w, \tag{21}$$

where E_t denotes the rational expectation conditional on state variables at the end of period

[7] While I do not model fiscal sources of inflation, under certain conditions a shock to inflation expectations due to fiscal policy can act similarly to a shift to the Phillips curve (Hazell, Herreno, Nakamura, and Steinsson (2022), Bianchi, Faccini, and Melosi (2023)). Up to the distinction between wage and price inflation, supply shocks are also isomorphic to shifts to potential output unrecognized by the central bank, in which case x_t is the output gap perceived by consumers and the central bank, and the actual output gap is $x_t + \frac{1}{\kappa} v_{\pi,t}$.

t. Hence, while financial assets are priced with rational inflation expectations, wage-setters' expectations are more sluggish, capturing the idea that markets are more sophisticated and attentive to macroeconomic dynamics than individual wage-setters. A similar assumption has been used by Bianchi, Lettau, and Ludvigson (2022a). A long-standing Phillips curve literature has found that adaptive inflation expectations and a strongly backward-looking Phillips curve are needed to capture the empirical persistence of inflation (Fuhrer and Moore (1995), Fuhrer (1997)).[8] If $\rho^{\pi,0}$ is the backward-looking component under rational inflation expectations ($\zeta = 0$), the backward- and forward-looking Phillips curve parameters equal:

$$\rho^{\pi} = \rho^{\pi,0} + \zeta - \rho^{\pi,0}\zeta, \;\; f^{\pi} = 1 - \rho^{\pi}. \tag{22}$$

Ten-year survey inflation expectations are modeled similarly as a weighted average of a moving average of inflation over the past ten years and the rational forecast, with the weight on past inflation given by ζ.

Equilibrium price inflation equals wage inflation minus productivity growth, which depends on the output gap:

$$\pi_t \;\; = \;\; \pi_t^w - (1-\phi)x_{t-1}. \tag{23}$$

In the calibrated model, ϕ is close to one, and price and wage inflation are very similar.[9] The reason to assume sticky wages rather than sticky prices is simply that with these assumptions a consumption claim (Abel (1990)) is identical to a claim to firm profits.[10]

[8] Consistent with this older literature, a quickly growing literature has documented deviations from rationality (Coibion and Gorodnichenko (2015), Bianchi, Ludvigson, and Ma (2022b)) and excess dependence of expectations on lagged inflation (Malmendier and Nagel (2016)).

[9] In reality, the link between price and wage inflation is arguably less close than in the model. Appendix G solves a model extension with an additional shock to the price-wage dynamics (23), showing that the results are unchanged even when the volatility of this shock is set to a high value. For parsimony, I hence abstract from shocks to wage-price dynamics in the baseline version of the model.

[10] It is also in line with Christiano, Eichenbaum, and Evans (1999) who find that sticky wages are more important for aggregate inflation dynamics than sticky prices. See also Favilukis and Lin (2016) who find that wage-setting frictions are important to ensure that a claim to firm profits behaves similarly to a claim to consumption in an asset pricing sense. Appendix E.4 shows that model implications are robust to setting wage and price inflation equal through $\phi = 1$.

2.5 Monetary Policy

Let i_t denote the log nominal risk-free rate available from time t to $t+1$. Monetary policy is described by the following rule (ignoring constants):

$$i_t = \rho^i i_{t-1} + \left(1-\rho^i\right)\left(\gamma^x x_t + \gamma^\pi \pi_t\right) + v_{i,t}, \tag{24}$$

$$v_t \sim N\left(0, \sigma_i^2\right). \tag{25}$$

While there is an active debate to what extent monetary policy can be described by rules versus discretion, rules such as (24) are often found to provide a good description of historical policy rates (Taylor (1993), Clarida, Gali, and Gertler (2000)). The monetary policy shock in (24) can be interpreted to represent the discretionary component of monetary policy.[11] The term $\gamma^x x_t + \gamma^\pi \pi_t$ denotes the central bank's interest rate target, to which it adjusts slowly. The parameters γ^x and γ^π represent monetary policy's long-term output gap and inflation weights. The monetary policy shock, $v_{i,t}$, is assumed to be uncorrelated with supply and demand shocks, serially uncorrelated, and conditionally homoskedastic. A positive monetary policy shock represents a surprise tightening of the short-term nominal policy rate, which mean-reverts at rate ρ^i.

2.6 Model Solution

The solution proceeds in two steps. Different from Campbell et al. (2020), I need x_{t-1} as an additional state variable because it enters the surplus consumption ratio dynamics (3), and the new demand shock means that it is no longer spanned by the other state variables. First, I solve for log-linear macroeconomic dynamics. Second, I use numerical methods to solve for highly non-linear asset prices. This is aided by the particular tractability of the surplus consumption dynamics, implying that the surplus consumption ratio is a state variable for asset prices but not for macroeconomic dynamics. I solve for the dynamics of the log-linear state vector

$$Y_t = [x_t, \pi_t^w, i_t]'. \tag{26}$$

The dynamics of these equilibrium objects are driven by the vector of exogenous shocks

$$v_t = [v_{x,t}, v_{\pi,t}, v_{i,t}], \tag{27}$$

[11] I do not model the zero lower bound, because I am interested in longer-term regimes, and a substantial portion of the zero lower bound period appears to have been governed by expectations of a swift return to normal (Swanson and Williams (2014)). The zero-lower-bound may however be important for more cyclical changes in bond-stock betas, as emphasized by Gourio and Ngo (2020), and I leave this to future research.

according to the consumption Euler equation (17), the Phillips curve (20), the monetary policy rule (24), and the wage-price inflation link (23). I solve for a minimum state variable equilibrium of the form

$$Y_t = BY_{t-1} + \Sigma v_t, \tag{28}$$

where B and Σ are $[3 \times 3]$ and $[3 \times 3]$ matrices, and v_t is the vector of structural shocks. I solve for the matrix B using Uhlig (1999)'s formulation of the Blanchard and Kahn (1980) method. In both calibrations, there exists a unique equilibrium of the form (28) with non-explosive eigenvalues. I acknowledge that, as in most New Keynesian models, there may be further equilibria with additional state variables or sunspots (Cochrane (2011)), but resolving these issues is beyond the scope of this paper. Note that equation (28) implies that macroeconomic dynamics are conditionally lognormal. The output gap-consumption link (5) therefore implies that equilibrium consumption surprises $\varepsilon_{c,t+1}$ are conditionally lognormal, as previously conjectured.

The key properties of endogenously time-varying risk premia can be illustrated with a simple analytic expression. Consider a one-period claim with log real payoff αc_t. For illustrative purposes consider α to be an exogenous constant, though in the full model it depends on the macroeconomic equilibrium. Denoting the log return on the one-period claim by $r_{1,t+1}^{c,\alpha}$, the risk premium—adjusted for a standard Jensen's inequality term—equals the conditional covariance between the negative log SDF and the log real asset payoff:

$$E_t\left[r_{1,t+1}^{c,\alpha} - r_t\right] + \frac{1}{2}Var\left(r_{1,t+1}^{c,\alpha}\right) = \mathrm{Cov}_t\left(-m_{t+1}, x_{t+1}\right) = \alpha\gamma\left(1+\lambda\left(s_t\right)\right)\sigma_c^2. \tag{29}$$

This expression shows that assets with risky real cash flows ($\alpha > 0$) require positive risk premia. Since $\lambda(s_t)$ is downward-sloping, risk premia on risky assets increase further after a series of bad consumption surprises. Conversely, assets with safe real cash flows ($\alpha < 0$) require negative risk premia that decrease after a series of bad consumption surprises. Because real cash flows on nominal bonds are inversely related to inflation, nominal bonds resemble a risky asset ($\alpha > 0$) if inflation is countercyclical (i.e. stagflations) but a safe asset ($\alpha < 0$) if inflation falls in bad times.

Because full asset prices are not one-period claims, I use numerical value function iteration to solve the recursions (13) through (10) while accounting for the new demand shock and the link between wage and price inflation (23). Asset prices have five state variables: three state variables in Y_t, the lagged output gap x_{t-1}, and the surplus consumption ratio s_t.

3 Empirical Analysis and Calibration Strategy

Table 1 lists the parameters for the calibrations and how they vary across subperiods.

3.1 Calibration Strategy

I calibrate the model separately for two subperiods, where I choose the 2001.Q2 break date from Campbell, Pflueger, and Viceira (2020). This date was chosen by testing for a break in the inflation-output gap relationship and did not use asset prices. I start the sample in 1979.Q4, when Paul Volcker was appointed as Fed chairman. I end the sample in 2019.Q4 prior to the pandemic, leaving the analysis of how shocks changed during the pandemic period for a separate discussion. I do not account for the possibility that agents might have anticipated a change in regime.[12] The model is calibrated to macroeconomic moments, with only the inflation expectations parameter set to match Campbell and Shiller (1991)-type bond excess return predictability. The calibration does not match bond-stock betas directly but uses them as additional moments, because the solution for macroeconomic dynamics is much faster than the solution for asset prices. Section 4.4 provides a simple reverse-engineering exercise showing that a high supply shock volatility is needed to explain the bond-stock betas in the first subperiod starting from the calibration for the second subperiod.

3.2 Invariant Parameters

The calibration proceeds in three steps. First, I set some parameters to invariant values following the literature, shown in Panel B of Table 1. The expected consumption growth rate, utility curvature, the risk-free rate, and the persistence of the surplus consumption ratio (θ_0) are from Campbell and Cochrane (1999), who found that a utility curvature of $\gamma = 2$ gives an empirically reasonable equity Sharpe ratio and set θ_0 to match the quarterly persistence of the equity price-dividend ratio in the data. The output gap-consumption link parameter $\phi = 0.99$ is chosen to maximize the empirical correlation between stochastically detrended real GDP and the output gap from the Bureau of Economic Analysis. I choose a slightly higher value because the correlation between the output gap and stochastically detrended real GDP is flat over a range of values ($corr = 76\%$ at $\phi = 0.93$ vs. $corr = 73\%$ at $\phi = 0.99$), and higher ϕ minimizes the gap between price and wage inflation and hence simplifies the model. I calibrate $\theta_1 - \phi$ and hence the Euler equation exactly as in Pflueger and Rinaldi (2022), where the habit parameters θ_1 and θ_2 were chosen to replicate the hump-

[12] Cogley and Sargent (2008) show that an approximation with constant transition probabilities often provides a good approximation of fully Bayesian decision rules.

Table 1: Calibration Parameters

Panel A: Period-Specific Parameters		1979.Q4-2001.Q1	2001.Q2-2019.Q4
MP Inflation Coefficient	γ^{π}	1.35	1.10
		(0.53)	(0.06)
MP Output Coefficient	γ^{x}	0.50	1.00
		(0.74)	(0.30)
MP Persistence	ρ^{i}	0.54	0.80
		(0.18)	(0.04)
Vol. Demand Shock	σ_x	0.01	0.59
		(0.32)	(0.02)
Vol. PC Shock	σ_π	0.58	0.07
		(0.04)	(0.00)
Vol. MP Shock	σ_i	0.55	0.07
		(0.14)	(0.06)
Adaptive Inflation Expectations	ζ	0.6	0.0
		(0.51)	(2.67)
Leverage	δ	0.50	0.66

Panel B: Invariant Parameters		
Consumption Growth	g	1.89
Utility Curvature	γ	2
Risk-Free Rate	$\bar{r}$	0.94
Persistence Surplus Cons.	θ_0	0.87
Backward-Looking Habit	θ_1	-0.84
PC slope	κ	0.0062
Consumption-output gap	ϕ	0.99

Consumption growth and the real risk-free rate are in annualized percent. The standard deviation σ_x is in percent, and the standard deviations σ_π and σ_i are in annualized percent. The Phillips curve slope κ and the monetary policy parameters γ^{π}, γ^{x} and ρ^{i} are in units corresponding to the output gap in percent, and inflation and interest rates in annualized percent. Standard errors are computed using the delta method with details given in Appendix F.

shaped response of output to an identified monetary policy shock in the data. The second habit parameter, θ_2 is implied and set to ensure that the backward- and forward-looking components in the Euler equation sum up to one. Because the model impulse responses to a monetary policy shock are invariant to the shock volatilities and vary little with monetary policy and Phillips curve parameters, I effectively match habit preferences to the output response to an identified monetary policy shock. I set the slope of the Phillips curve to $\kappa = 0.0062$ based on Hazell, Herreno, Nakamura, and Steinsson (2022), who also find it to be stable over time. Appendix Table A4 shows that asset pricing implications are robust to choosing a different utility curvature γ, consumption-output gap link ϕ, and Phillips curve slope κ.

3.3 Period-Specific Shock Volatilities and Monetary Policy

The second step chooses period-specific monetary policy parameters and shock volatilities through Simulated Methods of Moments. Let $\hat{\Psi}$ denote the vector of twelve (13 for the second subperiod) empirical target moments, and $\Psi(\sigma_x, \sigma_\pi, \sigma_i, \gamma^x, \gamma^\pi, \rho^i; \zeta)$ the vector of model moments computed analogously on model-simulated data. I choose subperiod-specific monetary policy parameters γ^x, γ^π, and ρ^i and shock volatilities σ_x, σ_π, and σ_i while holding the inflation expectations parameter constant at $\zeta = 0$ to minimize the objective function:

$$\left\| \frac{\hat{\Psi} - \Psi(\sigma_x, \sigma_\pi, \sigma_i, \gamma^x, \gamma^\pi, \rho^i; \zeta = 0)}{SE(\hat{\Psi})} \right\|^2. \tag{30}$$

The vector of target moments $\hat{\Psi}$ includes the moments shown in Figure 2, plus the standard deviations of annual real consumption growth, the annual change in the fed funds rate, and the annual change in survey ten-year inflation expectations shown in the bottom panel of Table 2.[13] The vector of empirical standard errors $SE(\hat{\Psi})$ is computed via the delta method and Newey-West standard errors with h lags. I match many more empirical moments than I have parameters, so this is a demanding calibration objective. The rationale for including several lags from Figure 2 is that, for example, the negative empirical inflation-output gap relationship (i.e. stagflation) in the top-left panel is clearest at a 6-8 quarter lag horizon.

[13]Because I match three cross-relationships (output-inflation, output-fed funds, inflation-fed funds) at three different horizons (one, three, and seven quarters) and three volatilities, this step of the calibration procedure effectively chooses six parameters to fit $3 \times 3 + 3 = 12$ (13 for the second subperiod) moments. For the second calibration period when wage inflation data is easily available, I also estimate the specification $(z_t, y_t) = (x_t, \pi_t^w)$ and target the difference $a_{1,h}^{x,\pi} - a_{1,h}^{x,\pi^w}$. I include only one horizon for wage inflation to avoid over-weighting inflation moments by including many nearly identical moments. The grid search procedure is relatively simple and draws 50 random values for (γ^x, γ^π, ρ^i, σ_x, σ_π, σ_i) and picks the combination with the lowest objective function. I repeat this algorithm until convergence, meaning that the grid search result no longer changes starting from the calibrated values for each subperiod calibration. The only parameter value that reaches the externally set upper bound is $\gamma^x = 1$ for the 2000s calibration. I regard this as a plausible upper bound based on economic priors. Empirical ten-year CPI inflation expectations are from the Survey of Professional Forecasters after 1990 and from Blue Chip before that, available from the Philadelphia Fed.

Rather than picking different lags for different variables I include all lags for all variables, effectively averaging across different lead and lag horizons. Because the model is relatively parsimonious, the model cross-correlations should be expected to be matched on average but not at every lag.

The resulting subperiod calibrations shown in Table 1 are intuitive. The 1980s calibration features volatile supply shocks, volatile monetary policy shocks, and small demand shocks. The monetary policy rule in this period is characterized by a high inflation weight, a low output gap weight, and low inertia. Conversely, the 2000s calibration features volatile demand shocks, small supply and monetary policy shocks, and a less inflation-centric but more inertial monetary policy rule. The changing volatility of supply shocks across calibrations is primarily pinned down by the changing empirical inflation-output comovement in Panel A of Figure 2. The volatility of monetary policy shocks is roughly determined by the output-policy rate comovement shown in Panel B, with a higher monetary policy shock volatility making this relationship more negative in the 1980s calibration. The monetary policy inflation weight and inertia parameter are mostly determined by the empirical inflation-policy rate relationship in Panel C, which became flatter and more inertial during the second subperiod.

A simple newspaper count exercise supports the high perceived volatility of supply shocks for the 1980s calibration and a much smaller volatility of supply shocks for the 2000s calibration.[14] Whether one interprets the demand shock as an increase in financial frictions or as an expected growth shock, it is empirically plausible that its volatility increased from the first subperiod to the second subperiod. The standard deviation of the Gilchrist and Zakrajšek (2012) credit spread doubled between the first and the second subperiods in the data (0.54% vs. 1.06%). The standard deviation of expectations of one-year earnings growth similarly more than doubled.[15] The decline in the volatility of long-term inflation expectations from the 1980s to the 2000s is well-matched by the model, and the consumption growth and fed funds rate volatilities are roughly in line with the data.[16]

[14]For details of the newspaper count exercise see Appendix Figure A1. Fed Chairman Alan Greenspan argued in favor of supply shocks in 1996 saying "(...) powerful forces have evolved in the past few years to help contain inflationary tendencies. An ever-increasing share of our nation's workforce uses the tools of new technologies. Microchips embodied in physical capital make it work more efficiently, and sophisticated software adds to intellectual capital,"Semiannual monetary policy report, U.S. Senate July 18, 1996.

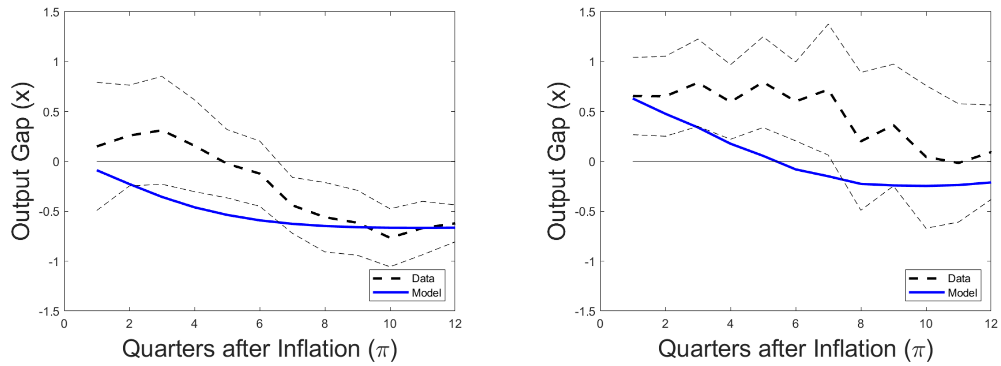
[15]Quarter-end credit spread data from https://www.federalreserve.gov/econres/notes/feds-notes/updating-the-recession-risk-and-the-excess-bond-premium-20161006.html. Quarterly data on one-year earnings growth expectations from De La'O and Myers (2021) ends in 2015.Q3 and was obtained from https://www.ricardodelao.com/data (accessed 12/12/2022).

[16]The model somewhat undershoots the volatility of changes in the fed funds rate in both periods, potentially due to monetary policy timing decisions about the very short-term that the model does not aim to capture and that are empirically less important for long-term asset prices (Bernanke and Kuttner (2005)).

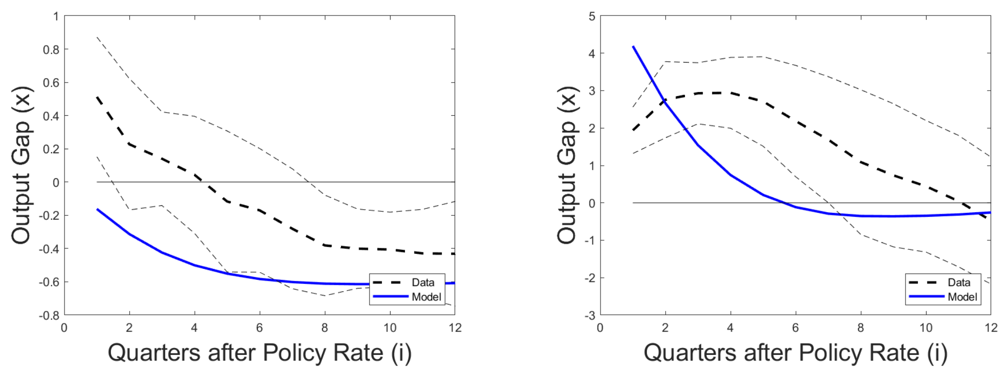
Figure 2: Local Projections for Inflation, Output Gap, and Fed Funds Rate

Panel A: Output Gap onto Lagged Price Inflation

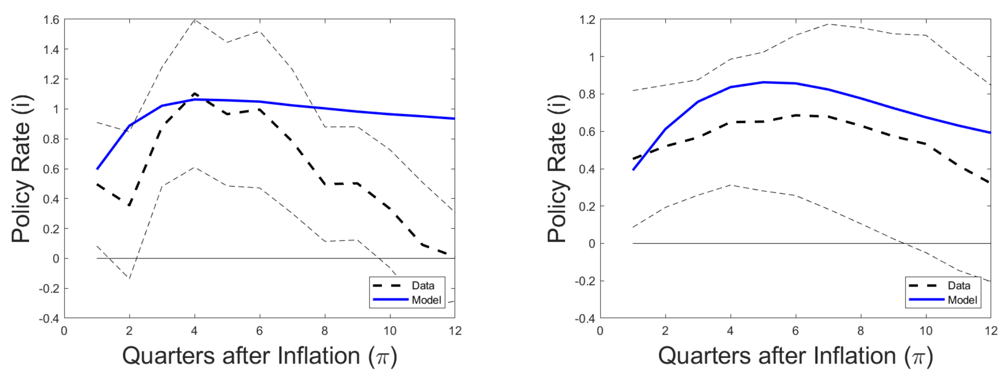
1979.Q4-2001.Q1 2001.Q2-2019.Q4



Panel B: Output Gap onto Lagged Policy Rate



Panel C: Policy Rate onto Lagged Price Inflation



This figure shows quarterly regressions of the form $z_{t+h} = a_{0,h} + a_{1,h} y_t + a_{2,h} y_{t-1} + \varepsilon_{t+h}$ and plots the regression coefficient $a_{1,h}$ on the y-axis against horizon h on the x-axis in the model vs. the data. Panel A uses the output gap on the left-hand side and GDP deflator inflation on the right-hand side, i.e. $z_t = x_t$ and $y_t = \pi_t$. Panel B uses the output gap on the left-hand side and the fed funds rate on the right-hand side, i.e. $z_t = x_t$ and $y_t = i_t$. Panel C uses the fed funds rate on the left-hand side and inflation on the right-hand side, i.e. $z_t = i_t$ and $y_t = \pi_t$. Black dashed lines show the regression coefficients in the data. Thin dashed lines show 95% confidence intervals for the data coefficients based on Newey-West standard errors with h lags. Blue solid lines show the corresponding model regression coefficients averaged across 100 independent simulations of length 1000.

Table 2: Quarterly Asset Prices and Macro Volatilities

	1979.Q4-2001.Q1		2001.Q2-2019.Q4	
Asset Prices: Stocks	Model	Data	Model	Data
Equity Premium	7.33	7.96	9.15	7.64
Equity Vol	14.95	16.42	19.29	16.80
Equity SR	0.49	0.48	0.47	0.45
AR(1) pd	0.96	1.00	0.93	0.84
1 YR Excess Returns on pd	-0.38	-0.01	-0.38	-0.50
1 YR Excess Returns on pd (R^2)	0.06	0.00	0.14	0.28
Asset Prices: Bonds				
Expected Bond Exc. Return	4.26	2.46	-0.84	-2.07
Return Vol.	15.82	14.81	2.12	9.28
Nominal Bond-Stock Beta	0.86	0.24	-0.09	-0.31
Real Bond-Stock Beta	0.05	0.08	-0.08	-0.06
1 YR Excess Return on Yield Spread*	1.26	2.55	-0.31	0.86
1 YR Excess Return on Yield Spread (R^2)	0.01	0.07	0.01	0.02
Macroeconomic Volatilities				
Std. Annual Cons. Growth*	0.76	1.15	1.59	1.15
Std Annual Change Fed Funds Rate*	1.64	2.26	0.65	1.40
Std. Annual Change 10-Year Subj. Infl. Forecast*	0.62	0.47	0.12	0.12

Moments that are explicitly targeted in the calibration procedure are noted with an asterisk. The expected bond excess return in the data starts in 1985 and is defined as the one-year expected return on an 11-year par bond with no Jensen's inequality adjustment using Blue Chip financial forecasters' 4-quarter forecast of 10-year bond yields following Piazzesi, Salomao, and Schneider (2015) and Nagel and Xu (2022). The model bond excess return is the steady-state excess return for a zero-coupon ten-year bond. Ten-year CPI inflation expectations are from the Survey of Professional Forecasters after 1990 and from Blue Chip before that. Long-term inflation forecast available from the Philadelphia Fed research website. Model ten-year inflation expectations are computed as. $\tilde{E}_t \pi_{t \to t+40} = \zeta \pi_{t-41 \to t-1} + (1-\zeta) E_t \pi_{t \to t+40}$, where E_t denotes rational expectations.

The change in the shock volatilities in Table 1 is consistent with the long-standing macroeconomics DSGE literature with regime switches. Liu et al. (2011) and Bianchi and Ilut (2017) estimate regime-switching DSGE models with two volatility regimes with samples that start significantly earlier and end earlier than the one in this paper, and detect a shift towards a lower volatility of all shocks around 1980. My first subperiod starts at the time of this shift, which is why naturally my approach does not pick up on their shift in volatility regimes. The more reduced-form approach of Sims and Zha (2006) allows for multiple regime switches and also detects a shift in relative volatilities in 2000, supporting the break date and shock calibrations in Table 1.

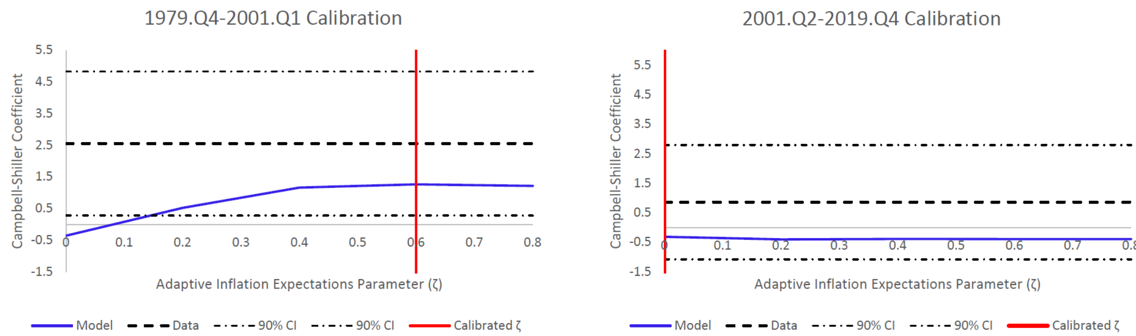
3.4 Adaptiveness of Inflation Expectations and Leverage

I choose the adaptive inflation expectations parameter $\zeta \in \{0, 0.6\}$ to match the empirical evidence on bond excess return predictability for each subperiod, while holding all other parameters constant at their values chosen in the second step. This step is kept separate because solving for asset prices is orders of magnitude slower than solving for macroeconomic dynamics.[17]

Figure 3 shows the predictability of bond excess returns from the yield spread at different values for the inflation expectations parameter, ζ. The model-implied Campbell-Shiller coefficients in Figure 3 indicate that we can reject fully rational inflation expectations for the 1980s subperiod. Of course, the distance to the macroeconomic dynamics is affected by varying ζ in this separate step. However, viewed more broadly within the econometric literature on inflation dynamics the macroeconomic fit improves, as setting $\zeta = 0.6$ in the 1980s calibration makes the policy rate response in the left plot of Panel C more persistent, in line with empirical evidence of a strong persistent inflation component during the 1980s (Stock and Watson (2007)). Appendix Figure A2 shows the macroeconomic model fit with $\zeta = 0$ vs. $\zeta = 0.6$. Intuitively, when inflation is highly persistent the expectations hypothesis term in the yield spread cancels, the yield spread predicts future bond excess returns, and the Campbell-Shiller coefficient is positive. The link between bond excess return predictability and the persistence of inflation is reminiscent of an older empirical literature that has documented that the expectations hypothesis holds in time periods and countries where interest rates are less persistent (Mankiw, Miron, and Weil (1987), Hardouvelis (1994)). It is also consistent with Cieslak and Povala (2015)'s evidence that removing trend inflation uncovers

[17]The objective function minimized in this step is equation (30) plus the squared standardized difference between the model and data Campbell-Shiller bond return predictability coefficient with a weight of 100. Inflation forecast error regressions along the lines of Coibion and Gorodnichenko (2015) further support partially adaptive inflation expectations in the 1980s and rational inflation expectations in the 2000s (see Appendix Table A3).

Figure 3: Model Campbell-Shiller Predictability by Inflation Expectations



This figure shows the model Campbell-Shiller bond excess return predictability regression coefficient b from a regression of the form $xr^{\$}_{n,t\rightarrow t+4} = a + b\left(y^{\$}_{n,t} - y^{\$}_{1,t}\right) + \varepsilon_t$ using quarterly overlapping observations and $n = 40$ quarters in the model and in the data. On the x-axis is the parameter determining the adaptiveness of inflation expectations, ζ, which determines the backward-looking component of the PC through (22) in the model. All other parameters are as in Table 1. The data coefficient is shown as a black dashed line with 90% confidence intervals based on Newey-West standard errors with 4 lags.

time-varying risk premia in the yield curve. Bond risks therefore contribute to the understanding of forward- vs. backward-looking Phillips curves (Fuhrer (1997)) and expectations in New Keynesian models (Gabaix (2020)).

The leverage parameter effectively scales up stock returns but leaves all other model implications unchanged. I set it to roughly match the volatility of equity returns in the data. The model does not require high leverage, with $\delta = 0.5$ for the 1980s calibration corresponding to a debt-to-assets ratio of 50%, and $\delta = 0.66$ for the 2000s calibration corresponding to a debt-to-assets ratio of 33%.

3.5 Asset Pricing Implications

The top panels of Table 2 show that asset pricing habit preferences generate quantitatively plausible time-varying risk premia and volatilities in stocks and bonds. The model matches the high equity Sharpe ratio, equity volatility, stock excess return predictability, and the persistence of price-dividend ratios, which would not be possible in a model with constant risk aversion (Mehra and Prescott (1985)). Appendix Tables A1 and A2 show that time-varying risk premia are responsible for the vast majority of model stock and bond return variation, and drive about 90% of bond-stock covariances.

The middle panel in Table 2 shows that the model explains the key pattern of bond-stock betas, even though they were not explicitly targeted in the calibration. The model-implied nominal bond beta is strongly positive and much larger than the real bond beta in the 1980s calibration, but negative and close to the real bond beta in the 2000s calibration, similar to the data.[18] Model-implied nominal Treasury bond excess returns are volatile in the 1980s calibration and much less volatile in the 2000s calibration, again similar to the data.[19] Steady-state expected bond excess returns in the model are closely related to betas, and switch sign from positive in the 1980s calibration to negative in the 2000s calibration. This pattern is consistent with empirical expected bond excess returns based on professional survey forecasts, constructed following Piazzesi, Salomao, and Schneider (2015) and Nagel and Xu (2022).

The 1980s calibration generates a positive regression coefficient of ten-year nominal bond excess returns with respect to the lagged slope of the yield curve, as in the data and targeted in the calibration. On the other hand, the 2000s calibration does not generate any such bond excess return predictability, which is also in line with a much weaker and statistically insignificant relationship in the data. In unreported results, I find that the model does not generate return predictability in real bond excess returns. This is broadly in line with the empirical findings of Pflueger and Viceira (2016), who find stronger evidence for predictability in nominal than real bond excess returns after adjusting for time-varying liquidity.

4 Counterfactual Analysis and Economic Mechanism

What would it take for bonds to become similarly risky as in the stagflationary 1980s, and what would this tell us about the economy and monetary policy? In this Section, I show how nominal and real bond betas change in the model as I vary the uncertainty of macroeconomic shocks and the monetary policy rule. The adaptiveness of wage-setters' inflation expectations is found to matter little for bond-stock betas.

4.1 Counterfactual Betas 1980s vs. 2000s

Figure 4 shows that nominal bond betas remain negative but real bond betas increase in the presence of shock volatilities similar to the 1980s, provided that the monetary policy framework is more output-focused, less inflation-focused, and more inertial than during the 1980s.

[18] Figure 8 shows that exactly matching the nominal bond-stock beta in the first subperiod starting from the second subperiod also requires a high volatility of supply shocks, further supporting that a high volatility of supply shocks is consistent with financial moments in the first subperiod.

[19] My model is consistent with Duffee (2011)'s evidence of low volatility of inflation expectations relative to bond yields ("inflation variance ratios"), as he also finds that habit models give reasonable implications due to their volatile risk premia. Inflation variance ratios in my model range between 1/3 to 1/2.

Figure 4: Counterfactual Bond-Stock Betas

This figure shows model-implied nominal and real bond betas while changing parameter groups one at a time. Panel A sets all parameters to the 1979.Q4-2001.Q1 calibration unless stated otherwise. It reports bond betas while setting the following parameters to their averages of the 1979.Q4-2001.Q1 and 2001.Q2-2019.Q4 calibrations: "Shocks" (σ_x, σ_π, and σ_i), "MP All" (ρ^i, γ^x and γ^π), "MP Inertia" (ρ^i), "MP Output/Inflation Weights" (γ^x and γ^π), and "Inflation Expectations" (ζ). Panel B does the reverse exercise, holding all parameter values constant at the 2001.Q2-2019.Q4 baseline.

Panel A starts from the 1980s calibration and shows that changing either the shock volatilities or the monetary policy rule towards the 2000s calibration flips nominal bonds from risky (i.e. positive nominal bond beta) to safe (i.e. zero or negative nominal bond beta).[20] Said differently, the model does not imply positive nominal bond-stock betas unless it has both: 1980s-style shock volatilities and a 1980s-style monetary policy rule. However, different from the 2000s calibration, the real bond-stock beta in the "MP All" counterfactual in Panel A is positive. The real bond-stock beta in this counterfactual is positive because supply and monetary policy shocks dominate, moving the output gap inversely to the real rate along the Euler equation (17). The next two columns in Panel A show that changes in monetary policy inertia (ρ^i) and the long-term inflation and output weights (γ^x, γ^π) both act in the same direction, but that the output and inflation weights are more important quantitatively. The last column shows that the adaptiveness of wage-setters inflation expectations (ζ) has little effect on bond-stock betas.

Panel B of Figure 4 shows the central result: Starting from the 2000s calibration none of the changes to individual parameter groups have the power to flip the sign of nominal bond betas. Most tellingly, the "Shocks" column implies that even if the shock volatilities were to resemble the 1980s, an inertial and more output-focused monetary policy rule as in the 2000s would keep nominal bond-stock betas negative. Real bond-stock betas in this counterfactual are pushed up by more volatile supply and monetary policy shocks. This is directionally similar to the "MP All" column in Panel A, which combines even more volatile supply and monetary policy shocks with a 2000s-style monetary policy rule. Overall, these counterfactuals indicate that positive nominal bond-stock betas and stagflations are not the result of fundamental economic shocks or monetary policy in isolation, but instead require the interaction to create a "perfect storm". This interpretation is reminiscent of the macroeconomics literature (Bernanke, Gertler, and Watson (1997), Primiceri (2006)), though this prior literature did not consider bond-stock betas. The last column in Panel B again shows that the adaptiveness of wage-setters inflation expectations is not priced in bond-stock betas.

My counterfactuals differ from Rudebusch and Swanson (2012)'s counterfactuals for term premia. In their long-run risks model nominal term premia tend to be positive and are largest (i.e. nominal bonds are riskiest) when monetary policy price level targeting is weak. By contrast, here bond-stock betas can be positive or negative, in line with the data, and are highest when the monetary policy inflation coefficient is high. While resolving the debate between leading asset pricing models is beyond this paper, a perspective based on cyclical

[20] In some cases, the equilibrium may not exist if I move a parameter group all the way to the other calibration, so for comparability I move all parameter groups to the average of the 1980s and 2000s calibrations.

risk-bearing capacity seems useful as monetary policy has strong and immediate risk premium effects in the data (Bernanke and Kuttner (2005), Boyd, Hu, and Jagannathan (2005)).

4.2 Mechanism

I now illustrate the mechanism through impulse response functions for the macroeconomy and asset prices.

4.2.1 Macroeconomic Impulse Responses

Figure 5 shows model impulse responses for the output gap, nominal policy rate, and wage inflation to one-percentage-point demand, supply, and monetary policy shocks. Because of the structure of the model, the macroeconomic impulse responses preserve the intuition of a standard log-linearized three-equation New Keynesian model for given parameter values, but parameter values are partly chosen to match evidence on bond excess return predictability.

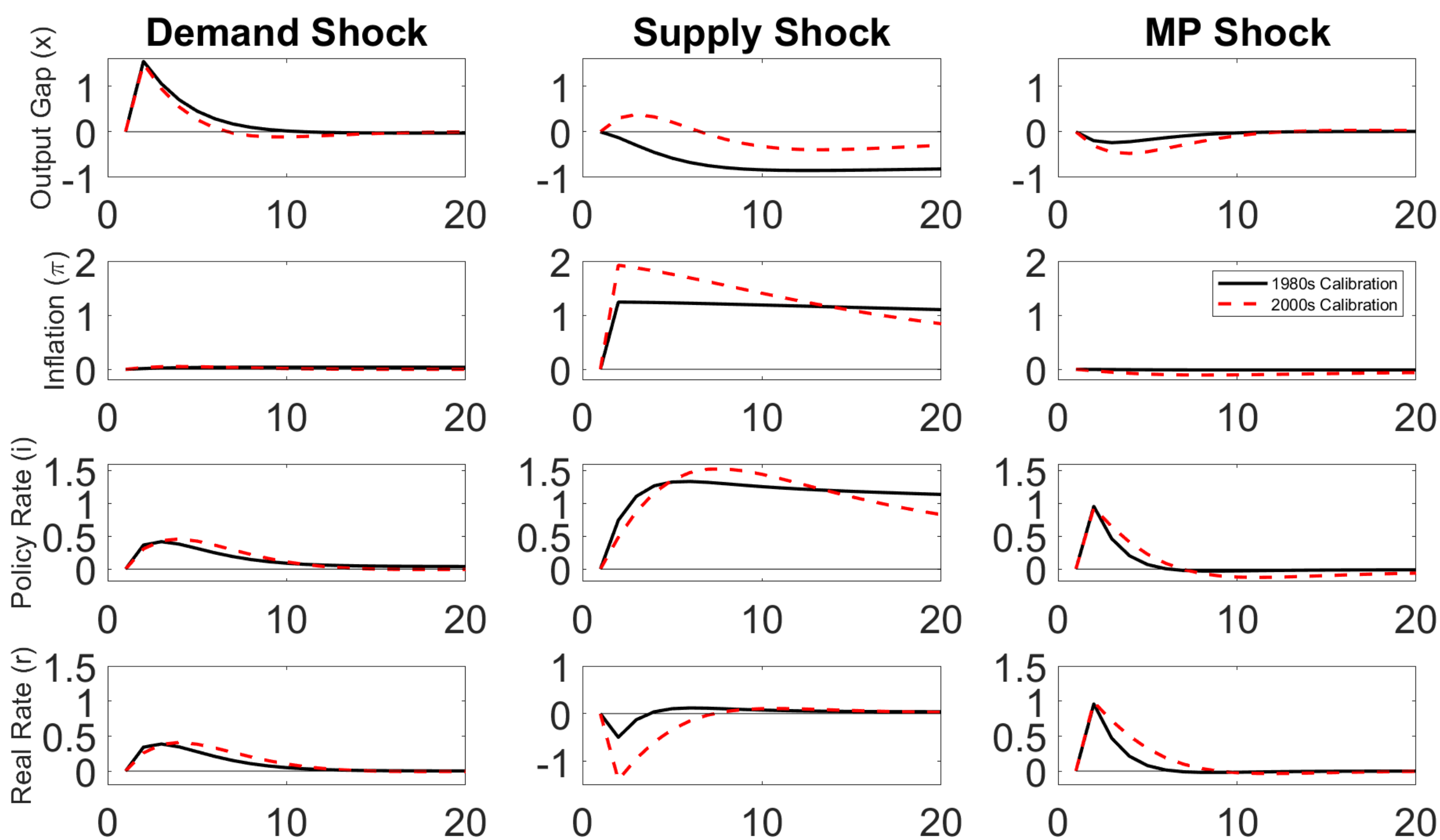
The first column in Figure 5 shows that demand shocks move the output gap, the policy rate, and inflation in the same direction, as if the economy moves along a stable Phillips curve. The responses are similar for the 1980s and 2000s calibrations, though of course demand shocks are more important in the 2000s calibration.

The middle column shows the interaction between supply shocks and systematic monetary policy. For the 1980s calibration, a positive supply shock leads to an immediate and persistent jump in inflation, a rapid increase in the nominal policy rate, and a large and persistent decline in the output gap—a stagflation. The inflation response is more persistent in the 1980s calibration due to backward-looking wage-setter inflation expectations. By contrast, for the 2000s calibration, a monetary policy rule that prescribes little immediate tightening in response to such a shock implies a real rate substantially below steady-state for several quarters. As a result, the output gap follows a much more moderate s-shaped path—a "soft landing". There is, however, a trade-off, as can be seen from the higher peak inflation response in the 2000s calibration.

Finally, the third column in Figure 5 shows intuitive responses to monetary policy shocks. A positive monetary policy shock tends to lower the output gap in a hump-shaped fashion and leads to a small and delayed fall in inflation, in line with the empirical evidence from identified monetary policy shocks (Ramey (2016)). The responses to a monetary policy shock are similar across the 1980s and 2000s calibrations.

Taken together, the macroeconomic impulse responses show that both supply shock uncertainty and a high monetary policy inflation coefficient are needed to generate stagflations and negative inflation-output gap comovement. By contrast, inflation and the output gap

Figure 5: Model Macroeconomic Impulse Responses



This figure shows model impulse responses for the output gap (%,), inflation (ann %), nominal policy rate (ann. %), and real one-quarter interest rate (ann. %). The impulse in the left column is a one-percentage-point demand shock, in the middle column is a one-percentage-point Phillips curve or supply shock, and in the right column is a one-percentage-point monetary policy shock. Impulse responses for the 1979.Q4-2001.Q1 calibration are shown in black, while the impulse responses for the 2001.Q2-2019.Q4 calibration are shown in red dashed.

comove little if monetary policy engineers a "soft landing" after an inflationary supply shock. Positive inflation-output gap comovement results after demand or monetary policy shocks. I next show how these macroeconomic dynamics shape bonds and stocks with endogenously time-varying risk premia.

4.2.2 Asset Price Impulse Responses

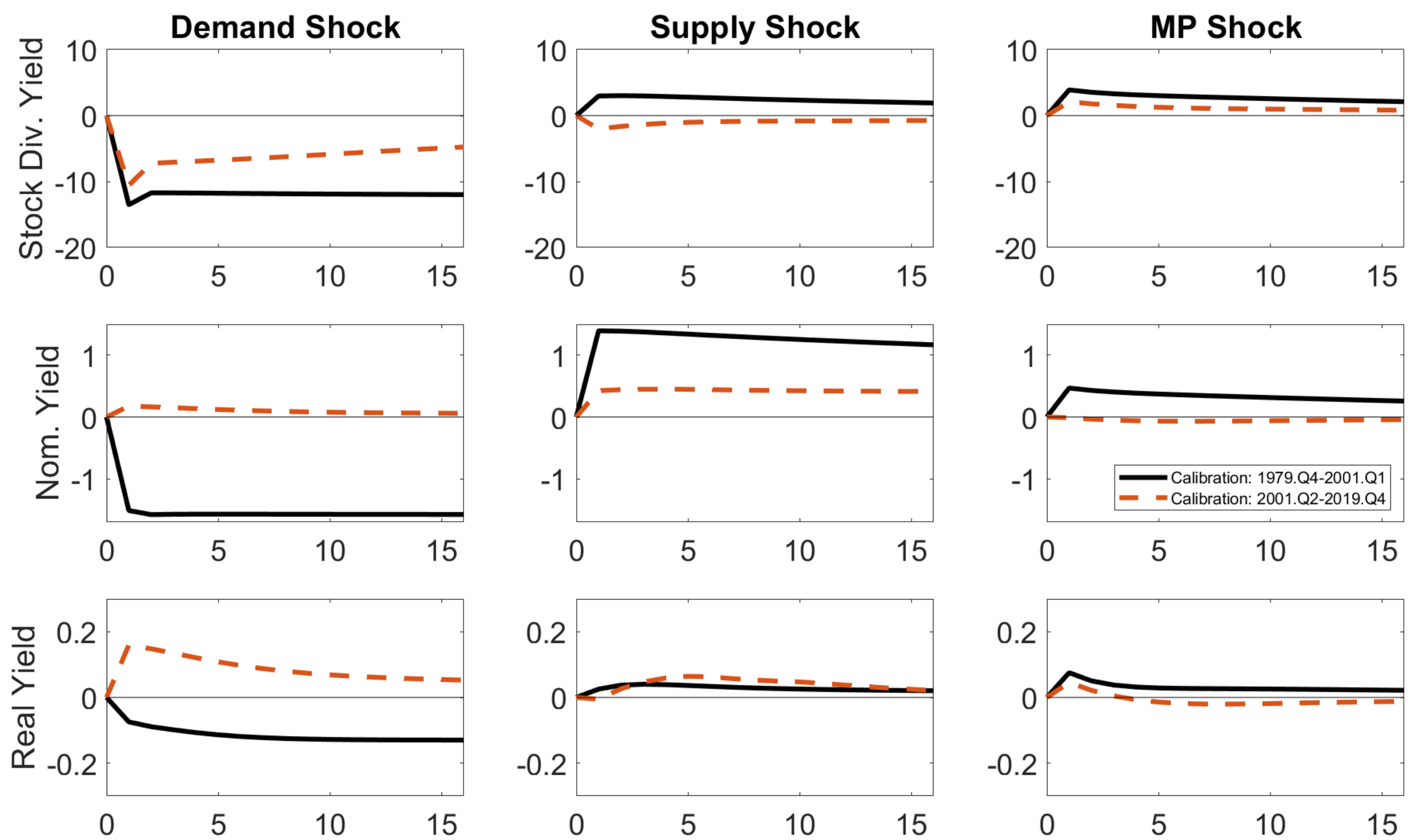
Since time-varying risk premia dominate the volatilities of asset prices in the model, risk premia are crucial for the responses of bonds and stocks to shocks. As a result, the responses of long-term bond yields in the model may even differ in sign from the expected path of policy rates. The macroeconomic equilibrium is nonetheless at the root of bond-stock comovements by determining whether bonds benefit or suffer from "flight-to-safety" when risk aversion rises.

Figure 6 shows impulse responses for the dividend yield of levered stocks (top row), ten-year nominal bond yield (middle row), and ten-year real bond yield (bottom row). Because dividend yields are inversely related to stock prices and bond yields are inversely related to bond prices, a shock that moves stock dividend yields and bond yields in the same direction tends to induce a positive bond-stock beta and vice versa.

The stock dividend yield in Figure 6 responds in the opposite direction from the output gap depicted in the top row of Figure 5. Adverse output gap and consumption news lower expected dividends and the surplus consumption ratio, raising risk aversion. The dividend yield hence rises, and stock prices fall more than the expected discounted value of future dividends. The top-middle panel shows that the dividend yield falls and stock prices rise in response to an adverse supply shock in the 2000s calibration, mirroring the initial output gap increase when monetary policy generates a "soft landing" after an adverse supply shock. This sign change in the model stock response is consistent with the empirical finding that "bad news" for the macroeconomy is often "good news" for the stock market, if investors price an accommodative monetary policy response (Boyd, Hu, and Jagannathan (2005), Elenev, Law, Song, and Yaron (2024)).

The responses of long-term nominal and real bond yields in Figure 6 differ substantially from their short-term counterparts in Figure 5 due to the endogenous nature of time-varying risk premia. For example, the left column in Figure 6 shows that long-term nominal bond yields decrease following a positive demand shock in the 1980s calibration but increase in the 2000s calibration, even though the corresponding policy rate paths in Figure 5 are very similar. This sign flip in the nominal bond yield response occurs because the mix of shocks and monetary policy lead nominal, and to a certain degree real bonds, to resemble positive-α assets in equation (29) in the 1980s calibration, but negative α assets in the 2000s calibration.

Figure 6: Model Asset Price Impulse Responses



This figure shows model impulse responses for the stock dividend yield, 10-year zero-coupon nominal bond yield, and 10-year zero-coupon real bond yield (all in ann. %). The 1979.Q4-2001.Q1 calibration is shown with black solid lines and the 2001.Q2-2019.Q4 calibration is shown with red dashed lines. The impulse in the left column is a one-percentage-point demand shock, in the middle column is a one-percentage-point Phillips curve or supply shock, and in the right column is a one-percentage-point monetary policy shock.

A positive demand shock raises consumption relative to habit, lowering risk aversion, and thereby making investors willing to pay more for risky nominal bonds in the 1980s calibration. However, nominal bonds have hedging value in the 2000s calibration, so lower risk aversion leads investors to be willing to pay less for nominal bonds after a positive demand shock.

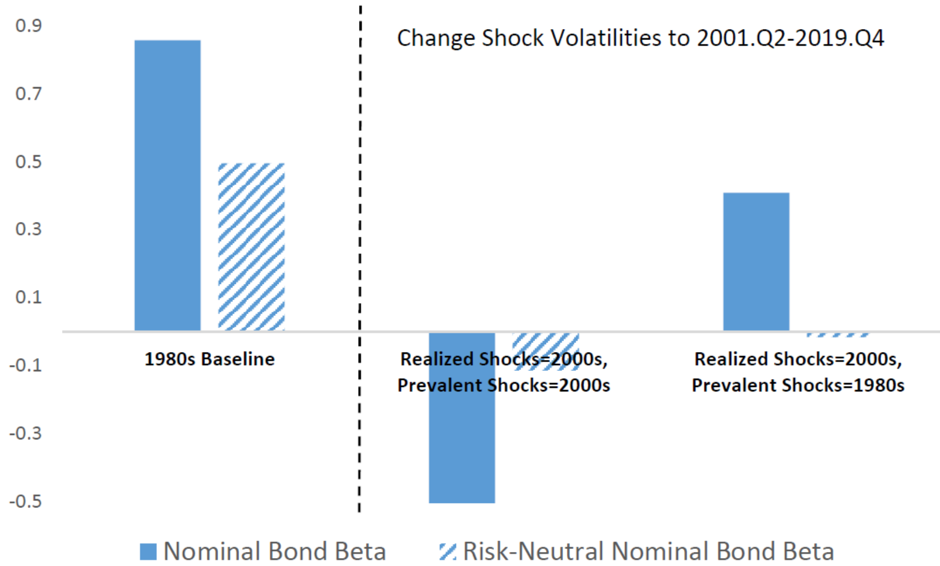
One might wonder whether "sentiment" shocks could turn bond-stock betas negative – if bonds benefit from "flight-to-safety" – or positive – if bonds are viewed as risky similar to stocks. One advantage of the present framework is that such "sentiment" shocks are endogenously linked to macroeconomic factors, reducing the degrees of freedom that the model has to understand changes in bond risks. Intuitively, bonds benefit from "flight-to-safety" only if monetary policy and the macroeconomic shock volatilities render their real cash flows safe, i.e. if bonds pay out when investors' marginal utility is high, such as in the 2000s calibration. In this case, a decrease in consumption and an increase in risk aversion lead investors to reduce their valuations of risky stocks, but raise the amount they are willing to pay for safe bonds. Conversely, if bonds' real cash flows are risky, such as in the 1980s calibration, an increase in investor risk aversion lowers the valuations of both bonds and stocks through risk premia, amplifying the positive bond-stock comovement. As a result, time-varying risk aversion is a quantitatively important driver of bond-stock betas within the model and turns bond-stock betas into forward-looking measures, but does not change the sign of bond-stock betas in equilibrium.

4.3 Counterfactual Bond Betas Prevalent vs. Realized Shocks

Figure 7 shows counterfactual bond-stock betas when out-of-equilibrium shocks are realized, finding that the priced equilibrium (or prevalent shocks) is more important for bond-stock betas than realized shocks. It shows that moving the distributions of both prevalent and realized shocks towards the 2000s calibration eliminates positive nominal bond-stock betas, similar to the top panel in Figure 4.[21] However, the picture looks different when only realized shocks are drawn from the 2000s distribution and the equilibrium is still priced as if shocks follow the 1980s distribution. In this case, the nominal bond-stock beta is positive even though the risk-neutral nominal bond beta (i.e. the beta with constant risk premia) is slightly negative. The mechanism goes back to the left column of Figure 6, where nominal bond yields and dividend yields comove positively after a surprise demand shock in the 1980s calibration. Endogenously time-varying risk premia therefore matter and imply that the macroeconomic equilibrium is priced in bond-stock betas.

[21] All other parameters are held constant at the 1980s calibration.

Figure 7: Nominal Bond Betas by Prevalent vs. Realized Shocks



This figure shows model-implied nominal bond betas (solid) and the betas of risk-neutral nominal bond returns with respect to the stock market (dashed) across prevalent and realized shock distributions. The leftmost bars set all parameter values to the 1979.Q4-2001.Q1 calibration. The middle bars change both the realized and prevalent shock volatilities to the 2001.Q2-2019.Q4 values, i.e. the equilibrium is recomputed at the 2001.Q2-2019.Q4 shock volatilities. The rightmost bars change only the realized but not the prevalent shock volatilities to their 2001.Q2-2019.Q4 values, i.e. equilibrium asset prices are not recomputed and only the simulated shocks drawn from the 2001.Q2-2019.Q4 distribution.

4.4 Replicating 1980s Bond-Stock Betas in the Model

One might wonder whether the high volatility of supply shocks in the 1980s calibration is supported by asset pricing moments during this period. Figure 8 shows that a high supply shock volatility is indeed needed to replicate the bond-stock betas observed in the first subperiod starting from the model calibration for the second subperiod. In addition, it shows that a monetary policy rule with a high inflation weight and low inertia – similar to the rule in the 1980s calibration – is also necessary.

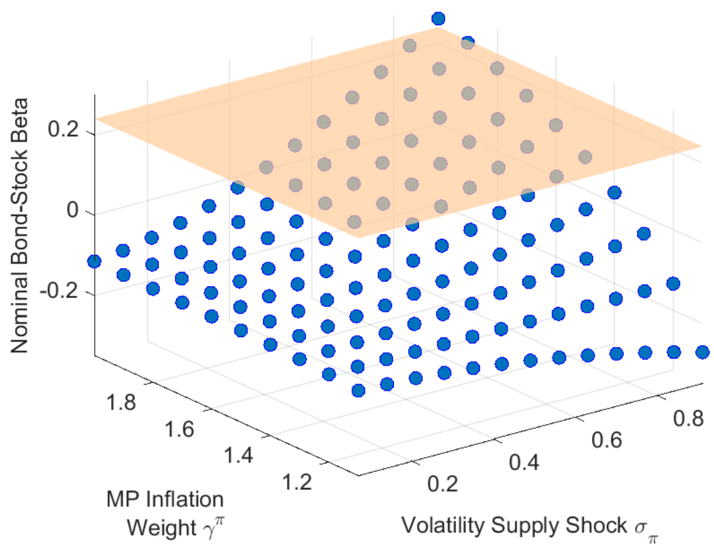
Figure 8 plots nominal bond-stock betas implied by the model starting from the 2000s calibration against an increasing supply shock standard deviation σ_π on the x-axis, and the monetary policy weight γ^π on the y-axis. The model-implied nominal bond-stock beta is depicted with blue circles on the z-axis, with a plane indicating the empirical 1980s nominal bond-stock beta. Dependence on the monetary policy inertia coefficient ρ^i is also depicted, with $\rho^i = 0.54$ in Panel A and $\rho^i = 0.80$ in Panel B.

Figure 8 shows that a high volatility of supply shocks is necessary to explain the empirically observed nominal bond-stock betas within the model. Consistent with the notion that a "perfect storm" is needed to make bonds risky again, it further shows that a high supply shock volatility is necessary but not sufficient to replicate the high nominal bond-stock betas of the 1980s. In both Panels A and B, the model-implied bond-stock beta is highest in the back-right corner, corresponding to a high monetary policy inflation weight and a high volatility of supply shocks. In Panel A, where the monetary policy rule is assumed to have only moderate inertia as plausible for the 1980s, the model can replicate the 1980s bond-stock beta if the supply shock volatility is high at $\sigma_\pi \geq 0.9$ and the monetary policy inflation weight is also high at $\gamma^\pi \geq 2.0$.[22] The model nominal bond-stock betas in Panel B are smaller than the observed nominal bond-stock betas during the 1980s, indicating that lower monetary policy inertia is also necessary to explain bond risks in this earlier period. Overall, reverse-engineering the nominal bond-stock betas observed during the 1980s from the model calibration for the second subperiod provides additional support to the notion that investors viewed supply shocks as highly volatile during this period.

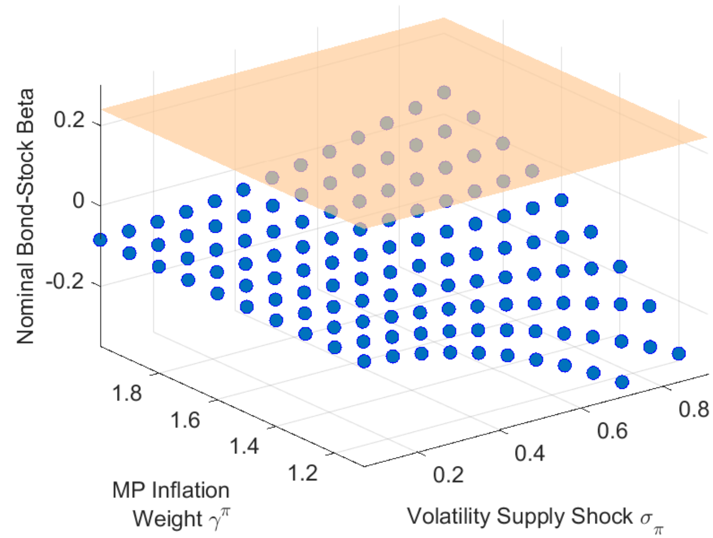
[22] Because Figure 8 starts from the 2000s calibration and hence includes a volatile demand shock, this pushes nominal bond-stock betas downwards. As a result, the supply shock volatility required to explain the positive nominal bond-stock beta in Figure 8 is even larger than the supply shock volatility for the 1980s calibration, which features negligible demand shock volatility.

Figure 8: Model-Implied Nominal Bond-Stock Beta vs. 1980s Bond-Stock Beta

Panel A: MP Inertia $\rho^i = 0.54$



Panel B: MP Inertia $\rho^i = 0.80$

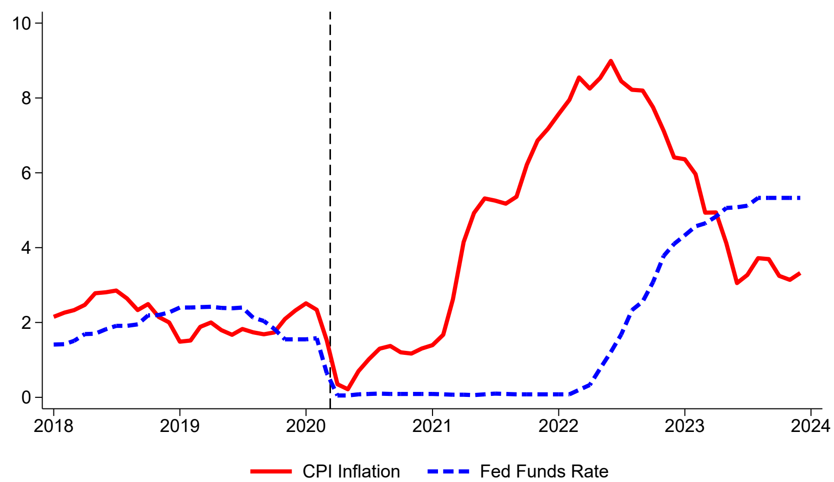


Note: This figure shows model ten-year nominal bond-stock betas (blue circles) against the standard deviation of supply shocks σ_π on the x-axis and the MP inflation weight γ^π on the y-axis while holding the monetary policy inertia coefficient constant at $\rho^i = 0.54$ (Panel A) and $\rho^i = 0.80$ (Panel B), respectively. All other parameters are set to their values in the 2000s calibration in Table 1. The orange plane indicates the empirical ten-year nominal bond-stock beta for the 1979.Q4-2001.Q1 subperiod.

5 Post-Pandemic Bond Risks and Monetary Policy

This Section applies the model to analyze bond-stock betas from daily returns during the recent post-pandemic period. This period was marked by heightened concerns over supply shocks and an inflationary surge unseen since the 1980s. Dividing the analysis into an early post-pandemic (until 2023.Q2) and a late post-pandemic period (2023.Q3-Q4), I find that the model can explain negative nominal and positive real bond-stock betas during the early post-pandemic with a low inflation coefficient and 1980s-style shocks. However, for the later post-pandemic the model requires a high monetary policy inflation coefficient with 1980s-style shocks, explaining positive nominal, real, and breakeven (i.e. nominal-minus-real) bond betas during this last part of the sample.

Figure 9: Timeline of Post-Pandemic Inflation and Monetary Policy



Note: This figure shows CPI headline inflation for all urban consumers (CPIAUCSL) over the past twelve months (ann, %) and the effective federal funds rate (DFF, ann, %) for the sample 01/01/2018 through 31/12/2023. Monthly data is from the St. Louis Fred. A vertical line indicates March 11, 2020 (Covid).

Figure 9 depicts the timeline of inflation and the federal funds rate from 2018 to 2023. Inflation began to accelerate sharply in 2021. However, monetary policy initially remained passive, with the federal funds rate held at the zero-lower-bound.[23] The Fed lifted off from the zero-lower-bound in March 2022, with the hiking cycle plateauing in July 2023 at a policy rate of 5.25–5.5%. Bauer, Pflueger, and Sunderam (2024b) use rich survey data to argue that this late rise in the policy rate led to a late rise in the monetary policy inflation coef-

[23]For example, Fed Chairman Powell reiterated the Fed's commitment to a near-zero rate, despite headline CPI running at 5-6% annually in his 2021 Jackson Hole speech. Jerome Powell, "Monetary Policy in the Time of COVID", Speech at the Jackson Hole Economic Symposium, August 27, 2021.

ficient as perceived by professional forecasters, with the perceived monetary policy inflation coefficient peaking in the second half of 2023.[24] Macroeconomic developments hence suggest dividing the post-pandemic period into two distinct phases: An early post-pandemic phase characterized by a low Fed response and a later phase featuring a substantially more active Fed response to inflation.

Table 3: Model-Implied Post-Pandemic Monetary Policy and Shocks

	Early Post-Pandemic 2020.Q2-2023.Q2		Late Post-Pandemic 2023.Q3-2023.Q4	
Bond-Stock Betas	Data	Model	Data	Model
Nominal Bond-Stock Beta	-0.01	-0.04	0.49	0.42
Real Bond-Stock Beta	0.05	0.12	0.39	0.14
Model Parameters				
MP Inflation Coefficient γ^π	1.3		2	
Shocks	1979.Q4-2001.Q1		1979.Q4-2001.Q1	

Note: This table shows nominal and real bond-stock betas from daily data for the early and late post-pandemic periods, together with the best fit from the model. The monetary policy inflation weight is allowed to vary over $\gamma^\pi \in [1.1, 1.2, ..., 2.0]$. The shock volatilities are allowed to vary over the 1980s calibration, the 2000s calibration, or $(\sigma_x, \sigma_\pi, \sigma_i) = (0.00, 0.00, 1.19)$. The following parameters are held constant: $\zeta = 0$, $\gamma^x = 0.5$, and $\rho^i = 0.8$. All other parameter values are as Table 1. Nominal and real bond betas in the data are computed from daily bond and stock log returns.

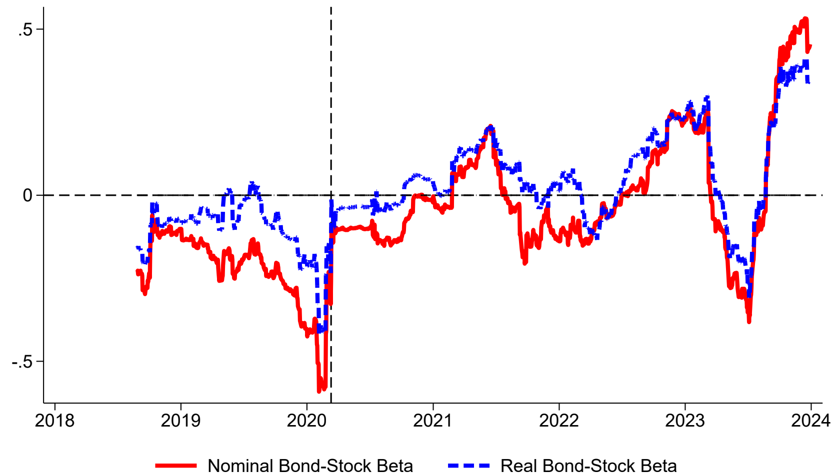
Figure 10 shows rolling bond-stock betas computed using daily returns for the post-pandemic period. It shows that despite parallels between the 1980s and 2020s inflation, bond risks evolved differently during the recent post-pandemic period.[25] Panel A shows that throughout 2021 and 2022, nominal bond betas broadly remained negative, even at the inflationary peak, while real bond betas turned slightly positive. Panel B shows that during this period nominal bond-stock betas were either significantly lower than real bond-stock betas or at least not statistically significantly different. Since breakeven (or nominal-

[24]Since the forecast horizon in Bauer, Pflueger, and Sunderam (2024b) is shorter, the magnitudes of the estimated perceived inflation coefficients are not directly comparable to those priced in bond-stock betas over the lifetime of the bond. Bocola et al. (2024) also provide evidence that the perceived monetary policy inflation coefficient was low during the earlier post-pandemic period 2021-2022.

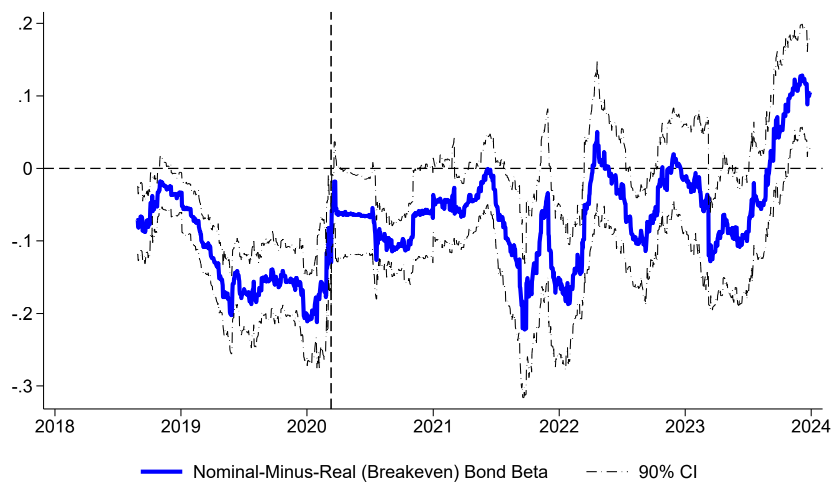
[25]To isolate variation in bond risk over shorter time intervals, bond-stock betas are computed using 120-day rolling window regressions of daily log bond returns onto daily log stock returns. The estimates in Panel A are roughly consistent with the lower-frequency estimates in Figure 1, though of course betas from daily data are more volatile.

Figure 10: Post-Pandemic Bond-Stock Betas from Daily Data

Panel A: Real and Nominal Bond Betas



Panel B: Breakeven Betas



Note: Panel A shows betas from regressing daily ten-year nominal and TIPS bond log returns onto daily US equity log returns over 120-trading day backward-looking rolling windows for the sample 01/01/2018 through 31/12/2023. Zero-coupon yield curves are from Gürkaynak, Sack, and Wright (2007, 2010). A vertical line indicates March 11, 2020 (Covid). 90% confidence intervals based on heteroskedasticity robust standard errors are shown in dashed. Panel B shows the difference between the nominal and real (TIPS) bond-stock beta with 90% confidence intervals based on Huber-White heteroskedasticity robust standard errors.

minus-real bond) returns are inversely related to inflation expectations, this indicates that the inflationary dynamics priced in bond-stock betas were still procyclical. This outcome aligns with the counterfactual combining 1980s shock volatilities with a monetary policy rule reflecting the 1980s-2000s average (see Figure 4 Panel A, "Shocks" column).

However, in the second half of 2023, Figure 1, Panel B shows that breakeven betas did turn significantly positive, suggesting that the priced cyclicality of inflation expectations had changed. I interpret these patterns as providing out-of-sample confirmation of the key model predictions if supply shocks were a significant concern throughout the post-pandemic period, but the priced monetary policy rule changed from a low inflation coefficient in the early post-pandemic to a high inflation coefficient in the later post-pandemic.

Table 3 reports nominal and real bond-stock betas using daily log returns for the early and late post-pandemic periods, and the best fit from the model starting from the 2000s calibration. I divide the post-pandemic period into two distinct phases: An early post-pandemic phase (2020.Q2–2023.Q2) characterized by a low or negative breakeven beta, and a later phase (2023.Q3–2023.Q4) featuring a significantly positive breakeven-stock beta. As discussed, this segmentation also lines up roughly with the increase in the monetary policy inflation coefficient as perceived by professionals and investors. I use the model to find the monetary policy inflation weight and the macroeconomic shocks that provide the closest fit for the empirically observed bond-stock betas. To account for strongly anchored long-term inflation expectations during the recent period and to stack the deck against finding a role for volatile long-term inflation expectations, wage-setters' inflation expectations are assumed to be rational in Table 3.[26] I vary only the monetary policy inflation weight γ^π for this exercise, as a higher value of γ^π in the presence of supply shocks acts similarly to lower inertia ρ^i or a lower output gap weight γ^x, and varying all three monetary policy parameters would not necessarily be identified. The implied γ^π in Table 3 hence should be interpreted broadly as monetary policy "hawkishness".

I find that the model matches the changes in nominal and real bond-stock betas with a shift from a low to a high inflation coefficient in monetary policy. For the entire post-pandemic period, the model requires that investors priced similar shock volatilities to the 1980s calibration, in line with anecdotal evidence of significant supply shock concerns.[27] The 1980s shock volatilities also feature highly volatile monetary policy shocks, complementing

[26]I minimize an objective function that is the sum of squared differences between the data and model moments listed in the top panel in Table 3. To save on computational complexity the shock volatilities are allowed to take only three sets of possible values, the 1980s calibration, the 2000s calibration, or dominant monetary policy shocks. I set $(\sigma_x, \sigma_\pi, \sigma_i) = (0.00, 0.00, 1.19)$ for the third set of shock volatilities. However, the precise magnitude of the volatility of monetary policy shocks is not crucial here, since bond-stock betas are a ratio and therefore reflect the relative volatility of monetary policy shocks compared to other shocks. Monetary policy parameters other than γ^π are held constant at conventional values, $\gamma^x = 0.5$, and $\rho^i = 0.8$. Scatter plots of model nominal and real bond-stock betas for all considered shock volatilities and monetary policy parameters are shown in Appendix Figure A5.

[27]The similarities in supply shock concerns between the 1980s and the recent period are evident in Appendix Figure A1, which shows related article counts in the Wall Street Journal. The literature on recent inflation dynamics also attributes a significant role to supply shocks (Reis (2022), Di Giovanni, Kalemli-Özcan, Silva, and Yildirim (2022), Gagliardone and Gertler (2023), Rubbo (2022)).

high supply shock uncertainty by further driving up nominal and real bond-stock betas. However, monetary policy shocks by themselves cannot explain the negative nominal bond stock beta in the early post-pandemic period, or the positive breakeven beta in the late post-pandemic period, and therefore the model favors the 1980s mix of shocks with both volatile supply and monetary policy shocks.

Overall, the model's implied sequence of monetary policy coefficients and macroeconomic uncertainty aligns well with the timeline of post-pandemic events. The post-pandemic period therefore provides out-of-sample evidence that supply shock uncertainty combined with inflation-focused monetary policy are both essential to turn bonds into a risky asset class as in the 1980s. As inflationary supply shocks ease and monetary policy recalibrates towards lower inflation responsiveness, the model hence suggests that one might expect bond-stock betas to moderate.

6 Conclusion

A New Keynesian asset pricing model with countercyclical risk-bearing capacity shows that the interaction between supply shocks and inflation-focused monetary policy leads to positive nominal bond-stock betas, as observed during the stagflationary 1980s. Conversely, a combination of 1980s-style shocks with a more output-focused and inertial monetary policy rule leads to a "soft landing", which is priced in negative nominal bond-stock betas, and positive real bond-stock betas. However, bond-stock betas do not price realized shocks in this model, but instead the expected equilibrium mix of shocks going forward.

The mechanism works through the monetary policy trade-off between inflation and output after a supply shock and endogenously time-varying risk premia. An adverse supply shock in the model generally moves inflation expectations up and output down, which leads to simultaneous falls in nominal bond and stock prices. However, monetary policy can alter these implications and engineer a "soft landing" if the central bank keeps nominal rates sufficiently steady and allows the real rate to fall. A less inflation-focused or inertial monetary policy rule mitigates the positive bond-stock comovement that would otherwise result from supply shocks. By contrast, demand shocks move output and inflation up and down together independently of the monetary policy rule, implying negative nominal and real bond-stock betas, as observed during the pre-pandemic 2000s.

Time-varying risk premia generate predictability in stocks and bonds, and imply that bond-stock betas price the distribution of shocks in equilibrium rather than past realized shocks. When investors are surprised by realized demand shocks but bonds and stocks are priced as if 1980s shocks are prevalent, the model nominal bond-stock beta is similarly positive as in the 1980s calibration. Intuitively, bond and stock returns are dominated by time-varying risk premia, and in a 1980s-type equilibrium nominal bonds' real cash flows are stock-like, so risk premia in nominal bonds and stocks also move together.

The framework in this paper provides an intuitive interpretation of post-pandemic developments in bond-stock betas, implying that investors initially priced a low monetary policy inflation coefficient during the initial inflation run-up, but a much higher inflation coefficient towards the peak of the latest hiking cycle. More broadly, when the economy is driven by volatile supply shocks, nominal bond stock betas in the model emerge as a forward-looking indicator of "soft landings". This analysis suggests that further research on financial market comovements and their connection to drivers of the macroeconomy is likely to be fruitful.

References

Abel, Andrew B. 1990. "Asset prices under habit formation and catching up with the Joneses." *American Economic Review* 80 38–42.

Albuquerque, Rui, Martin Eichenbaum, Victor Xi Luo, and Sergio Rebelo. 2016. "Valuation risk and asset pricing." *Journal of Finance* 71 (6): 2861–2904.

Baele, Lieven, Geert Bekaert, and Koen Inghelbrecht. 2010. "The determinants of stock and bond return comovements." *Review of Financial Studies* 23 (6): 2374–2428.

Bauer, Michael D, Carolin E Pflueger, and Adi Sunderam. 2024a. "Perceptions about monetary policy." *Quarterly Journal of Economics* 139 (4): 2227–2278.

Bauer, Michael D, and Eric T Swanson. 2023. "A reassessment of monetary policy surprises and high-frequency identification." *NBER Macroeconomics Annual* 37 (1): 87–155.

Bauer, Michael, Carolin Pflueger, and Adi Sunderam. 2024b. "Changing Perceptions and Post-Pandemic Monetary Policy." *Proceedings of 2024 Jackson Hole Economic Symposium, forthcoming.*

Beaudry, Paul, and Franck Portier. 2006. "Stock prices, news, and economic fluctuations." *American Economic Review* 96 (4): 1293–1307.

Bernanke, Ben S, and Mark Gertler. 2001. "Should central banks respond to movements in asset prices?" *American Economic Review: Papers and Proceedings* 91 (2): 253–257.

Bernanke, Ben S, Mark Gertler, and Mark Watson. 1997. "Systematic monetary policy and the effects of oil price shocks." *Brookings Papers on Economic Activity* 1997 (1): 91–157.

Bernanke, Ben S, and Kenneth N Kuttner. 2005. "What Explains the Stock Market's Reaction to Federal Reserve Policy?" *Journal of Finance* 60 (3): 1221–1257.

Bianchi, Francesco, Renato Faccini, and Leonardo Melosi. 2023. "A fiscal theory of trend inflation." *Quarterly Journal of Economics, forthcoming.*

Bianchi, Francesco, and Cosmin Ilut. 2017. "Monetary/fiscal policy mix and agents' beliefs." *Review of economic Dynamics* 26 113–139.

Bianchi, Francesco, Martin Lettau, and Sydney C Ludvigson. 2022a. "Monetary policy and asset valuation." *Journal of Finance* 77 (2): 967–1017.

Bianchi, Francesco, Sydney C Ludvigson, and Sai Ma. 2022b. "Belief distortions and macroeconomic fluctuations." *American Economic Review* 112 (7): 2269–2315.

Bianchi, Francesco, Sydney C Ludvigson, and Sai Ma. 2022c. "Monetary-based asset pricing: A mixed-frequency structural approach." National Bureau of Economic Research wp30072.

Bianchi, Javier, and Guido Lorenzoni. 2021. "The prudential use of capital controls and foreign currency reserves." *Handbook of International Economics Vol. V.*

Blanchard, Olivier Jean, and Charles M Kahn. 1980. "The solution of linear difference models under rational expectations." *Econometrica* 1305–1311.

Bocola, Luigi, Alessandro Dovis, Kasper Jørgensen, and Rishabh Kirpalani. 2024. "Bond market views of the Fed." NBER Working Paper wp32620.

Bordalo, Pedro, Nicola Gennaioli, Rafael LaPorta, and Andrei Shleifer. 2022. "Belief overreaction and stock market puzzles." Working paper, Oxford, Bocconi, Brown University, and Harvard University.

Boyd, John H, Jian Hu, and Ravi Jagannathan. 2005. "The stock market's reaction to unemployment news: Why bad news is usually good for stocks." *Journal of Finance* 60 (2): 649–672.

Caballero, Ricardo J, and Alp Simsek. 2020. "A risk-centric model of demand recessions and macroprudential policy." *Quarterly Journal of Economics* forthcoming.

Caballero, Ricardo J, and Alp Simsek. 2022. "A monetary policy asset pricing model." NBER Working Paper wp30132.

Campbell, John Y, and John H Cochrane. 1999. "By force of habit: A consumption-based explanation of aggregate stock market behavior." *Journal of Political Economy* 107 205–251.

Campbell, John Y, Carolin Pflueger, and Luis M Viceira. 2020. "Macroeconomic drivers of bond and equity risks." *Journal of Political Economy* 128 (8): 3148–3185.

Campbell, John Y, and Robert J Shiller. 1991. "Yield spreads and interest rate movements: A bird's eye view." *Review of Economic Studies* 58 (3): 495–514.

Campbell, John Y, Robert J Shiller, and Luis M Viceira. 2009. "Understanding Inflation-Indexed Bond Markets.." *Brookings Papers on Economic Activity* 79–138.

Campbell, John Y, Adi Sunderam, and Luis M Viceira. 2017. "Inflation Bets or Deflation Hedges? The Changing Risks of Nominal Bonds." *Critical Finance Review* 6 (2): 263–301.

Chan, Yeung Lewis, and Leonid Kogan. 2002. "Catching Up with the Joneses: Heterogeneous Preferences and the Dynamics of Asset Prices." *Journal of Political Economy* 110 (6): 1255–1285.

Chernov, Mikhail, Lars A Lochstoer, and Dongho Song. 2021. "The real explanation of nominal bond-stock puzzles." National Bureau of Economic Research Working Paper, wp 29085.

Christiano, Lawrence H, Martin Eichenbaum, and Charles L Evans. 2005. "Nominal rigidities and the dynamic effects of a shock to monetary policy." *Journal of Political Economy* 113 (1): 1–45.

Christiano, Lawrence J, Martin Eichenbaum, and Charles L Evans. 1999. "Monetary policy shocks: What have we learned and to what end?" *Handbook of Macroeconomics* 1 65–148.

Cieslak, Anna, Michael McMahon, and Hao Pang. 2024. "Did I make myself clear? The Fed and the market in the post-2020 framework period." Working Paper.

Cieslak, Anna, and Pavol Povala. 2015. "Expected returns in Treasury bonds." *Review of Financial Studies* 28 (10): 2859–2901.

Cieslak, Anna, and Annette Vissing-Jorgensen. 2021. "The economics of the Fed put." *Review of Financial Studies* 34 (9): 4045–4089.

Clarida, Richard, Jordi Gali, and Mark Gertler. 2000. "Monetary policy rules and macroeconomic stability: Evidence and some theory." *Quarterly Journal of Economics* 115 (1): 147–180.

Cochrane, John H. 2011. "Determinacy and identification with Taylor rules." *Journal of Political Economy* 119 (3): 565–615.

Cogley, Timothy, and Thomas J Sargent. 2008. "Anticipated utility and rational expectations as approximations of Bayesian decision making." *International Economic Review* 49 (1): 185–221.

Coibion, Olivier, and Yuriy Gorodnichenko. 2015. "Information rigidity and the expectations formation process: A simple framework and new facts." *American Economic Review* 105 (8): 2644–2678.

David, Alexander, and Pietro Veronesi. 2013. "What ties return volatilities to fundamentals and price valuations?" *Journal of Political Economy* 121 682–746.

De La'O, Ricardo, and Sean Myers. 2021. "Subjective cash flow and discount rate expectations." *Journal of Finance* 76 (3): 1339–1387.

Di Giovanni, Julian, Sebnem Kalemli-Özcan, Alvaro Silva, and Muhammed A Yildirim. 2022. "Global supply chain pressures, international trade, and inflation." National Bureau of Economic Research Working Paper, wp30240.

Drechsler, Itamar, Alexi Savov, and Philipp Schnabl. 2018. "A model of monetary policy and risk premia." *Journal of Finance* 73 (1): 317–373.

Du, Wenxin, Joanne Im, and Jesse Schreger. 2018. "The US Treasury premium." *Journal of International Economics* 112 167–181.

Duffee, Gregory R. 2011. "Information in (and not in) the term structure." *Review of Financial Studies* 24 (9): 2895–2934.

Elenev, Vadim, Tzuo-Hann Law, Dongho Song, and Amir Yaron. 2024. "Fearing the Fed: How Wall Street reads Main Street." *Journal of Financial Economics* 153 103790.

Fang, Xiang, Yang Liu, and Nikolai Roussanov. 2022. "Getting to the core: Inflation risks within and across asset classes." National Bureau of Economic Research Working Paper wp30169.

Favilukis, Jack, and Xiaoji Lin. 2016. "Wage rigidity: A quantitative solution to several asset pricing puzzles." *Review of Financial Studies* 29 (1): 148–192.

Fuhrer, Jeff, and George Moore. 1995. "Inflation persistence." *Quarterly Journal of Economics* 110 (1): 127–159.

Fuhrer, Jeffrey C. 1997. "The (un) importance of forward-looking behavior in price specifications." *Journal of Money, Credit, and Banking* 29 338–350.

Gabaix, Xavier. 2020. "A behavioral New Keynesian model." *American Economic Review* 110 (8): 2271–2327.

Gabaix, Xavier, and Matteo Maggiori. 2015. "International liquidity and exchange rate dynamics." *Quarterly Journal of Economics* 130 (3): 1369–1420.

Gagliardone, Luca, and Mark Gertler. 2023. "Oil Prices, Monetary Policy and Inflation Surges." National Bureau of Economic Research wp31263.

Gately, Dermot. 1986. "Lessons from the 1986 oil price collapse." *Brookings Papers on Economic Activity* 1986 (2): 237–284.

Gilchrist, Simon, and John V Leahy. 2002. "Monetary policy and asset prices." *Journal of Monetary Economics* 49 (1): 75–97.

Gilchrist, Simon, and Egon Zakrajšek. 2012. "Credit spreads and business cycle fluctuations." *American Economic Review* 102 (4): 1692–1720.

Goodfriend, Marvin. 2007. "How the world achieved consensus on monetary policy." *Journal of Economic Perspectives* 21 (4): 47–68.

Gourio, François, and Phuong Ngo. 2020. "Risk premia at the ZLB: A macroeconomic interpretation." Working Paper, Federal Reserve Bank of Chicago.

Gourio, François, and Phuong Ngo. 2022. "Downward Nominal Rigidities and Term Premia Preliminary." Working Paper, Chicago Federal Reserve and Cleveland State University.

Greenwood, Jeremy, Zvi Hercowitz, and Gregory W Huffman. 1988. "Investment, capacity utilization, and the real business cycle." *American Economic Review* 402–417.

Gürkaynak, Refet S, Brian Sack, and Jonathan H Wright. 2007. "The US Treasury Yield Curve: 1961 to the Present." *Journal of Monetary Economics* 54 (8): 2291–2304.

Gürkaynak, Refet S, Brian Sack, and Jonathan H Wright. 2010. "The TIPS Yield Curve and Inflation Compensation." *American Economic Journal: Macroeconomics* 2 (1): 70–92.

Hardouvelis, Gikas A. 1994. "The term structure spread and future changes in long and short rates in the G7 countries: Is there a puzzle?" *Journal of Monetary Economics* 33 (2): 255–283.

Hazell, Jonathon, Juan Herreno, Emi Nakamura, and Jón Steinsson. 2022. "The slope of the Phillips curve: Evidence from US states." *Quarterly Journal of Economics* 137 1299–1344.

Hazell, Jonathon, and Stephan Hobler. 2024. "Do Deficits Cause Inflation? A High Frequency Narrative Approach."

He, Zhiguo, Stefan Nagel, and Zhaogang Song. 2022. "Treasury inconvenience yields during the COVID-19 crisis." *Journal of Financial Economics* 143 (1): 57–79.

Itskhoki, Oleg, and Dmitry Mukhin. 2021. "Exchange rate disconnect in general equilibrium." *Journal of Political Economy* 129 (8): 2183–2232.

Jiang, Zhengyang, Arvind Krishnamurthy, and Hanno Lustig. 2021. "Foreign safe asset demand and the dollar exchange rate." *Journal of Finance* 76 (3): 1049–1089.

Jurado, Kyle, Sydney C Ludvigson, and Serena Ng. 2015. "Measuring uncertainty." *American Economic Review* 105 (3): 1177–1216.

Kekre, Rohan, and Moritz Lenel. 2021. "The flight to safety and international risk sharing." National Bureau of Economic Research Working Paper wp29238.

Kekre, Rohan, and Moritz Lenel. 2022. "Monetary policy, redistribution, and risk premia." *Econometrica* 90 (5): 2249–2282.

Kekre, Rohan, Moritz Lenel, and Federico Mainardi. 2024. "Monetary policy, segmentation, and the term structure." NBER Working Paper wp32324.

Kozak, Serhiy. 2022. "Dynamics of bond and stock returns." *Journal of Monetary Economics* 126 188–209.

Kozak, Serhiy, and Shrihari Santosh. 2020. "Why do discount rates vary?" *Journal of Financial Economics* 137 (3): 740–751.

Krishnamurthy, Arvind, and Annette Vissing-Jorgensen. 2012. "The aggregate demand for Treasury debt." *Journal of Political Economy* 120 (2): 233–267.

Kung, Howard. 2015. "Macroeconomic linkages between monetary policy and the term structure of interest rates." *Journal of Financial Economics* 115 (1): 42–57.

Laarits, Toomas. 2022. "Precautionary savings and the stock-bond covariance." Working Paper, NYU Stern School of Business.

Li, Erica XN, Tao Zha, Ji Zhang, and Hao Zhou. 2022. "Does fiscal policy matter for stock-bond return correlation?" *Journal of Monetary Economics* 128 20–34.

Liu, Zheng, Daniel F Waggoner, and Tao Zha. 2011. "Sources of macroeconomic fluctuations: A regime-switching DSGE approach." *Quantitative Economics* 2 (2): 251–301.

Lucas, Robert E. Jr. 1988. "On the Mechanics of Economic Development." *Journal of Monetary Economics* 22 3–42.

Malmendier, Ulrike, and Stefan Nagel. 2016. "Learning from inflation experiences." *Quarterly Journal of Economics* 131 (1): 53–87.

Mankiw, N Gregory, Jeffrey A Miron, and David N Weil. 1987. "The adjustment of expectations to a change in regime: A study of the founding of the Federal Reserve." *American Economic Review* 77 358–374.

Mehra, Rajnish, and Edward C Prescott. 1985. "The equity premium: A puzzle." *Journal of monetary Economics* 15 (2): 145–161.

Nagel, Stefan, and Zhengyang Xu. 2022. "Dynamics of subjective risk premia." National Bureau of Economic Research working paper wp29803.

Pflueger, Carolin E, and Luis M Viceira. 2016. "Return predictability in the Treasury market: Real rates, inflation, and liquidity." In *Handbook in Fixed-Income Securities*, edited by Veronesi, Pietro, Wiley, New Jersey.

Pflueger, Carolin, and Gianluca Rinaldi. 2022. "Why does the Fed move markets so much? A model of monetary policy and time-varying risk aversion." *Journal of Financial Economics* 146 71–68.

Pflueger, Carolin, Emil Siriwardane, and Adi Sunderam. 2020. "Financial market risk perceptions and the macroeconomy." *Quarterly Journal of Economics* 135 (3): 1443–1491.

Piazzesi, Monika, Juliana Salomao, and Martin Schneider. 2015. "Trend and Cycle in Bond Premia." Working Paper, Stanford University and University of Minnesota.

Piazzesi, Monika, and Martin Schneider. 2006. "Equilibrium yield curves." *NBER Macroeconomics Annual* 21 389–472.

Primiceri, Giorgio E. 2006. "Why inflation rose and fell: Policy-makers' beliefs and US postwar stabilization policy." *Quarterly Journal of Economics* 121 (3): 867–901.

Ramey, Valerie A. 2016. "Macroeconomic shocks and their propagation." *Handbook of Macroeconomics* 2 71–162.

Reis, Ricardo. 2022. "The burst of high inflation in 2021–22: How and why did we get here?", Working paper, London School of Economics.

Rotemberg, Julio J. 1982. "Monopolistic price adjustment and aggregate output." *Review of Economic Studies* 49 (4): 517–531.

Rubbo, Elisa. 2022. "Networks, Phillips curves, and monetary policy." Working Paper, University of Chicago.

Rudebusch, Glenn D, and Eric T Swanson. 2012. "The bond premium in a DSGE model with long-run real and nominal risks." *American Economic Journal: Macroeconomics* 4 (1): 105–43.

Sims, Christopher A, and Tao Zha. 2006. "Were there regime switches in US monetary policy?" *American Economic Review* 96 (1): 54–81.

Smets, Frank, and Rafael Wouters. 2007. "Shocks and frictions in US business cycles: A Bayesian DSGE approach." *American Economic Review* 97 586–606.

Song, Dongho. 2017. "Bond market exposures to macroeconomic and monetary policy risks." *Review of Financial Studies* 30 (8): 2761–2817.

Stock, James H, and Mark W Watson. 2007. "Why has US inflation become harder to forecast?" *Journal of Money, Credit and Banking* 39 3–33.

Swanson, Eric T. 2021. "A macroeconomic model of equities and real, nominal, and defaultable debt." Working paper, University of California, Irvine.

Swanson, Eric T, and John C Williams. 2014. "Measuring the effect of the zero lower bound on medium-and longer-term interest rates." *American Economic Review* 104 (10): 3154–85.

Taylor, John B. 1993. "Discretion versus policy rules in practice." *Carnegie-Rochester Conference Series on Public Policy* 39 195–214.

Uhlig, Harald. 1999. "A toolkit for analysing nonlinear dynamic stochastic models easily." In *Computational Methods for the Study of Dynamic Economics*, edited by Marimon, Ramon, and Andrew Scott 30–61, Oxford University Press.

APPENDIX - FOR ONLINE PUBLICATION

Appendix: Back to the 1980s or Not? The Drivers of Inflation and Real Risks in Treasury Bonds

December 2024

A Newspaper Count Evidence of Inflationary Supply Shocks

This Section presents a measure of the perceived relevance of different macroeconomic shocks for inflation from a completely separate data source, namely a simple newspaper article count. Figure A1 shows that between 15 and 30 percent of Wall Street Journal articles that mention inflation also mention the keyword "supply". Variations of this metric, e.g. scaling by the number of articles mentioning "inflation" and "demand" or instead counting articles in other major newspapers, such as the New York Times, yield similar patterns over time in unreported results.

We see that this share rises through the 1960s and 1970s, peaking at 26% in the 1980s. Figure A1 hence suggests that while supply shocks were prominent in investors' minds during the 1960s and 1970s, their prominence persisted and even slightly increased during the 1980s, supporting a high volatility of supply shocks for the first subperiod calibration in the main paper.

Figure A1 shows that the share of inflation news articles mentioning supply declined after the 1980s, reaching a trough of 15% during the 2010s. Hence, this simple news article count supports the shock calibrations for the two subperiods in the paper, with volatile supply shocks during the 1980s and 1990s but volatile demand shocks during the pre-pandemic 2010s. The spike in news articles mentioning "supply" during the post-pandemic period in Figure A1 further suggests that supply shocks were of substantial concern during the post-pandemic period, similarly to the 1980s.

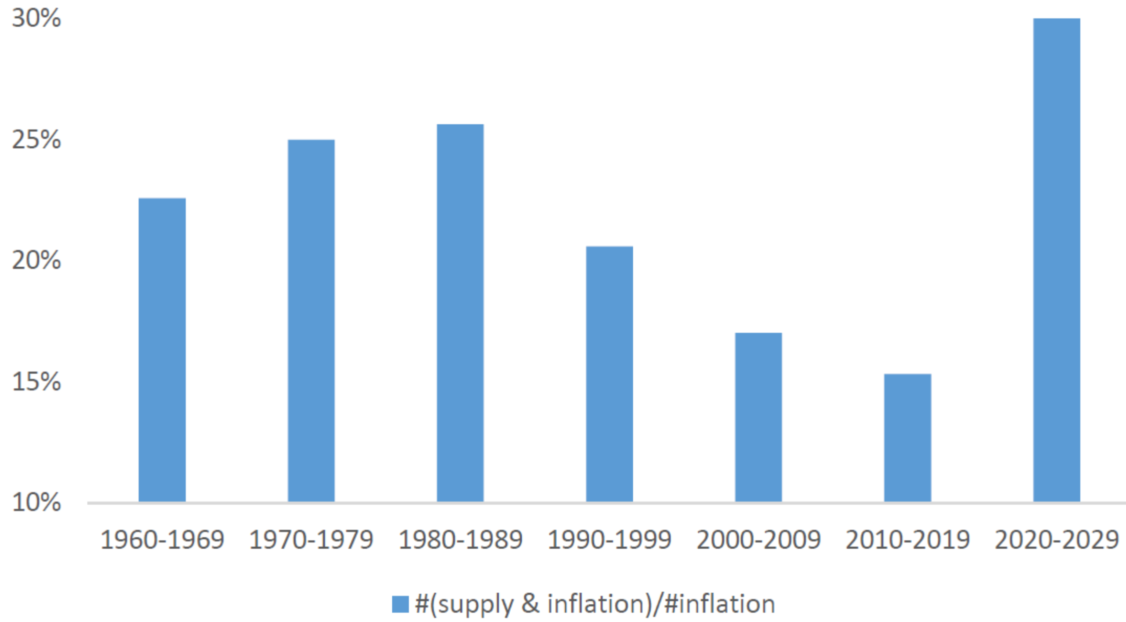
B Supply Side Microfoundations

B.1 Final good

A final consumption good is produced by a representative perfectly competitive firm from a continuum of differentiated goods $Y_{i,t}$:

$$Y_t = \left(Y_{i,t}^{\frac{\epsilon_p - 1}{\epsilon_p}} \right)^{\frac{\epsilon_p}{\epsilon_p - 1}}. \tag{A1}$$

Figure A1: Relative Frequency of Supply and Inflation in WSJ Articles



This figure shows the number of results from the ProQuest Historical Newspapers search "publication(Wall Street Journal) AND inflation AND supply" divided by the number of results from the ProQuest Historical Newspapers search "publication(Wall Street Journal) AND inflation" for each decade starting in 1960. The start date for the search is January 1, 1960 and the end date is September 19, 2024.

The constant $\epsilon_p > 1$ is the elasticity of substitution across intermediate goods. The resulting demand for the differentiated good i is downward-sloping in its product price $P_{i,t}$:

$$Y_{i,t} = Y_t \left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon_p}. \tag{A2}$$

The aggregate price level is given by

$$P_t = \left(\int_0^1 P_{i,t}^{-(\epsilon_p - 1)} di\right)^{-\frac{1}{\epsilon_p - 1}}. \tag{A3}$$

B.2 Intermediate good producers

Intermediate goods firm i produces according to a Cobb-Douglas production function with constant returns to scale

$$Y_{i,t} = A_t N_{i,t}, \tag{A4}$$

where productivity equals A_t and N_t is the supply of the aggregate labor index. Each firm takes the downward-sloping demand schedule as given (A3) and may therefore choose a different amount of the aggregate labor index. With the final good equation (A1) aggregate output equals

$$Y_t = A_t N_t \tag{A5}$$

where

$$N_t = \int_0^1 N_{i,t} di. \tag{A6}$$

The aggregate resource constraint is simple because there is no real investment and consumption equals output:

$$C_t = Y_t. \tag{A7}$$

Following Lucas (1988) we assume that productivity depends on past skills gained by all agents, and depends on past market labor, n_{t-1}:

$$a_t = \nu + a_{t-1} + (1-\phi)n_{t-1}, \tag{A8}$$

where $0 \leq \phi \leq 1$ and $\nu > 0$ are constants. The assumption (A8) ensures that potential output increases with past output. The process(A8) can equivalently be interpreted as a simple endogenous capital stock, similarly to Woodford (2003, Chapter 5), if a fixed proportion of market labor each period is used to produce investment goods with a constant-returns-to-scale technology, and the total amount of labor is scaled accordingly.

Intermediate firm i' real profit in period t equals

$$Pr_{i,t} = \frac{P_{i,t}}{P_t} Y_{i,t} - \frac{W_t}{P_t} N_{i,t}, \tag{A9}$$

subject to the production function (A4), demand for differentiated goods (A2), and taking the wage W_t as given.

B.3 Employment agency

There is a continuum of monopolistically competitive households, each of which supplies a differentiated labor service, $L_{h,t}$, to the production sector. A representative employment agency aggregates households' labor hours according to a CES production technology with elasticity of substitution $\epsilon_w > 1$:

$$N_t = \left(\int_0^1 L_{h,t}^{\frac{\epsilon_w - 1}{\epsilon_w}} \right)^{\frac{\epsilon_w}{\epsilon_w - 1}} \tag{A10}$$

The agency produces the aggregate labor index, N_t, taking each household's wage rate, $W_{h,t}$ as given, and then sells it to the production sector at the unit cost W_t. The profit maximization of the employment agency is:

$$max_{L_{h,t}} W_t \left(\int_0^1 L_{h,t}^{\epsilon_w - 1} \epsilon_w \right) - \int_0^1 W_{h,t} L_{h,t} dh, \tag{A11}$$

which yields the following demand schedule for the labor hours of household h:

$$L_{h,t} = \left(\frac{W_{h,t}}{W_t} \right)^{-\epsilon_w} N_t. \tag{A12}$$

The wage index faced by intermediary producers is then given by

$$W_t = \left(\int_0^1 W_{h,t}^{1-\epsilon_w} \right)^{\frac{1}{1-\epsilon_w}}. \tag{A13}$$

B.4 Labor-leisure choice

Following the classic model of Greenwood et al. (1988), we assume that total consumption consists of a combination of market consumption and home production, given by:

$$C_{h,t}^{home} = A_t \left(1 - \frac{L_{h,t}^{1+\eta}}{1+\eta} \right) \tag{A14}$$

Home production has decreasing returns to scale as in Campbell and Ludvigson (2001), and the parameter η determines the elasticity of market labor supply. Household h's utility depends on market and home good consumption and the corresponding external habit levels H_t and H_t^{home}:

$$U_{h,t} = \frac{\left((C_{h,t} - H_t) + \left(C_{h,t}^{home} - H_t^{home} \right) \right)^{1-\gamma} - 1}{1-\gamma} \tag{A15}$$

We assume that home good habits are shaped by the aggregate consumption of home goods, so $H_t^{home} = C_t^{home}$ and in equilibrium home goods drop out of the utility function because all households end up choosing the same labor supply in equilibrium. Home production nonetheless matters for the wage-setting first-order condition, which depends on the marginal change in utility from choosing an off-equilibrium path labor supply. External market habit is described by the surplus consumption dynamics in the main paper.

B.5 Price- and wage-setting

We consider the simplified case with flexible product prices but sticky wages. Wage-setting frictions take the form of Rotemberg (1982). Specifically, we assume that wage-setters face a quadratic cost if they raise wages faster than past inflation. The indexing to past inflation is analogous to the indexing assumption in Smets and Wouters (2007) and Christiano et al. (2005). The cost of re-setting wages for household h in terms of aggregate output equals

$$Cost^h = \frac{\gamma_w}{2}\left(\frac{W_{h,t}}{W_{h,t-1}}/\frac{W_{t-1}}{W_{t-2}} - 1\right)^2 Y_t. \tag{A16}$$

We assume that wage-setting costs get rebated to households lump-sum, i.e. aggregate consumption is unaffected.

B.6 Profit first-order condition

Because product prices are flexible, intermediate firm i's profit becomes

$$Pr_{i,t} = Y_t\left(\left(\frac{P_{i,t}}{P_t}\right)^{-(\epsilon_p-1)} - \frac{W_t}{P_tA_t}\left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon_p}\right). \tag{A17}$$

Taking the first-order condition with respect to the relative price $\frac{P_{i,t}}{P_t}$ gives

$$\frac{P_{i,t}}{P_t} = \frac{\epsilon_p}{\epsilon_p - 1}\frac{W_t}{P_tA_t}. \tag{A18}$$

Because in equilibrium all firms end up choosing the same price, we have that the real wage equals

$$\frac{W_t}{P_t} = \frac{\epsilon_p - 1}{\epsilon_p}A_t. \tag{A19}$$

This means that due to partially monopolistic competition the real wage is compressed by a constant fraction relative to productivity and equilibrium profits of intermediary i are exactly proportional to aggregate output:

$$Pr_{i,t} = \frac{1}{\epsilon_p} Y_t. \tag{A20}$$

This is good because a consumption claim is the same as a claim to firm profits.

B.7 Wage-setting first-order condition with flexible wages

To derive the wage-setting first-order condition, we first start by understanding what happens if wages are flexible. In this case, the first-order condition equals:

$$\begin{aligned}
0 &= \frac{d\left(C_{h,t} + C_{h,t}^{home}\right)}{d\left(W_{h,t}/W_t\right)} && \text{(A21)}\\
&= \frac{d}{d\left(W_{h,t}/W_t\right)}\left[\frac{W_{h,t}}{P_t}\left(\frac{W_{h,t}}{W_t}\right)^{-\epsilon_w} N_t - \frac{A_t}{1+\eta}\left(\frac{W_{h,t}}{W_t}\right)^{-\epsilon_w(1+\eta)} N_t^{(1+\eta)}\right], && \text{(A22)}\\
&= \left[(-\epsilon_w + 1)\frac{W_t}{P_t} N_t \left(\frac{W_{h,t}}{W_t}\right)^{-\epsilon_w} + \epsilon_w A_t \left(\frac{W_{h,t}}{W_t}\right)^{-\epsilon_w(1+\eta)-1} N_t^{(1+\eta)}\right] && \text{(A23)}
\end{aligned}$$

Because all wage-setters choose the same flexible-wage wage, we can set $W_{h,t} = W_t$. It then follows that the flexible-wage real wage increases proportionately with productivity and increases with the total amount of labor supplied

$$\begin{aligned}
\frac{W_t^{flex}}{P_t} &= \frac{\epsilon_w}{\epsilon_w - 1} A_t N_t^{\eta}, && \text{(A24)}\\
&= \frac{\epsilon_w}{\epsilon_w - 1} A_t^{1-\eta} Y_t^{\eta} && \text{(A25)}
\end{aligned}$$

B.8 Sticky wage first-order condition

In the derivation of the wage Phillips curve we use the operator $\tilde{E}_t$ to denote the partially adaptive inflation expectations of wage-setters. With the quadratic wage-setting cost (A14) the first-order condition for wage-setting becomes

$$\begin{aligned}
0 = &\frac{d\left(C_{h,t} + C_{h,t}^{home}\right)}{d\left(\frac{W_{h,t}}{W_t}\right)} - \gamma_w \left(\frac{W_{h,t}}{W_{h,t-1}}\frac{W_{t-2}}{W_{t-1}} - 1\right)\frac{W_t}{W_{h,t-1}}\frac{W_{t-2}}{W_{t-1}} Y_t \\
&+ \gamma_w \tilde{E}_t M_{h,t+1}\left(\frac{W_{h,t+1}}{W_{h,t}}\frac{W_{t-1}}{W_t} - 1\right)\frac{W_{h,t+1}}{W_{h,t}}\frac{W_t}{W_{h,t}}\frac{W_{t-1}}{W_t} Y_{t+1},
\end{aligned} \tag{A26}$$

Since there is symmetry (i.e. all households face the same problem), we can drop the h index when solving for the aggregate wage.

B.9 Log-linearizing the first-order wage-setting condition

Denoting the flexible wage steady-state output by $\bar{Y}_t$, we have that

$$\bar{Y}_t = A_t \bar{N}, \tag{A27}$$

where the flexible-wage labor supply solves

$$\frac{\epsilon_p - 1}{\epsilon_p} = \frac{\epsilon_w}{\epsilon_w - 1}\bar{N}^{\eta}. \tag{A28}$$

Using lower case for logs and hats to denote deviations from the flexible-wage equilibrium, the log output gap equals

$$x_t \equiv \hat{y}_t = n_t - \bar{n}, \tag{A29}$$

$$= \hat{n}_t. \tag{A30}$$

The steady-state stochastic discount factor equals

$$\bar{M}_{t,t+1} = \beta exp\left(-\gamma g\right). \tag{A31}$$

For convenience we define the constant

$$\beta_g \equiv \beta exp(-(\gamma - 1)g). \tag{A32}$$

Letting $\pi_t^w = \log \frac{W_t}{W_{t-1}}$ denote nominal log wage inflation and taking a first-order approximation around $\pi^w = 0$, expression (A26) simplifies to:

$$\begin{aligned} 0 = & (-\epsilon_w + 1)\frac{W_t}{P_t}N_t + \epsilon_w A_t N_t^{(1+\eta)} - \gamma_w\left(\pi_t^w - \pi_{t-1}^w\right)Y_t \\ & +\gamma_w Y_t \tilde{E}_t M_{t+1}\left(\pi_{t+1}^w - \pi_t^w\right)\frac{Y_{t+1}}{Y_t} \end{aligned} \tag{A33}$$

Re-arranging:

$$\begin{aligned}
\epsilon_w - 1 &= \epsilon_w A_t \frac{P_t}{W_t} N_t^\eta - \gamma_w \left(\pi_t^w - \pi_{t-1}^w\right) \frac{P_t}{W_t}\frac{Y_t}{N_t} \\
&\quad +\beta\gamma_w \frac{P_t}{W_t}\frac{Y_t}{N_t}\tilde{E}_t M_{t+1}\left(\pi_{t+1}^w - \pi_t^w\right)\frac{Y_{t+1}}{Y_t}, && \text{(A34)} \\
&= \epsilon_w A_t \frac{P_t}{W_t} N_t^\eta - \gamma_w \left(\pi_t^w - \pi_{t-1}^w\right) \frac{P_t}{W_t}\frac{Y_t}{N_t} \\
&\quad +\gamma_w \frac{P_t}{W_t}\frac{Y_t}{N_t}\beta_g \tilde{E}_t \left(\pi_{t+1}^w - \pi_t^w\right), && \text{(A35)}
\end{aligned}$$

where in the last step we dropped second-order terms in M_{t+1} and output growth interacted with wage inflation. We next substitute the production function into (A35):

$$\begin{aligned}
(\epsilon_w - 1)\frac{W_t}{A_t P_t} &= \epsilon_w N_t^\eta - \gamma_w\left(\pi_t^w - \pi_{t-1}^w\right) + \beta^g \gamma_w \tilde{E}_t\left(\pi_{t+1}^w - \pi_t^w\right), && \text{(A36)} \\
& && \text{(A37)}
\end{aligned}$$

giving the wage Phillips curve

$$\pi_t^w = \frac{1}{1+\beta_g}\pi_{t-1}^w + \frac{\beta^g}{1+\beta_g}\tilde{E}_t \pi_{t+1}^w + \gamma_w^{-1}\left(\epsilon_w N_t^\eta - (\epsilon_w - 1)\frac{W_t}{A_t P_t}\right). \tag{A38}$$

Note that the term in parentheses is the wedge between the real productivity-adjusted wage and workers' productivity-adjusted disutility of labor. Because we have flexible product prices the real productivity-adjusted wage is constant and we can substitute in from (A19):

$$\pi_t^w = \frac{1}{1+\beta_g}\pi_{t-1}^w + \frac{\beta^g}{1+\beta_g}\tilde{E}_t \pi_{t+1}^w + \gamma_w^{-1}\left(\epsilon_w N_t^\eta - (\epsilon_w - 1)\frac{\epsilon_p - 1}{\epsilon_p}\right) \tag{A39}$$

In the flexible-wage equilibrium the term in parentheses is zero, giving the first-order log-linearization

$$\begin{aligned}
\epsilon_w N_t^\eta - (\epsilon_w - 1)\frac{\epsilon_p - 1}{\epsilon_p} &= \epsilon_w \bar{N}^\eta exp(\eta \hat{n}_t) - (\epsilon_w - 1)\frac{\epsilon_p - 1}{\epsilon_p}, && \text{(A40)} \\
&\approx \epsilon_w \bar{N}^\eta \eta \hat{n}_t, && \text{(A41)} \\
&= \epsilon_w \bar{N}^\eta \eta \hat{y}_t && \text{(A42)}
\end{aligned}$$

We therefore obtain the standard log-linearized wage Phillips curve

$$\pi_t^w = \frac{1}{1+\beta_g}\pi_{t-1}^w + \frac{\beta^g}{1+\beta_g}\tilde{E}_t \pi_{t+1}^w + \kappa \hat{y}_t, \tag{A43}$$

where the constant κ equals

$$\kappa = \gamma_w^{-1} \epsilon_w \bar{N}^\eta \eta \tag{A44}$$

Substituting in the adaptive inflation expectations assumption

$$\tilde{E}_t \pi_{t+1}^w = (1-\zeta) E_t \pi_{t+1}^w + \zeta \pi_{t-1}^w, \tag{A45}$$

gives the wage Phillips curve

$$\pi_t^w = \rho^\pi \pi_{t-1}^w + f^\pi E_t \pi_{t+1}^w + \kappa \hat{y}_t, \tag{A46}$$

where

$$\rho^\pi = \frac{1}{1+\beta_g} + \zeta - \frac{1}{1+\beta_g} \zeta, \tag{A47}$$

$$f^\pi = 1 - \rho^\pi. \tag{A48}$$

Phillips curve shocks to (A46), $v_{\pi,t}$, arise from making the degree of monopolistic wage-setting frictions ϵ_w or the marginal cost of providing labor outside the home η time-varying.

B.10 Price inflation

Product prices equal

$$P_t = \frac{\epsilon_p}{\epsilon_p - 1} \frac{W_t}{A_t}, \tag{A49}$$

so log price inflation equals (up to a constant)

$$\pi_t^p = \pi_t^w - \Delta a_t, \tag{A50}$$

$$= \pi_t^w - (1-\phi) \hat{y}_{t-1}, \tag{A51}$$

where the log deviation of real GDP from potential is the output gap, i.e. $x_t = \hat{y}_t$.

C Solution

In the absence of demand shocks the lagged output gap does not enter as a separate state variable because x_{t-1} can be expressed as a linear combination of the time-t state vector Y_t.

This is no longer possible in the presence demand shocks, thereby adding x_{t-1} as a new state variable for asset prices relative to Campbell, Pflueger and Viceira (2020).

C.1 Solving for macroeconomic dynamics

The full macroeconomic dynamics are determined by the Euler equation, the wage Phillips curve (A46) and the monetary policy rule, as well as the short-rate Fisher equation $r_t = i_t - E_t\pi^p_{t+1}$, the relationship between price and wage inflation (A51). The Euler equation is given by

$$x_t = f^x E_t x_{t+1} + \rho^x x_{t-1} - \psi\left(i_t - E_t\pi^p_{t+1}\right) + v_{x,t}, \tag{A52}$$

where

$$\rho^x = \frac{\theta_2}{\phi - \theta_1}, \tag{A53}$$

$$f^x = \frac{1}{\phi - \theta_1}, \tag{A54}$$

$$\psi = \frac{1}{\gamma(\phi - \theta_1)}, \tag{A55}$$

$$\theta_2 = \phi - 1 - \theta_1. \tag{A56}$$

The wage Phillips curve is given by

$$\pi^w_t = \rho^\pi \pi^w_{t-1} + f^\pi E_t\pi^w_{t+1} + \kappa x_t + v_{\pi,t}, \tag{A57}$$

The monetary policy rule is given by

$$i_t = \rho^i i_{t-1} + (1 - \rho^i)\left(\gamma^x x_t + \gamma^\pi \pi^p_t\right) + v_{i,t}, \tag{A58}$$

where $v_{x,t} = \frac{1}{\gamma(\phi-\theta_1)}\xi_t$ denotes the demand shock; $v_{\pi,t}$ is the supply shock; and $v_{i,t}$ is the monetary policy shock.

We want to find a solution of the form

$$Y_t = BY_{t-1} + \Sigma v_t, \tag{A59}$$

where the matrix B is $[3 \times 3]$, the matrix Σ is $[3 \times 3]$, and we work with the state vector

$$Y_t = [x_t, \pi^w_t, i_t]', \tag{A60}$$

and the shock vector

$$v_t = [v_{x,t}, v_{\pi,t}, v_{i,t}]'. \tag{A61}$$

Using the relationship (A51), we can write the macroeconomic dynamics in terms of the state vector Y_t:

$$\begin{aligned}
Y_{1,t} &= f^x E_t Y_{1,t+1} + \rho^x Y_{1,t-1} - \psi\left(Y_{3,t} - E_t Y_{2,t+1} + (1-\phi)Y_{1,t}\right) + v_{x,t}, && \text{(A62)}\\
Y_{2,t} &= f^\pi E_t Y_{2,t+1} + \rho^\pi Y_{2,t-1} + \kappa Y_{1,t} + v_{\pi,t}, && \text{(A63)}\\
Y_{3,t} &= \rho^i Y_{3,t-1} + (1-\rho^i)\left(\gamma^x Y_{1,t} + \gamma^\pi Y_{2,t} - \gamma^\pi(1-\phi)Y_{1,t-1}\right) + v_{i,t}. && \text{(A64)}
\end{aligned}$$

We can write this in matrix form:

$$0 = F E_t Y_{t+1} + G Y_t + H Y_{t-1} + M v_t,$$

where the matrices F, G and H are given by

$$\begin{aligned}
F &= \begin{bmatrix} \frac{f^x}{1+\psi(1-\phi)} & \frac{\psi}{1+\psi(1-\phi)} & 0 \\ 0 & f^\pi & 0 \\ 0 & 0 & 0 \end{bmatrix},\\
G &= \begin{bmatrix} -1 & 0 & -\frac{\psi}{1+\psi(1-\phi)} \\ \kappa & -1 & 0 \\ (1-\rho^i)\gamma^x & (1-\rho^i)\gamma^\pi & -1 \end{bmatrix},\\
H &= \begin{bmatrix} \frac{\rho^x}{1+\psi(1-\phi)} & 0 & 0 \\ 0 & \rho^\pi & 0 \\ -(1-\rho^i)(1-\phi)\gamma^\pi & 0 & \rho^i \end{bmatrix}.
\end{aligned}$$

The matrix M is $[3 \times 3]$ and equals:

$$M = \begin{bmatrix} \frac{1}{1+\psi(1-\phi)} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{A65}$$

Following Uhlig (1999), we solve for the generalized eigenvectors and eigenvalues of the matrix Ξ with respect to the matrix Δ, where

$$\Xi = \begin{bmatrix} -G & -H \\ I_3 & 0_3 \end{bmatrix}, \tag{A66}$$

$$\Delta = \begin{bmatrix} F & 0_3 \\ 0_3 & I_3 \end{bmatrix} \tag{A67}$$

To obtain a solution, we then pick three generalized eigenvalues $\lambda_1, \lambda_2, \lambda_3$ with generalized eigenvectors $[\lambda z_1', z_1']'$, $[\lambda_2 z_2', z_2]'$, and $[\lambda_3 z_3', z_3']'$. We denote the diagonal matrix of these eigenvalues by $\Lambda = diag\,(\lambda_1, \lambda_2, \lambda_3)$, and the matrix of the lower $[3 \times 1]$ portion of the eigenvectors by $\Omega = [z_1, z_2, z_3]$. The corresponding solutions for B and Σ are then given by:

$$B = \Omega\Lambda\Omega^{-1}, \tag{A68}$$

$$\Sigma = -\left[FB + G\right]^{-1} M. \tag{A69}$$

For both our calibrations, there exist exactly three generalized eigenvalues with absolute value less than one, and we pick the non-explosive solution corresponding to these three eigenvalues.

C.2 Rotated state vector

Our state space for solving for asset prices is five-dimensional: It consists of $\tilde{Z}_t$, which a scaled version of Y_t, the surplus consumption ratio relative to steady-state $\hat{s}_t$, and the lagged output gap x_{t-1}.

We next describe the definition of $\tilde{Z}_t$. To simplify the numerical implementation of the asset pricing recursions, we require that shocks to the scaled state vector $\tilde{Z}_t$ are independent standard normal and that the first dimension of the scaled state vector is perfectly correlated with consumption innovations. This rotation facilitates the numerical analysis, because it is easier to integrate over independent random variables. Aligning the first dimension of the scaled state vector with output gap innovations (and hence surplus consumption innovations) helps, because it allows us to use a finer grid to integrate numerically over this crucial dimension over which asset prices are most non-linear.

If the scaled state vector equals $\tilde{Z}_t = AY_t$ for some invertible matrix A, the dynamics of

$\tilde{Z}_t$ are given by:

$$\tilde{Z}_t = AY_t, \tag{A70}$$

$$\tilde{Z}_{t+1} = \underbrace{ABA^{-1}}_{\tilde{B}}\tilde{Z}_t + \underbrace{A\Sigma v_{t+1}}_{\epsilon_{t+1}}. \tag{A71}$$

We hence want a matrix, A, such that

$$Var\left(\epsilon_{t+1}\right) = A\Sigma\Sigma_v\Sigma' A', \tag{A72}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{A73}$$

Finding such a matrix A should in general be possible, because the matrix M and therefore $\Sigma\Sigma_v\Sigma'$ generally have rank three. We require that the first dimension of ϵ_{t+1} is perfectly correlated with the consumption shock. We can therefore find the three rows of A using the following steps:

1. Set $A_1 = \frac{e_1}{\sqrt{e_1\Sigma\Sigma_v\Sigma' e_1'}}$.

2. We use the MATLAB function *null* to compute the null space of $A_1\Sigma\Sigma_v\Sigma'$. Let n_2 denote the first vector in $null\left(A_1\Sigma\Sigma_v\Sigma'\right)$. We then define the second row of A as the normalized version of n_2:

$$A_2 = \frac{n_2}{\sqrt{n_2\Sigma\Sigma_v\Sigma' n_2'}}. \tag{A74}$$

3. Let n_3 denote the first vector in $null\left(A_1\Sigma\Sigma_v\Sigma', A_2\Sigma\Sigma_v\Sigma'\right)$. We then define the third row of A as the normalized version of n_3:

$$A_3 = \frac{n_3}{\sqrt{n_3\Sigma\Sigma_v\Sigma' n_3'}}. \tag{A75}$$

It is then straightforward to verify that equation (A73) holds for

$$A = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \end{bmatrix}. \tag{A76}$$

C.3 Asset pricing recursions

Before deriving the recursions for the numerical asset pricing computations, we derive a convenient form for the dynamics of the log surplus consumption ratio. We use e_i to denote a row vector with 1 in position i and zeros elsewhere. The matrix

$$\Sigma_M = e_1\Sigma \tag{A77}$$

denotes the loading of consumption innovations onto the vector of shocks v_t, where e_1 is a basis vector with a one in the first position and zeros everywhere else. The volatility of consumption surprises equals:

$$\sigma_c^2 = \Sigma_M \Sigma_v \Sigma_M'. \tag{A78}$$

To simplify notation, we define $\hat{s}_t$ as the log deviation of surplus consumption from its steady state. The dynamics of $\hat{s}_t$ are:

$$\hat{s}_t = s_t - \bar{s}, \tag{A79}$$

$$\hat{s}_t = \theta_0 \hat{s}_{t-1} + \theta_1 x_{t-1} + \theta_2 x_{t-2} + \lambda(\hat{s}_{t-1})\varepsilon_{c,t}, \tag{A80}$$

where with an abuse of notation we write:

$$\lambda(\hat{s}_t) = \lambda_0\sqrt{1-2\hat{s}_t} - 1, \hat{s}_t \leq s_{max} - \bar{s}, \tag{A81}$$

$$\lambda(\hat{s}_t) = 0, \hat{s}_t \geq s_{max} - \bar{s}. \tag{A82}$$

The steady-state surplus consumption sensitivity equals:

$$\lambda_0 = \frac{1}{\bar{S}}. \tag{A83}$$

In our calculations of bond prices, we repeatedly substitute out expected log SDF growth, which equals:

$$E_t\left[m_{t+1}\right] = \log\beta - \gamma E_t \Delta\hat{s}_{t+1} - \gamma E_t \Delta c_{t+1}, \tag{A84}$$

$$= -r_t + \xi_t - \frac{\gamma}{2}(1-\theta_0)(1-2\hat{s}_t), \tag{A85}$$

$$= -(e_3 - e_2 B + (1-\phi)e_1)Y_t + \xi_t - \frac{\gamma}{2}(1-\theta_0)(1-2\hat{s}_t) \tag{A86}$$

We often combine this with $r_t = \bar{r} + (e_3 - e_2 B)Z_t$ and $\hat{r}_t = (e_3 - e_2 B)Z_t$.

Including the constant, consumption growth is given by:

$$\Delta c_{t+1} = g + x_{t+1} - \phi x_t. \tag{A87}$$

The steady state real short-term interest rate at $x_t = 0$ and $s_t = \bar{s}$ is the same as in Campbell and Cochrane (1999):

$$\bar{r} = \gamma g - \frac{1}{2}\gamma^2 \sigma_c^2 / \bar{S}^2 - \log(\beta). \tag{A88}$$

The updating rule for the log surplus consumption ratio can then be written in terms of the state variables as:

$$\hat{s}_{t+1} = \theta_0 \hat{s}_t + \theta_1 e_1 A^{-1} \tilde{Z}_t + \theta_2 x_{t-1} + \lambda(\hat{s}_t)\varepsilon_{c,t+1}. \tag{A89}$$

C.3.1 Recursion for zero-coupon consumption claims

We now derive the recursion for zero-coupon consumption claims in terms of state variables $\tilde{Z}_t$, $\hat{s}_t$ and x_{t-1}. Let P^c_{nt}/C_t denote the price-dividend ratio of a zero-coupon claim on consumption at time $t+n$. The outline of our strategy here is that we first derive an analytic expression for the price-dividend ratio for P^c_{1t}/C_t. For $n \geq 1$ we guess and verify recursively that there exists a function $F_n(\tilde{Z}_t, \hat{s}_t, x_{t-1})$, such that

$$\frac{P^c_{nt}}{C_t} = F_n\left(\tilde{Z}_t, \hat{s}_t, x_{t-1}\right). \tag{A90}$$

The ex-dividend price-consumption ratio for a claim to all future consumption is then given by

$$\frac{P_t}{C_t} = F\left(\tilde{Z}_t, \hat{s}_t, x_{t-1}\right), \tag{A91}$$

where we define

$$F\left(\tilde{Z}_t, \hat{s}_t, x_{t-1}\right) = \sum_{n=1}^{\infty} F_n\left(\tilde{Z}_t, \hat{s}_t, x_{t-1}\right). \tag{A92}$$

We now derive the recursion of zero-coupon consumption claims in terms of state variables $\tilde{Z}_t$ and $\hat{s}_t$. The one-period zero coupon price-consumption ratio solves:

$$\frac{P^c_{1,t}}{C_t} = E_t\left[\frac{M_{t+1}C_{t+1}}{C_t}\right] \tag{A93}$$

We simplify

$$\begin{aligned}\frac{M_{t+1}C_{t+1}}{C_t} &= \beta\exp(-\gamma E_t\Delta\hat{s}_{t+1}-(\gamma-1)E_t\Delta c_{t+1}\\ &\quad -\gamma(\hat{s}_{t+1}-E_t s_{t+1})-(\gamma-1)(c_{t+1}-E_t c_{t+1})).\end{aligned}$$

Using the notation $f_n=\log(F_n)$, this gives the log one-period price-consumption ratio as:

$$\begin{aligned}f_1\left(\tilde{Z}_t,\hat{s}_t,x_{t-1}\right) &= \log\beta-\gamma[(\theta_0-1)\hat{s}_t+\theta_1 x_t+\theta_2 x_{t-1}]-(\gamma-1)[g+E_t x_{t+1}-\phi x_t]\\ &\quad +\frac{1}{2}\left(\gamma\lambda(\hat{s}_t)+(\gamma-1)\right)^2\sigma_c^2, \qquad\qquad (A94)\\ &= \log\beta-(\gamma-1)g-e_1[(\gamma\theta_1-\gamma\phi+\phi)I+(\gamma-1)B]A^{-1}\tilde{Z}_t\\ &\quad -\gamma(\theta_0-1)\hat{s}_t-\gamma\theta_2 x_{t-1}+\frac{1}{2}\left(\gamma\lambda(\hat{s}_t)+(\gamma-1)\right)^2\sigma_c^2 \qquad\qquad (A95)\end{aligned}$$

Next, we solve for f_n, $n\geq 2$ iteratively. Note that:

$$\frac{P^c_{nt}}{C_t}=\mathbb{E}_t\left[\frac{M_{t+1}C_{t+1}}{C_t}\frac{P^c_{n-1,t+1}}{C_{t+1}}\right]=\mathbb{E}_t\left[\frac{M_{t+1}C_{t+1}}{C_t}F_{n-1}\left(\tilde{Z}_{t+1},\hat{s}_{t+1},x_t\right)\right] \qquad (A96)$$

This gives the following expression for f_n:

$$\begin{aligned}f_n(\tilde{Z}_t,\hat{s}_t,x_{t-1}) &= \log\Big[\mathbb{E}_t\Big[\exp\Big(\log\beta-(\gamma-1)g-e_1[(\gamma\theta_1-\gamma\phi+\phi)I+(\gamma-1)B]A^{-1}\tilde{Z}_t\\ &\quad -\gamma(\theta_0-1)\hat{s}_t-\gamma\theta_2 x_{t-1}-(\gamma(1+\lambda(\hat{s}_t))-1)\sigma_c\epsilon_{1,t+1}\\ &\quad +f_{n-1}(\tilde{Z}_{t+1},\hat{s}_{t+1},x_t)\Big)\Big]\Big]. \qquad\qquad (A97)\end{aligned}$$

Here, $\epsilon_{1,t+1}$ denotes the first dimension of the shock ϵ_{t+1}.

C.3.2 Recursion for zero-coupon bond prices

We use $P^{\$}_{n,t}$ and $P_{n,t}$ to denote the prices of nominal and real n-period zero-coupon bonds. The strategy is to develop analytic expressions for one- and two-period bond prices. We then guess and verify recursively that the prices of real and nominal zero-coupon bonds with maturity $n\geq 2$ can be written in the following form:

$$P_{n,t} = B_n(\tilde{Z}_t,\hat{s}_t,x_{t-1}), \qquad (A98)$$

$$P^{\$}_{n,t} = B^{\$}_n(\tilde{Z}_t,\hat{s}_t,x_{t-1}), \qquad (A99)$$

where $B_n(\tilde{Z}_t,\hat{s}_t,x_{t-1})$ and $B^{\$}_n(\tilde{Z}_t,\hat{s}_t,x_{t-1})$ are functions of the state variables. As discussed in the main paper, we assume that the short-term nominal interest rate contains no risk premium, so the one-period log nominal interest rate equals $i_t = r_t + E_t\pi_{t+1}$. Taking account of the constants, one-period bond prices equal:

$$P^{\$}_{1,t} = \exp(-Y_{3,t} - \bar{r}), \tag{A100}$$

$$P_{1,t} = \exp(-Y_{3,t} + \mathbb{E}_t Y_{2,t+1} - (1-\phi)Y_{1,t} - \bar{r}). \tag{A101}$$

We next solve for longer-term bond prices including risk premia. Substituting in (A100) into the bond-pricing recursion gives:

$$P^{\$}_{2,t} = \exp(-\xi_t)\mathbb{E}_t\left[M_{t+1}P^{\$}_{1,t+1}\exp(-Y_{2,t+1} + (1-\phi)Y_{1,t})\right] \tag{A102}$$

$$= \exp(-\xi_t)\mathbb{E}_t\left[M_{t+1}\exp(-Y_{3,t+1} - Y_{2,t+1} + (1-\phi)Y_{1,t} - \bar{r})\right]. \tag{A103}$$

We can now verify that the two-period nominal bond price takes the form (A99):

$$\begin{aligned} B^{\$}_2(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= \exp\left(E_t\left(m_{t+1} - \xi_t - Y_{3,t+1} - Y_{2,t+1} + (1-\phi)Y_{1,t}\right) - \bar{r}\right) \\ &\quad \times\mathbb{E}_t\left[\exp\left(\left(-\gamma\left(\lambda(\hat{s}_t)+1\right)\Sigma_M - \underbrace{[(e_2+e_3)\Sigma]}_{v_{\$}}\right)v_{t+1}\right)\right]. \end{aligned} \tag{A104}$$

Here, we define the vector $v_{\$}$ to simplify notation. Taking logs, substituting out for $E_t m_{t+1}$, and using the definition for the sensitivity function $\lambda(\hat{s}_t)$, we get:

$$\begin{aligned} b^{\$}_2 &= -e_3[I+B]A^{-1}\tilde{Z}_t + \frac{1}{2}v_{\$}\Sigma_v v_{\$}\prime \\ &\quad +\gamma\left(\lambda(\hat{s}_t)+1\right)\Sigma_M\Sigma_v v_{\$}' - 2\bar{r}. \end{aligned} \tag{A105}$$

The closed-form solution for the two-period real bond price becomes

$$\begin{aligned} P_{2,t} &= \exp\left(E_t\left(m_{t+1} - \xi_t - Y_{3,t+1} - (1-\phi)Y_{1,t+1} + Y_{2,t+2}\right) - \bar{r}\right) \\ &\quad \times\mathbb{E}_t\left[\exp\left((-\gamma(\lambda(\hat{s}_t)+1)\Sigma_M - \underbrace{(e_3 + (1-\phi)e_1 - e_2B)\Sigma)}_{v_r}v_{t+1}\right)\right] \end{aligned} \tag{A106}$$

We define the vector v_r to simplify notation. Taking logs, substituting out for $E_t m_{t+1}$, and using the definition for $\lambda(\hat{s}_t)$ gives:

$$b_2(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = -v_r \left[I + B\right] A^{-1} \tilde{Z}_t + \frac{1}{2} v_r \Sigma_v v_r' + \gamma \left(\lambda(\hat{s}_t) + 1\right) \Sigma_M \Sigma_v v_r' - 2\bar{r}. \tag{A107}$$

For $n \geq 3$, we repeatedly substitute out for $E_t m_{t+1}$ to obtain the following recursion for nominal and real bond prices, respectively:

$$\begin{aligned} B_n^{\$}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= \mathbb{E}_t \left[\exp\left(m_{t+1} - \xi_t - Y_{2,t+1} + (1-\phi) Y_{1,t} + b_{n-1}^{\$}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, B^{\$} x_t)\right)\right] \\ &= \mathbb{E}_t \left[\exp\left(-\bar{r} - e_3 A^{-1} \tilde{Z}_t - \frac{\gamma}{2}(1-\theta_0)(1-2\hat{s}_t) \right.\right. \\ &\quad \left.\left. -\gamma(1+\lambda(\hat{s}_t))\sigma_c \epsilon_{1,t+1} - e_2 A^{-1} \epsilon_{t+1} + b_{n-1}^{\$}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t)\right)\right]. \end{aligned} \tag{A108}$$

The value function iteration for real bond prices then becomes

$$\begin{aligned} B_n(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= \mathbb{E}_t \left[\exp\left(m_{t+1} - \xi_t + b_{n-1}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t)\right)\right] \\ &= \mathbb{E}_t \left[\exp\left(-\bar{r} - (e_3 - e_2 B + (1-\phi) e_1) A^{-1} \tilde{Z}_t - \frac{\gamma}{2}(1-\theta_0)(1-2\hat{s}_t) \right.\right. \\ &\quad \left.\left. -\gamma(1+\lambda(\hat{s}_t))\sigma_c \epsilon_{1,t+1} + b_{n-1}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t)\right)\right]. \end{aligned} \tag{A109}$$

C.3.3 Computing returns

The log return on the consumption claim equals:

$$r_{t+1}^c = \log\left(\frac{P_{t+1}^c + C_{t+1}}{P_t^c}\right), \tag{A110}$$

$$= \Delta c_{t+1} + \log\left(\frac{1 + \frac{P_{t+1}^c}{C_{t+1}}}{\frac{P_t^c}{C_t}}\right). \tag{A111}$$

Real and nominal log bond yields equal:

$$y_{n,t} = -\frac{1}{n} b_{n,t}, \tag{A112}$$

$$y_{n,t}^{\$} = -\frac{1}{n} b_{n,t}^{\$}. \tag{A113}$$

Real log bond returns equal:

$$r_{n,t+1} = b_{n-1,t+1} - b_{n,t}. \tag{A114}$$

Nominal log bond returns equal:

$$r^{\$}_{n,t+1} = b^{\$}_{n-1,t+1} - b^{\$}_{n,t}. \quad \text{(A115)}$$

Real and nominal bond log excess returns then equal:

$$xr_{n,t+1} = r_{n,t+1} - r_t, \quad \text{(A116)}$$
$$xr^{\$}_{n,t+1} = r^{\$}_{n,t+1} - i_t. \quad \text{(A117)}$$

C.3.4 Levered stock prices and returns

We note that the price of the levered equity claim is δP^c_t, so the price-dividend ratio equals:

$$\frac{P^\delta_t}{D^\delta_t} = \delta \frac{C_t}{D^\delta_t} \frac{P^c_t}{C_t}. \quad \text{(A118)}$$

Using the expression

$$D^\delta_{t+1} = P^c_{t+1} + C_{t+1} - (1-\delta) P^c_t \exp(r_t) - \delta P^c_t, \quad \text{(A119)}$$

and

$$P^\delta_t = \delta P^c_t \quad \text{(A120)}$$

gives the gross return on levered stocks:

$$\left(1 + R^\delta_{t+1}\right) = \frac{D^\delta_{t+1} + P^\delta_{t+1}}{P^\delta_t}, \quad \text{(A121)}$$
$$= \frac{1}{\delta} \frac{P^c_{t+1} + C_{t+1} - (1-\delta) P^c_t \exp(r_t)}{P^c_t}, \quad \text{(A122)}$$
$$= \frac{1}{\delta}\left(1 + R^c_{t+1}\right) - \frac{1-\delta}{\delta} \exp(r_t). \quad \text{(A123)}$$

Log stock excess returns then equal:

$$xr^\delta_{t+1} = r^\delta_{t+1} - r_t. \quad \text{(A124)}$$

To mimic firms' dividend smoothing in the data, we report simulated moments for the price of equities dividend by dividends smoothed over the past 64 quarters:

$$P_t^\delta / \left(\frac{1}{64} (D_t^\delta + D_{t-1}^\delta + ... + D_{t-63}^\delta) \right). \tag{A125}$$

C.4 Risk-premium decomposition

We use the superscript ^{rn} for risk-neutral, superscript ^{cf} for cash flow, and ^{rp} for risk premium. Risk-neutral valuations are expected cash flows discounted with the risk-neutral discount factor, given by:

$$M_{t+1}^{rn} = \exp(-(r_t - \xi_t)). \tag{A126}$$

Note that since we are not interested in risk-neutral bond and stock prices, but only a decomposition of returns, multiplying M_{t+1}^{rn} by a constant discount rate does not matter. For any zero-coupon claim it would shift risk-neutral returns merely by a constant and therefore leave our decomposition into risk-neutral and risk-premium components unaffected. For a claim to all future consumption or stock returns, a constant discount rate could theoretically shift the weights between nearer-term consumption claims and longer-term consumption claims, and therefore change risk-neutral returns. However, since consumption growth is stationary we have found that this makes very little different to risk-neutral stock returns in any of our numerical applications.

C.4.1 Risk-neutral zero-coupon bond prices

We use analogous recursions to solve for risk-neutral bond prices. One-period risk-neutral bond prices are given exactly as before by equations (A100) and (A101). For $n > 1$, we guess and verify that the prices of real and nominal risk-neutral zero-coupon bonds with maturity n can be written in the following form

$$P_{n,t}^{rn} = B_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}), \tag{A127}$$

$$P_{n,t}^{\$,rn} = B_n^{\$,rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}). \tag{A128}$$

for some functions $B_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1})$ and $B_n^{\$,rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1})$.

We derive the two-period risk-neutral nominal bond price analytically:

$$P_{2,t}^{\$,rn} = \exp(-\xi_t)\mathbb{E}_t \left[M_{t+1}^{rn} P_{1,t+1}^{\$,rn} \exp(-Y_{2,t+1} + (1-\phi)Y_{1,t}) \right] \tag{A129}$$

$$= \exp(-r_t)\mathbb{E}_t \left[\exp(-Y_{3,t+1} - Y_{2,t+1} + (1-\phi)Y_{1,t} - \bar{r}) \right]. \tag{A130}$$

We can hence verify that the two-period risk-neutral nominal bond price takes the form (A99)

$$b_2^{\$,rn} = -e_3\left[I+B\right]A^{-1}\tilde{Z}_t + \frac{1}{2}v_{\$}\Sigma_v v_{\$}\prime - 2\bar{r} \tag{A131}$$

Here, the vector $v_{\$}$ is identical to the case with risk aversion. Comparing expressions (A131) and (A105) shows that they agree when $\gamma = 0$. We similarly solve for 2-period real bond prices in closed form:

$$\begin{aligned} P_{2,t}^{rn} &= \exp\left(-Y_{3,t} + \mathbb{E}_t Y_{2,t+1} - (1-\phi)Y_{1,t} - \bar{r}\right) \times \exp\left(\mathbb{E}_t\left(-Y_{3,t+1} + \mathbb{E}_{t+1}Y_{2,t+2} + (1-\phi)Y_{1,t+1} - \bar{r}\right)\right) \\ &\times \mathbb{E}_t\left[\exp\left(-\underbrace{(e_3 + (1-\phi)e_1 - e_2 B)\Sigma}_{v_r} v_{t+1}\right)\right]. \end{aligned} \tag{A132}$$

The vector v_r is again identical to the case with risk aversion. Taking logs gives:

$$b_2^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = -(e_3 + (1-\phi)e_1 - e_2 B)\left[I+B\right]A^{-1}\tilde{Z}_t + \frac{1}{2}v_r\Sigma_v v_r' - 2\bar{r}. \tag{A133}$$

We note that the risk-neutral bond prices (A133) and bond prices with risk aversion (A107) are identical when the utility curvature parameter γ equals zero.

For $n \geq 3$ the n-period risk neutral nominal and real bond prices satisfy the following recursions, respectively:

$$B_n^{\$,rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = \mathbb{E}_t\left[\exp\left(-\bar{r} - e_3 A^{-1}\tilde{Z}_t - e_2 A^{-1}\epsilon_{t+1} + b_{n-1}^{\$}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t)\right)\right] \tag{A134}$$

$$B_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = \mathbb{E}_t\left[\exp\left(-\bar{r} - (e_3 + (1-\phi)e_1 - e_2 B)A^{-1}\tilde{Z}_t + b_{n-1}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t)\right)\right] \tag{A135}$$

C.4.2 Risk-neutral zero-coupon consumption claims

Next, we derive recursive solutions for the risk-neutral prices of zero-coupon consumption claims. Let $P_{nt}^{c,rn}/C_t$ denote the risk-neutral price-dividend ratio of a zero-coupon claim on consumption at time $t+n$. The risk-neutral price-consumption ratio of a claim to the entire stream of future consumption equals:

$$\frac{P_t^{c,rn}}{C_t} = \sum_{n=1}^{\infty} \frac{P_{nt}^{c,rn}}{C_t}. \tag{A136}$$

For $n \geq 1$, we guess and verify there exists a function $F_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1})$, such that

$$\frac{P_{nt}^{c,rn}}{C_t} = F_n^{rn}\left(\tilde{Z}_t, \hat{s}_t, x_{t-1}\right). \tag{A137}$$

We start by deriving the analytic expression for F_1^{rn}. The one-period risk-neutral zero-coupon price-consumption ratio solves

$$\begin{aligned} \frac{P_{1,t}^{c,rn}}{C_t} &= \mathbb{E}_t\left[M_{t+1}^{rn}\frac{C_{t+1}}{C_t}\right] && \text{(A138)} \\ &= \exp(-r_t + u_t)\mathbb{E}_t\left[\frac{C_{t+1}}{C_t}\right] && \text{(A139)} \\ &= \exp(-\gamma\mathbb{E}_t x_{t+1} + \gamma(\phi-\theta_1)x_t - \gamma\theta_2 x_{t-1} - \bar{r})\mathbb{E}_t\left[\frac{C_{t+1}}{C_t}\right] && \text{(A140)} \end{aligned}$$

Using (A87) to substitute for consumption growth, we can derive the following analytic expression for f_1^{rn}:

$$f_1^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = g - \bar{r} - e_1[(\gamma\theta_1 - \gamma\phi + \phi)I - (1-\gamma)B]A^{-1}\tilde{Z}_t - \gamma\theta_2 x_{t-1} + \frac{1}{2}\sigma_c^2. \tag{A141}$$

Next, we solve for f_n, $n \geq 2$ iteratively:

$$\frac{P_{nt}^{c,rn}}{C_t} = \exp\left(-\gamma\mathbb{E}_t Y_{1,t+1} + \gamma(\phi-\theta_1)Y_{1,t} - \gamma\theta_2 x_{t-1} - \bar{r}\right)\mathbb{E}_t\left[\frac{C_{t+1}}{C_t}F_{n-1}^{rn}\left(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t\right)\right] \tag{A142}$$

This gives the following expression for f_n^{rn}:

$$\begin{aligned} f_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) = \log\Big[\mathbb{E}_t\Big[\exp\Big(&-\gamma\mathbb{E}_t Y_{1,t+1} + \gamma(\phi-\theta_1)Y_{1,t} - \gamma\theta_2 x_{t-1} - \bar{r} \\ &+g - \phi Y_{1,t} + \mathbb{E}_t Y_{1,t+1} + \sigma_c\epsilon_{1,t+1} \\ &+f_{n-1}^{rn}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t)\Big)\Big]\Big]. \end{aligned} \tag{A143}$$

Finally, we re-write $f_{n,t}^{rn}$ as an expectation involving $f_{n-1,t+1}^{rn}$, the state variables $\tilde{Z}_t$, and period $t+1$ shocks:

$$f_n^{rn}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) \;=\; \log\Big[\mathbb{E}_t\Big[\exp\Big(e_1[(1-\gamma)B - (\phi - \gamma(\phi-\theta_1))I]A^{-1}\tilde{Z}_t - \gamma\theta_2 x_{t-1} + g - \bar{r} + \sigma_c\epsilon_{1,t+1} + f_{n-1}^{rn}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t)\Big)\Big]\Big]. \tag{A144}$$

C.5 Risk-neutral returns

We plug risk-neutral price-consumption ratios and bond prices into equations (A111) through (A117). This gives risk-neutral returns on the consumption claim, risk-neutral log excess bond returns, and risk-neutral bond yields. We then substitute risk-neutral returns on the consumption claim into (A123)-(A124) to obtain risk-neutral log excess stock returns.

D Microfoundations for bond preference shock

In this Section, I show that the bond preference shock ξ_t can be microfounded either as a safety shock or an optimism or expected growth shock.

D.1 Safety shock microfoundations

For simplicity, I follow Bianchi and Lorenzoni (2021) and model private loans as intermediated by intermediaries with time-varying convex intermediation costs. A model of government bonds in the utility function as in Krishnamurthy and Vissing-Jorgensen (2012) or Kekre and Lenel (2022) would act similarly, driving a wedge between the loan rate faced by private households and the government bond yield.

Households can borrow or invest in the private loan market and the stock market, but they cannot directly invest in the government bond market. The private loan market is intermediated by a sequence of short-lived intermediaries, who require repayment after one period. Households are able to take loans analogous to n-period nominal or inflation indexed bonds at time t, which require a time $t+1$ nominal repayment of $P^{\$}_{n-1,t+1}$ or $P_{n-1,t+1}exp(\pi_t)$ to the intermediary, respectively. If households wish to hold these n-period nominal or inflation-indexed private loans to maturity, they can roll them over using a sequence of n short-lived intermediaries. The assumption of short-lived intermediaries simplifies the exposition, as it ensures that future shocks to intermediation capacity are not priced in a separate risk premium.

Let $Q_{n,t}$ and $Q^{\$}_{n,t}$ denote the quantity of n-period private loans intermediated. Intermediaries are assumed to be able to go long and short in the government bond market and in the private loan market. So if intermediaries are at all risk averse, it is optimal to completely hedge the period $t+1$ cash flows in period $t+1$. Similarly to Bianchi and Lorenzoni (2021), generation t intermediaries are assumed to face convex intermediation costs in the quantity intermediated

$$\frac{1}{\omega_t}\Omega\left(\sum_{n=1}^{\infty}\hat{P}_{n,t}Q_{n,t}+\sum_{n=1}^{\infty}\hat{P}^{\$}_{n,t}Q^{\$}_{n,t}\right), \tag{A145}$$

where ω_t is a shock to intermediation capacity. The net profits of generation t intermediaries are given by

$$\sum_{n=1}^{\infty}\left(\hat{P}_{n,t}-P_{n,t}\right)Q_{n,t}+\sum_{n=1}^{\infty}\left(\hat{P}^{\$}_{n,t}-P^{\$}_{n,t}\right)Q^{\$}_{n,t}-\frac{1}{\omega_t}\Omega\left(\sum_{n=1}^{\infty}\hat{P}_{n,t}Q_{n,t}+\sum_{n=1}^{\infty}\hat{P}^{\$}_{n,t}Q^{\$}_{n,t}\right). \tag{A146}$$

The intermediaries' first-order condition for intermediating the n-period real private loan becomes

$$\hat{P}_{n,t}-P_{n,t} = \frac{1}{\omega_t}\Omega'\left(\sum_{n=1}^{\infty}\hat{P}_{n,t}Q_{n,t}+\sum_{n=1}^{\infty}\hat{P}^{\$}_{n,t}Q^{\$}_{n,t}\right)\hat{P}_{n,t}. \tag{A147}$$

The intermediaries' first-order condition for the n-period nominal private loan becomes

$$\hat{P}^{\$}_{n,t}-P^{\$}_{n,t} = \frac{1}{\omega_t}\Omega'\left(\sum_{n=1}^{\infty}\hat{P}_{n,t}Q_{n,t}+\sum_{n=1}^{\infty}\hat{P}^{\$}_{n,t}Q^{\$}_{n,t}\right)\hat{P}^{\$}_{n,t}. \tag{A148}$$

Simplifying further to quadratic adjustment costs of the form $\Phi(x)=\frac{c}{2}x^2$ these expressions simplify to

$$P_{n,t} = \left(1-\frac{c\left(\sum_{n=1}^{\infty}\hat{P}_{n,t}Q_{n,t}+\sum_{n=1}^{\infty}\hat{P}^{\$}_{n,t}Q^{\$}_{n,t}\right)}{\omega_t}\right)\hat{P}_{n,t}, \tag{A149}$$

$$= \left(1-\frac{c\left(\sum_{n=1}^{\infty}\hat{P}_{n,t}Q_{n,t}+\sum_{n=1}^{\infty}\hat{P}^{\$}_{n,t}Q^{\$}_{n,t}\right)}{\omega_t}\right)E_t\left[M_{t+1}P_{n-1,t+1}\right], \tag{A150}$$

$$P^{\$}_{n,t} = \left(1-\frac{c\left(\sum_{n=1}^{\infty}\hat{P}_{n,t}Q_{n,t}+\sum_{n=1}^{\infty}\hat{P}^{\$}_{n,t}Q^{\$}_{n,t}\right)}{\omega_t}\right)\hat{P}^{\$}_{n,t}, \tag{A151}$$

$$= \left(1-\frac{c\left(\sum_{n=1}^{\infty}\hat{P}_{n,t}Q_{n,t}+\sum_{n=1}^{\infty}\hat{P}^{\$}_{n,t}Q^{\$}_{n,t}\right)}{\omega_t}\right)E_t\left[M_{t+1}exp(-\pi_t)P^{\$}_{n-1,t+1}\right], \tag{A152}$$

where the last step follows from households' first order condition with respect to the inflation-indexed and nominal n-period private loan. Defining $exp(-\xi_t) = \left(1 - \frac{c\left(\sum_{n=1}^{\infty} \hat{P}_{n,t} Q_{n,t} + \sum_{n=1}^{\infty} \hat{P}^{\$}_{n,t} Q^{\$}_{n,t}\right)}{\omega_t}\right)$, the bond asset pricing recursions (13) and (14) in the main paper follow.

D.2 Expected growth shock microfoundations

An alternative for the bond preference shock is as a shock to expected growth, that is not subsequently not realized, along the lines of Bordalo et al. (2020) and Beaudry and Portier (2006). Assume that $v_{a,t}$ is an iid shock to expected productivity growth:

$$\tilde{E}_t \Delta a_{t+1} = (1-\phi)x_t + v_{a,t}. \tag{A153}$$

Here, $v_{a,t}$ is a shock to expected productivity, that is subsequently not realized. Let $\tilde{E}$ denote the subjective expectations including this expected productivity shock, and E denote the expectations of a rational outside observer. Then the bond pricing Euler equation for a real n-period bond becomes

$$\begin{aligned}
P_{n,t} &= \beta \tilde{E}_t \left[exp\left(-\gamma \Delta c_{t+1} - \gamma \Delta s_{t+1}\right) P_{n-1,t+1}\right], && \text{(A154)}\\
&= exp\left(-\gamma v_{a,t}\right) \tilde{E}_t \left[exp\left(-\gamma\left(x_{t+1} - \phi x_t\right) - \gamma \Delta s_{t+1}\right) P_{n-1,t+1}\right], && \text{(A155)}\\
&= exp(-\gamma v_{a,t}) E_t \left[exp\left(-\gamma\left(x_{t+1} - \phi x_t\right) - \gamma \Delta s_{t+1}\right) P_{n-1,t+1}\right], && \text{(A156)}\\
&= exp(-\gamma v_{a,t}) E_t \left[M_{t+1} P_{n-1,t+1}\right], && \text{(A157)}
\end{aligned}$$

where M_{t+1} is the SDF in terms of ex-post realized consumption and surplus consumption. Similarly, for an n-period nominal bond

$$P_{n,t} = exp(-\gamma v_{a,t}) E_t \left[M_{t+1} exp(-\pi_{t+1}) P^{\$}_{n-1,t+1}\right]. \tag{A158}$$

Defining $exp(-\xi_t) = exp(-\gamma v_{a,t})$ the bond pricing recursions in equation (13) and (14) in the main paper go through.

All that remains to check is that stock pricing recursions also go through. For this, I consider a levered claim to consumption with leverage parameter $\frac{1}{\delta}$ as in Abel (1990). The main paper uses a slightly more complicated model for leverage with very similar properties, while also keeping dividends and consumption cointegrated. The advantage of the simple Abel (1990) leverage specification is that it can show that stock prices are exactly unchanged, whereas with the cointegrated leverage specification stock prices might change slightly, though due to its similarity to the Abel (1990) specification any changes are plausibly quantitatively insignificant.

The value function iteration for a levered consumption claim then equals:

$$\frac{P_{n,t}^{\delta,c}}{C_t^{1/\delta}} = \tilde{E}_t\left[exp\left(-\gamma\Delta c_{t+1}-\gamma\Delta s_{t+1}\right)\left(\frac{C_{t+1}}{C_t}\right)^{1/\delta}\frac{P_{n-1,t+1}^{\delta,c}}{C_{t+1}^{1/\delta}}\right]. \tag{A159}$$

If $\gamma=\frac{1}{\delta}$, as in the 1980s calibration in the main paper, any shocks to expected potential output growth drop out from the right-hand-side

$$\frac{P_{n,t}^{\delta,c}}{C_t^{1/\delta}} = \tilde{E}_t\left[exp\left(-\gamma\Delta s_{t+1}\right)\frac{P_{n-1,t+1}^{\lambda,c}}{C_{t+1}^{\delta}}\right], \tag{A160}$$

$$= E_t\left[exp\left(-\gamma\Delta s_{t+1}\right)\frac{P_{n-1,t+1}^{\lambda,c}}{C_{t+1}^{\delta}}\right], \tag{A161}$$

$$= E_t\left[M_{t+1}\left(\frac{C_{t+1}}{C_t}\right)^{1/\delta}\frac{P_{n-1,t+1}^{\delta,c}}{C_{t+1}^{1/\delta}}\right], \tag{A162}$$

i.e. the standard pricing equation for a levered consumption claim takes the same form as in the main paper with Abel (1990) leverage.

Table A4 shows that setting $\gamma=1$ and/or $\phi=1$ leaves the asset pricing properties in Table 2 in the main paper unchanged, further addressing concerns that the simplification to Abel (1990) leverage might somehow matter for the results. With $\gamma=1$, the shock $v_{a,t}$ drops out of the asset pricing Euler equation for the unlevered consumption claim, as can easily be seen by setting $\gamma=\delta=1$ in equation (A162). Levered consumption claims, as modeled in the main paper, are then also exactly unchanged in the presence of $v_{a,t}$. Setting $\phi=1$ is also informative, as in this case price inflation equals wage inflation in equation (23) in the main paper, so there is no concern that shocks to expected productivity growth could lead to a wedge between price and wage inflation.

E Additional Model Results

E.1 The Importance of Time-Varying Risk Premia for Bond-Stock Return Comovements

Tables A1 and A2 show that time-varying risk premia are the dominant parts of model stock and bond returns, and especially of model bond-stock return comovements in the model, analogously to Table 5 in Campbell et al. (2020). Starting in the bottom-left of Table A2, we see that the overall covariance between nominal bond excess returns and stock returns in the model is 95.66 in the 1979.Q4-2001.Q1 calibration. However, the covariance of risk-neutral nominal bond excess returns and risk-neutral stock excess returns in the same calibration is only 8.72, i.e. less than 10% of the overall covariance. The rest is made up of the covariance between time-varying risk premia in stock excess returns and risk-neutral bond excess returns (11.47), time-varying risk premia in bond excess returns with risk-neutral stock excess returns (27.62), and the covariance between time-varying risk premia in bond excess returns and time-varying risk premia in stock excess returns (47.84).

The other panels confirm this pattern that time-varying risk premia represent the lion's share of model bond-stock comovements, and amplify the comovement of risk-neutral bond and stock excess returns by factors of 10 or greater.

E.2 Varying Adaptiveness of Wage-Setter Inflation Expectations

The adaptiveness of wage-setters' inflation expectations drives how long inflation remains high after an adverse supply shock. Figure A2 shows the model fit for the macroeconomic target moments with $\zeta = 0$ for both subperiods. As expected the setting $\zeta = 0.6$ to match the excess bond return predictability in the data reduces the macroeconomic fit, as seen in Figure 2 in the main paper. However, the top-left plot looks very similar under these two parameterizations, so the choice of rational vs. adaptive wage-setter inflation expectations does not affect the model's implication of "stagflationary" dynamics in the 1980s. What changes is mostly the persistence of inflation, which is visible in a more persistent policy rate response to an inflation innovation in the bottom-left plot of Figure 2. However, a volatile persistent component in inflation for the 1980s has been found in numerous papers in the time series econometrics literature (Stock and Watson (2007)), so the higher inflation persistence with $\zeta = 0.6$ is empirically plausible, especially if the data used for Figure 2 is subject to noise that understates the true persistent component in inflation during the 1980s.

Figures A3 inspects the mechanism for how ζ changes the macroeconomic dynamics by looking at model macroeconomic impulse responses to demand, supply and monetary policy shocks. It shows that the inflation response to a supply shock is much less persistent when $\zeta = 0$ than when $\zeta = 0.6$. The strong persistent component in the model inflation dynamics for the 1980s calibration matters for the predictability of bond excess returns. It is the reason that supply shocks move the slope of the term structure in the same direction as bond risk premia in the middle column in the top row in Figure A4. Hence, the adaptiveness of wage-setters' inflation expectations does have important implications for Campbell-Shiller bond excess return predictability in the model, as shown in Figure 3 in the main paper.

Figure 4 in the main paper, however, shows that nominal bond betas change little with the adaptiveness of wage-setters' inflation expectations. Intuitively, more adaptive wage-

Table A1: Decomposing Model Stock and Bond Return Variances

Panel A: Stock Return Variance

	1979.Q4-2001.Q1			2001.Q2-2019.Q4		
	Risk Neutral	Risk Premium	Total	Risk Neutral	Risk Premium	Total
Risk Neutral	2.16	8.51	10.67	16.31	32.92	49.23
Risk Premium	8.51	55.86	64.36	32.92	93.00	125.92
Total	10.67	64.36	75.03	49.23	125.92	175.15

Panel B: Nominal Bond Return Variance

	1979.Q4-2001.Q1			2001.Q2-2019.Q4		
	Risk Neutral	Risk Premium	Total	Risk Neutral	Risk Premium	Total
Risk Neutral	38.19	46.58	84.77	0.48	0.68	1.16
Risk Premium	46.58	62.60	109.18	0.68	1.13	1.80
Total	84.77	109.18	193.96	1.16	1.80	2.96

Panel C: Real Bond Return Variance

	1979.Q4-2001.Q1			2001.Q2-2019.Q4		
	Risk Neutral	Risk Premium	Total	Risk Neutral	Risk Premium	Total
Risk Neutral	0.05	0.08	0.13	0.15	0.30	0.45
Risk Premium	0.08	0.17	0.26	0.30	0.66	0.96
Total	0.13	0.26	0.39	0.45	0.96	1.41

This table decomposes model stock and bond returns into risk neutral and risk premium components, and reports their variances.

Table A2: Decomposing Model Bond-Stock Return Covariance

Panel A: Nominal Bond-Stock Return Covariance

	1979.Q4-2001.Q1			2001.Q2-2019.Q4		
Bonds↓/Stocks→	Risk Neutral	Risk Premium	Total	Risk Neutral	Risk Premium	Total
Risk Neutral	8.72	11.47	20.19	-1.07	-2.36	-3.43
Risk Premium	27.62	47.84	75.46	-3.67	-8.27	-11.93
Total	36.34	59.32	95.66	-4.74	-10.62	-15.36

Panel B: Real Bond-Stock Return Covariance

	1979.Q4-2001.Q1			2001.Q2-2019.Q4		
Bonds↓/Stocks→	Risk Neutral	Risk Premium	Total	Risk Neutral	Risk Premium	Total
Risk Neutral	0.11	0.34	0.45	-0.95	-2.10	-3.05
Risk Premium	1.21	2.88	4.09	-3.25	-7.37	-10.63
Total	1.32	3.21	4.54	-4.20	-9.48	-13.68

This table decomposes model stock and bond returns into risk neutral and risk premium components, and reports their covariances similarly. Panel A decomposes the covariance between model nominal bond excess returns and stock excess returns. Panel B decomposes the covariance between model real bond excess returns and stock excess returns.

setter inflation expectations imply a smaller initial but more persistent inflation response to a supply shock, with offsetting effects on bond-stock comovements in the model.

E.3 Inflation Forecast Error Regressions

Additional empirical moments from inflation surveys provide external empirical support for partially adaptive inflation expectations in the 1980s and rational inflation expectations in the 2000s, which are calibrated to bond excess return predictability. Table A3 runs the well-known test for the rationality of inflation expectations of Coibion and Gorodnichenko (2015):

$$\pi_{t+3} - \tilde{E}_t \pi_{t+3} = a_0 + a_1 \left(\tilde{E}_t \pi_{t+3} - \tilde{E}_{t-1} \pi_{t+3} \right) + \varepsilon_{t+3}. \tag{A163}$$

Here, a tilde denotes potentially subjective inflation expectations. If expectations are full information rational the forecast error on the left-hand side of (A163) should be unpredictable, and the coefficient a_1 should equal zero. The empirical specification follows Coibion and Gorodnichenko (2015), using the Survey of Professional Forecasters four-quarter and three-quarter GDP deflator inflation forecasts to compute forecast revisions. The first column in Table A3 uses a long sample 1968.Q4-2001.Q1 and confirms their well-known empirical result. An upward revision in inflation forecasts tends to predict positive forecast errors. The second and third columns run the same empirical regressions for the 1979.Q4-2001.Q1 and 2001.Q2-2019.Q4 subperiods. I find that for both subperiods the evidence becomes insignificant. While this is potentially due to the smaller sample size and weaker statistical power, the change in a_1 from 1968.Q4-2001.Q1 to 2001.Q2-2019.Q4 is statistically significant, consistent with more forward-looking inflation expectations in the 2000s. The last two columns of the table show that the model matches this broad pattern in the predictability of inflation forecast errors documented in the data.[1]

E.4 Calibration Robustness to Different Parameter Values

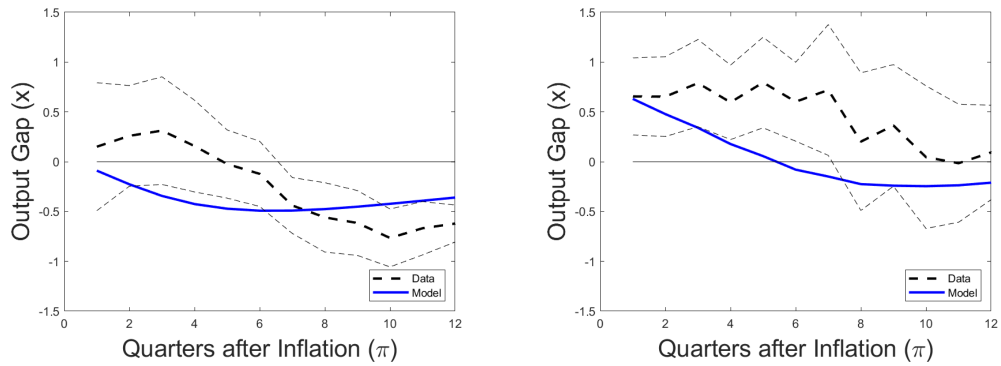
Table A4 shows that the main model implications for bond-stock betas are not sensitive to choosing different values for the utility curvature, the consumption-output gap link ϕ, the adaptiveness of inflation expectations ζ, or the slope of the Phillips curve κ. Implied nominal

[1] The literature has not reached an agreement on whether inflation expectations have become more or less rational over time. On the one hand, Bianchi, Ludvigson and Ma (2022) find less inflation forecast error predictability post-1995, and Davis (2012) shows that inflation expectations have become less responsive to oil prices shocks in recent decades. However, Coibion and Gorodnichenko (2015) and Maćkowiak and Wiederholt (2015) provide evidence and a model of decreasing attention to inflation as economic volatility declined during the 1990s.

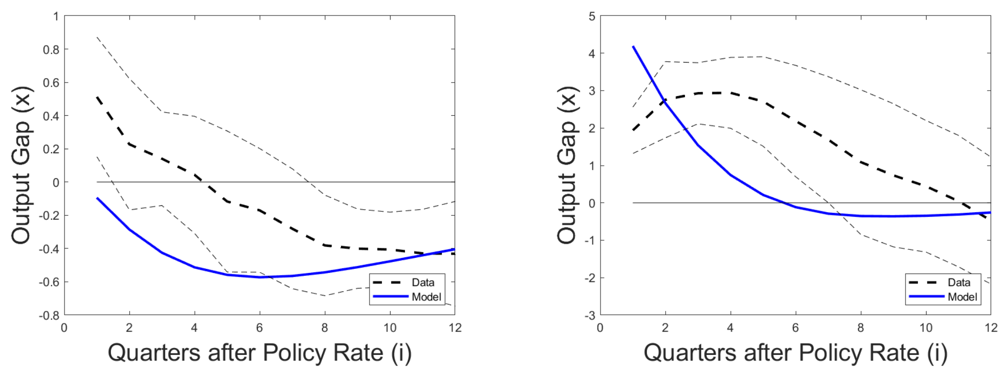
Figure A2: Empirical Output Gap, Inflation, and Policy Rate Dynamics Pre- vs. Post-2001 with Rational Inflation Expectations

Panel A: Output Gap onto Lagged Price Inflation

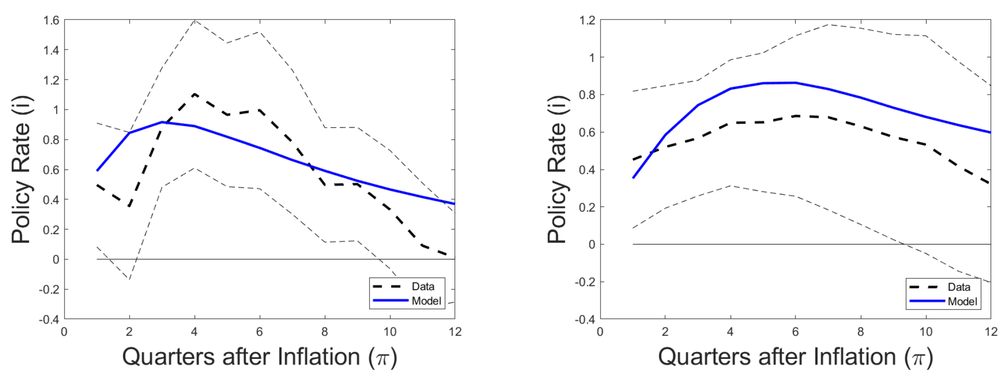
1979.Q4-2001.Q1 2001.Q2-2019.Q4



Panel B: Output Gap onto Lagged Policy Rate



Panel C: Policy Rate onto Lagged Price Inflation



This figure is analogous to Figure 2 in the main paper, but it sets $\zeta = 0$ for both calibrations.

bond-stock betas are positive for the 1980s calibration and negative for the 2000s calibration and all moments are qualitatively or even quantitatively unchanged.

The case with $\phi = 1$ is of interest because this switches off the difference between price and wage inflation (equation (23) in the main paper). Setting $\phi = 1$ leaves all asset pricing and macroeconomic moments qualitatively and quantitatively unchanged. This tells us that the model properties are unaffected whether we assume sticky wages or sticky prices.

While the parameterization in the main paper with $\gamma = 2$ is chosen to replicate the output impulse response to an identified monetary policy shock in the data, the case with $\gamma = 1$ and $\phi = 1$ is of interest because it is a simple case where bond preference shocks are exactly isomorphic to optimism or growth shocks. Changing the utility curvature parameter to $\gamma = 1$ implies a positive nominal bond-stock for the 1980s calibration and a negative nominal bond-stock beta for the 2000s calibration. The nominal bond-stock beta for the 1980s calibration is somewhat smaller in magnitude than in the baseline. While the Campbell-Shiller bond excess return predictability coefficient is still positive and within a 90% confidence interval of the empirical coefficient, it is also smaller. The reason is that with a larger EIS, monetary policy is more effective at reducing inflation, leading to a less persistent inflation process and hence less bond excess return predictability.

Setting $\zeta = 0$ (i.e. rational inflation expectations) changes the Campbell-Shiller excess bond return predictability coefficient to zero for the 1980s calibration, consistent with the comparative statics in Figure 3 in the main paper, but leaves all other asset pricing moments unchanged. This tells us that ζ is pinned down by the predictability of bond excess returns without affecting any of the other model implications.

Finally, the rightmost column sets the slope of the Phillips curve to a higher value following Rotemberg and Woodford (1997). The nominal and real bond-stock betas are unchanged from the main calibration and all other moments look similar. The only slight change is in the value of the 1980s Campbell-Shiller bond excess return predictability coefficient, which is smaller for this model parameterization. The reason is that a higher slope of the Phillips curve means that supply shocks lead to a stronger output response, and monetary policy is more powerful in stabilizing inflation by driving down the output gap, leading to a less persistent inflation process. As in the main paper, the Campbell-Shiller bond excess return predictability coefficient is therefore informative for the Phillips curve more broadly. Overall, the implications for nominal and real bond-stock betas are robust to varying the preference, consumption-output gap link, inflation rationality, and Phillips curve parameters.

E.5 Counterfactual Bond Betas by Monetary Policy $\times$ Economic Shocks

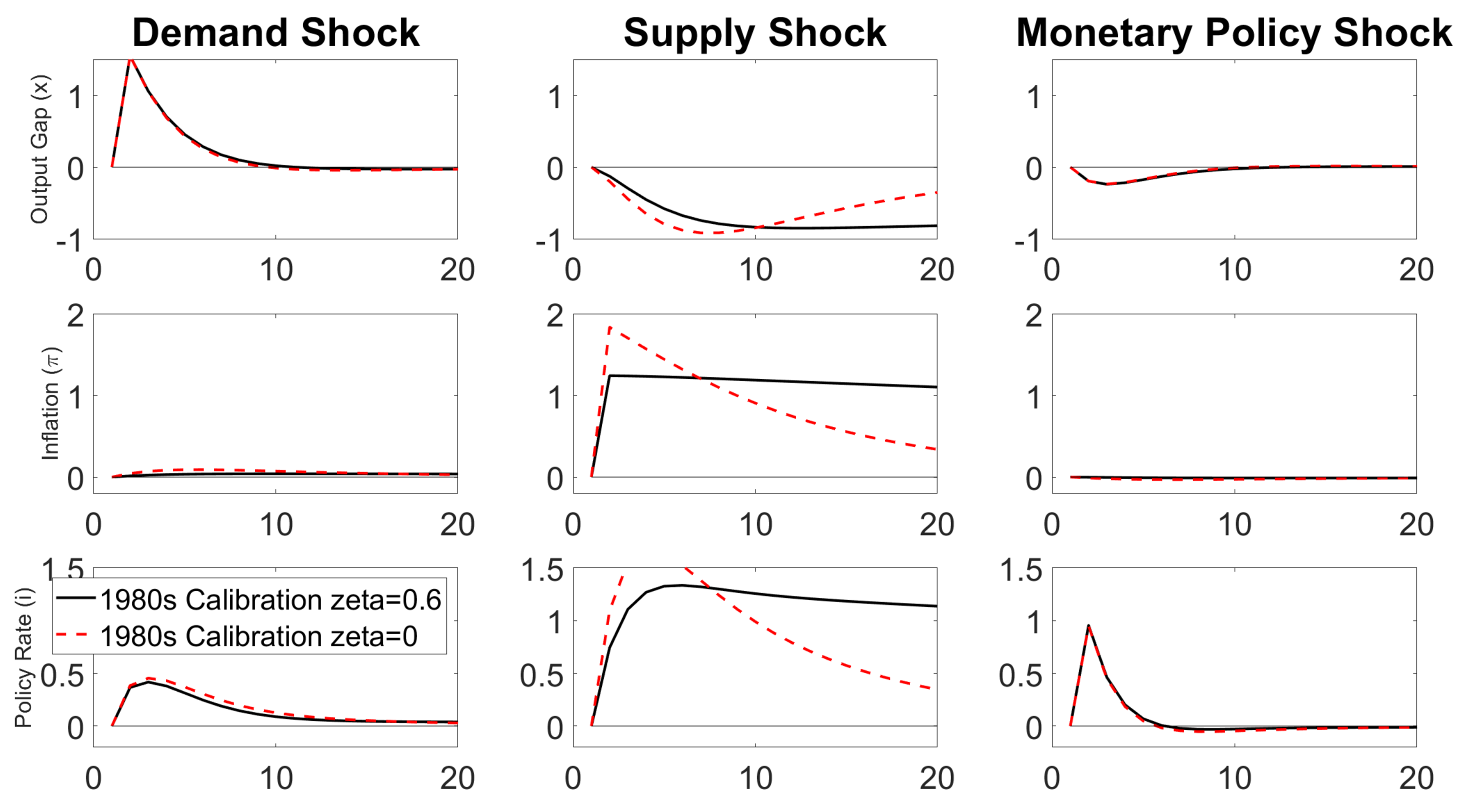
Figure A5 shows model-implied nominal and real bond-stock betas for three different shock regimes against the monetary policy inflation weight γ^{π}, and provides the model nominal and real bond-stock betas underlying the best-fit exercise in Table 3 in the main paper. Each marker shows the model-implied bond beta for a combination of γ^π, γ^x, and ρ^i while holding all other parameters constant. The shock volatilities in Panel A are set to the 1979.Q4-2001.Q1 calibration in Table 1 in the main paper, the shock volatilities in Panel B are set to the 2001.Q2-2019.Q4 calibration in Table 1. Panel C considers what happens when the economy is dominated by monetary policy shock uncertainty, setting the volatilities of demand and supply shocks to zero and assuming $\sigma_i = 1.19\%$, which generates a volatility of annual real consumption growth of 1.6% at monetary policy parameters $\gamma^\pi = 1.5$, $\gamma^x = 0.5$ and $\rho^i = 0.8$.

Table A3: Inflation Forecast Error Regressions by Subperiod

	Data			Model	
$\tilde{E}_t\pi_{t+3} - \tilde{E}_{t-1}\pi_{t+3}$	0.926***	0.433	-0.310	1.43	-0.01
	(0.34)	(0.32)	(0.43)		
Const.	-0.114	-0.795***	-0.046		
	(0.28)	(0.20)	(0.18)		
N	126	87	71		
R-sq	0.09	0.03	0.00		
Sample	1968.Q4-2001.Q1	1979.Q4-2001.Q1	2001.Q2-2019.Q4	1979.Q4-2001.Q1	2001.Q2-2019.Q4

This table estimates Coibion and Gorodnichenko (2015) regressions of the form $\pi_{t+3} - \tilde{E}_t\pi_{t+3} = a_0 + a_1\left(\tilde{E}_t\pi_{t+3} - \tilde{E}_{t-1}\pi_{t+3}\right) + \varepsilon_{t+3}$ using quarterly GDP deflator inflation forecasts from the Survey of Professional Forecasters. Newey-West standard errors with 4 lags in parentheses. Model subjective n-quarter inflation expectations are computed assuming that inflation expectations are a weighted average of rational expectations and past average inflation $\tilde{E}_t\pi_{t+n} = \zeta\pi_{t-n-1\to t-1} + (1-\zeta)E_t\pi_{t+n}$

Figure A3: Model Macroeconomic Impulse Responses by Adaptive Inflation Expectations for 1980s Calibration



This figure compares the structural macroeconomic impulse responses for the 1980s calibration with $\zeta = 0$ and $\zeta = 0.6$.

Table A4: Model and Data Moments Robustness

Panel A: 1979.Q4-2001.Q1 Calibration

	Baseline	$\phi = 1$	$\gamma = 1$	$\gamma = 1, \phi = 1$	$\zeta = 0$	$\kappa = 0.019$
Stocks						
Equity Premium	7.33	6.91	5.87	5.69	7.14	7.03
Equity Vol	14.95	13.95	15.16	14.43	14.40	14.19
Equity SR	0.49	0.50	0.39	0.39	0.50	0.50
AR(1) pd	0.96	0.95	0.95	0.95	0.95	0.95
1 YR Excess Returns on pd	-0.38	-0.43	-0.41	-0.43	-0.41	-0.42
1 YR Excess Returns on pd (R2)	0.06	0.07	0.05	0.05	0.06	0.06
Bonds						
Return Vol.	15.82	15.39	12.89	12.59	10.12	12.66
Nominal Bond-Stock Beta	0.86	0.85	0.42	0.37	0.65	0.71
Real Bond-Stock Beta	0.05	0.06	0.05	0.05	0.09	0.06
1 YR Excess Returns on slope	1.26	1.24	0.25	0.24	-0.36	0.15
1 YR Excess Returns on slope (R2)	0.01	0.01	0.00	0.00	0.00	0.00
Macroeconomic Volatilities						
Std. Annual Cons. Growth	0.76	0.73	0.89	0.87	0.98	0.75
Std. Annual Change Fed Funds Rate	1.64	1.63	1.62	1.62	2.15	1.65
Std. Annual Change 10-Year Subj. Infl. Forecast	0.73	0.71	0.64	0.62	0.87	0.58

This table shows robustness results for the calibration moments shown in Table 2 in the main paper. Panel A sets all parameters to their 1979.Q4-2001.Q1 calibration values and changes the indicated parameters one at a time. Panel B sets all parameters to their 2001.Q2-2019.Q4 calibration values and changes the indicated parameters one at a time.

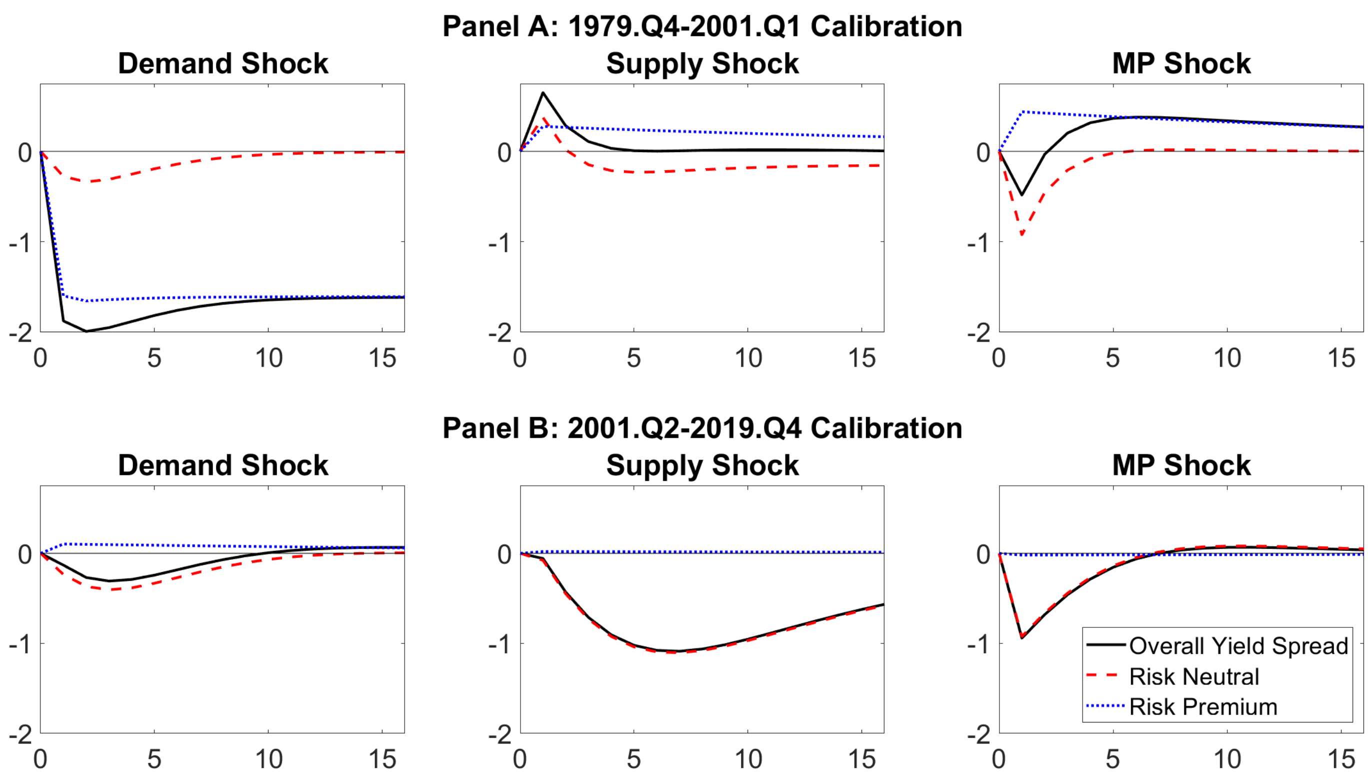
Model and Data Moments Robustness (continued)

Panel B: 2001.Q2-2019.Q4 Calibration

	Baseline	$\phi = 1$	$\gamma = 1$	$\gamma = 1$, $\phi = 1$	$\zeta = 0$	$\kappa = 0.019$
Stocks						
Equity Premium	9.15	9.15	5.69	5.68	9.15	9.17
Equity Vol	19.29	19.29	15.69	15.68	19.29	19.32
Equity SR	0.47	0.47	0.36	0.36	0.47	0.47
AR(1) pd	0.93	0.93	0.95	0.95	0.93	0.93
1 YR Excess Returns on pd	-0.38	-0.38	-0.32	-0.32	-0.38	-0.38
1 YR Excess Returns on pd (R2)	0.14	0.14	0.10	0.10	0.14	0.14
Bonds						
Return Vol.	2.12	2.16	1.42	1.44	2.62	1.89
Nominal Bond-Stock Beta	-0.09	-0.09	-0.07	-0.07	-0.10	-0.09
Real Bond-Stock Beta	-0.08	-0.08	-0.07	-0.07	-0.08	-0.08
1 YR Excess Returns on slope	-0.31	-0.31	-0.16	-0.15	-0.39	-0.27
1 YR Excess Returns on slope (R2)	0.01	0.01	0.00	0.00	0.00	0.01
Macroeconomic Volatilities						
Std. Annual Cons. Growth	1.59	1.62	1.44	1.47	1.58	1.62
Std. Annual Change Fed Funds Rate	0.65	0.68	0.56	0.58	0.61	0.71
Std. Annual Change 10-Year Subj. Infl. Forecast	0.12	0.12	0.10	0.10	0.09	0.06

This table shows robustness results for the calibration moments shown in Table 2 in the main paper. Panel A sets all parameters to their 1979.Q4-2001.Q1 calibration values and changes the indicated parameters one at a time. Panel B sets all parameters to their 2001.Q2-2019.Q4 calibration values and changes the indicated parameters one at a time.

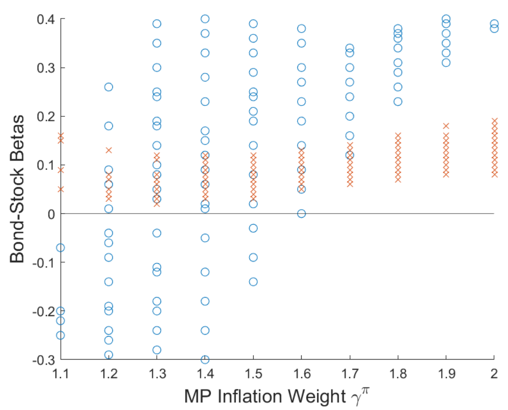
Figure A4: Model Yield Spread Impulse Responses



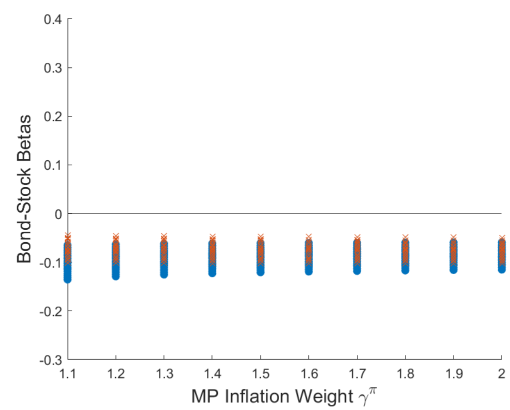
This figure shows model impulse responses for the yield spread, i.e. the right-hand-side of the Campbell-Shiller regressions in Table 2 in the main paper. The yield spread is defined as the ten-year nominal zero-coupon Treasury bond yield minus the nominal risk-free rate. It is decomposed into risk-neutral (expectations hypothesis) and risk premium components, see also Figure 6 in the main paper. The top row shows impulse responses for the 1979.Q4-2001.Q1 calibration and the bottom row shows impulse responses for the 2001.Q2-2019.Q4 calibration. The impulse in the left column is a one-percentage-point demand shock, in the middle column is a one-percentage-point cost-push or supply shock, and in the right column is a one-percentage-point monetary policy shock.

Figure A5: Scatter Bond Betas by Shocks and Monetary Policy

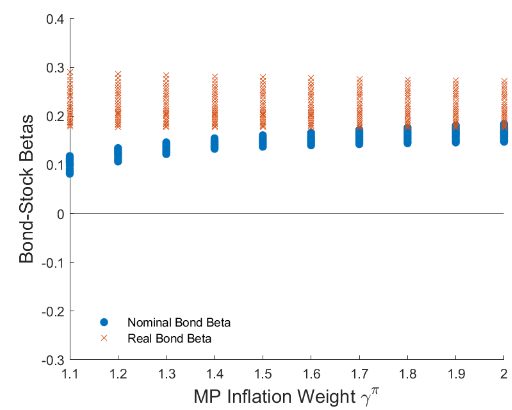
Panel A: 1979.Q4-2001.Q1



Panel B: 2001.Q2-2019.Q4



Panel C: MP Shocks



Note: This figure shows scatter plots of counterfactual ten-year nominal bond-stock betas (blue circles) and real bond betas (red asterisks) against the monetary policy inflation weight, γ^π on the x-axis. Panel A sets the volatilities of shocks to the 1980s column of Table 1 in the main paper, where supply shocks are dominant $(\sigma_x, \sigma_\pi, \sigma_i) = (0.01, 0.58, 0.55)$. Panel B sets the shock volatilities to the 2000s column in Table 1 in the main paper, where demand shocks are dominant $(\sigma_x, \sigma_\pi, \sigma_i) = (0.59, 0.07, 0.07)$. Panel C sets the shock volatilities to a calibration where monetary policy surprises are dominant $(\sigma_x, \sigma_\pi, \sigma_i) = (0.00, 0.00, 1.19)$. Each marker corresponds to a different combination of monetary policy parameters, taken from a grid on $\gamma^x \in [0.4, 1.1]$, $\gamma^\pi \in [1.1, 2]$, $\rho^i \in [0.5, 0.9]$. The inflation expectations parameter is set to zero $\zeta = 0$.

Figure A5 shows that real bond betas are positive when supply shocks are dominant, negative when demand shocks are dominant, and again positive when monetary policy shocks are dominant. The level of real bond-stock betas therefore helps pin down the volatility of demand shocks in Table 3 in the main paper.

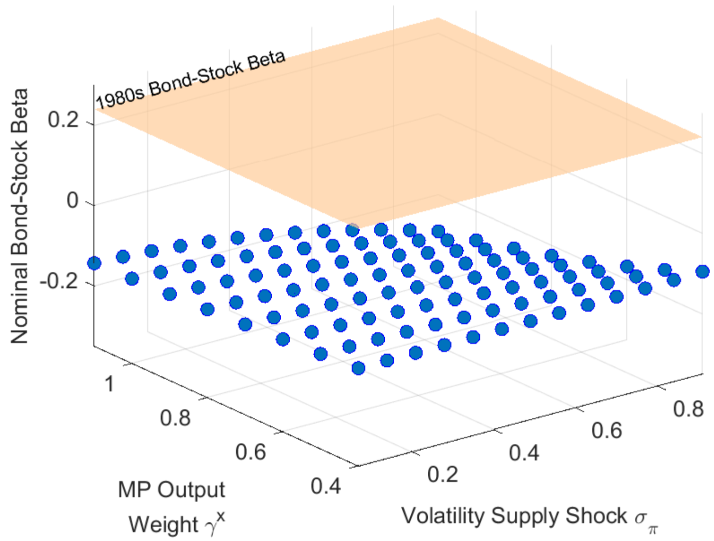
Comparing Panels A and C shows that supply and monetary policy are complementary in driving up real bond-stock betas, but they have different implications for nominal bond betas. When monetary policy shocks are dominant, as in Panel C, both real and nominal bond betas are positive and nominal bond betas are slightly smaller than real bond betas (i.e. breakeven betas are negative). The reason is that a contractionary monetary policy shock has a small but negative inflation effect in the model. By contrast, when supply shocks are dominant, as in Panel A, nominal bond betas can differ substantially from real bond betas and can be either positive or negative. Panel A shows that in the presence of supply shocks, nominal bond betas are positive when monetary policy reacts strongly to inflation (γ^{π} high), but negative when monetary policy reacts relatively weakly to inflation (γ^{π} close to one).

E.6 Additional Results for Replicating 1980s Bond-Stock Betas in the Model

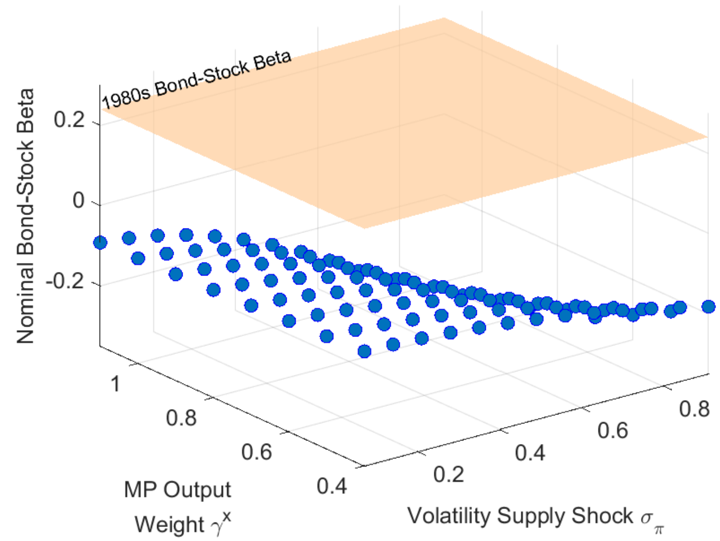
Figure A6 is analogous to Figure 8 in the main paper, but it varies the output gap weight γ^x rather than the inflation weight γ^{π}.

Figure A6: Model-Implied Nominal Bond-Stock Beta against γ^x

Panel A: MP Inertia $\rho^i = 0.54$



Panel B: MP Inertia $\rho^i = 0.80$



Note: This figure shows model ten-year nominal bond-stock betas (blue circles) against the standard deviation of supply shocks σ_π on the x-axis and the MP output gap weight γ^x on the y-axis while holding the monetary policy inertia coefficient constant at $\rho^i = 0.54$ (Panel A) and $\rho^i = 0.80$ (Panel B), respectively. All other parameters are set to their values in the 2000s calibration in Table 1 in the main paper. The orange plane indicates the empirical ten-year nominal bond-stock beta for the 1979.Q4-2001.Q1 subperiod.

F Details for Delta Method Standard Errors

This Section provides details for the Delta method used to construct standard errors in Table 2 in the main paper. Standard errors are computed separately for each of the calibration steps, taking all other parameters as given. Let $\hat{\Psi}$ denote the vector of twelve (13 for the second subperiod) empirical target moments, and $\Psi(\sigma_x, \sigma_\pi, \sigma_i, \gamma^x, \gamma^\pi, \rho^i; \zeta)$ the vector of model moments computed analogously on model-simulated data. I choose subperiod-specific monetary policy parameters γ^x, γ^π, and ρ^i and shock volatilities σ_x, σ_π, and σ_i while holding the inflation expectations parameter constant at $\zeta = 0$ to minimize the objective function:

$$\left\| \frac{\hat{\Psi} - \Psi(\sigma_x, \sigma_\pi, \sigma_i, \gamma^x, \gamma^\pi, \rho^i; \zeta = 0)}{SE(\hat{\Psi})} \right\|^2 . \tag{A164}$$

Let $\hat{H}_{SE}$ is the numerical Jacobian

$$\hat{H}_{SE} = \Delta \left(\Psi / SE \left(\hat{\Psi} \right) \right), \tag{A165}$$

i.e. the vector of derivatives of the normalized objective function with respect to each parameter. I calculate this derivative numerically by using a derivative step of 0.05 for most parameters (0.005 for σ_x to avoid setting it to negative values) and a simulation length of 100, as in the SMM estimation. I then average the simulated value for $\hat{H}$ across 20000 independent simulations to reduce simulation noise. Because the weighting function in the GMM estimation is the identity (after normalizing all moments into z-scores using the standard errors of the empirical moments computed in the data), standard GMM results imply that the variance-covariance matrix of the parameters is

$$\hat{V}_{params} = \hat{M}_{SE}^{-1} \hat{H}_{SE} \hat{V} \hat{H}'_{SE} \hat{M}_{SE}^{-1'} . \tag{A166}$$

Here $\hat{V}$ is the variance-covariance matrix of the target moments, and $\hat{M}_{SE} = \hat{H}'_{SE} \hat{H}_{SE}$. Under the simplifying assumption that the target moments are uncorrelated, $\hat{V}$ collapses to the identity and the expression for $\hat{V}_{params}$ becomes

$$\hat{V}_{params} = (\hat{H}_{SE} \hat{H}'_{SE})^{-1} . \tag{A167}$$

The standard errors reported in Table 2 do not make this simplifying assumption, instead computing $\hat{V}$ as the correlation matrix across 1000 separate simulations of the model, where again each simulation has sample length 100. The standard errors reported in Table 2 in the main paper are then obtained by substituting into (A166) and

$$SE(\sigma_x, \sigma_\pi, \sigma_i, \gamma^x, \gamma^\pi, \rho^i) \quad = \quad diag\left(\sqrt{\hat{V}_{params}}\right). \tag{A168}$$

A joint hypothesis test for $\gamma_x = 0.5$ and $\gamma_\pi = 1.1$ is run by comparing $[\gamma_x - 0.5, \gamma_\pi - 1.1]\, \hat{V}_{params}(1:2, 1:2)^{-1}\, [\gamma_x - 0.5, \gamma_\pi - 1.1]' \sim \chi^2_2$, where $\hat{V}_{params}(1:2, 1:2)$ is the upper $[2 \times 2]$ block of $\hat{V}_{params}$.

The standard errors for ζ is obtained similarly, holding all other parameter values fixed. The relevant moment is the Campbell-Shiller coefficient b^{CS}. By standard GMM algebra, the delta method standard error of ζ is

$$SE(\zeta) \quad = \quad \frac{SE\left(\hat{b}^{SE}\right)}{\left\{\frac{db^{CS}}{d\zeta}\right\}^{-1}}, \tag{A169}$$

i.e. the ratio of the standard error of the Campbell-Shiller coefficient in the data to the slope of the model coefficient with respect to ζ.

For the 1980s period, the empirical Newey-West standard error with 4 lags equals $SE(\hat{b}^{SE}) = 1.38$. The numerical derivative is computed as $\frac{db^{CS}}{d\zeta} = \frac{b^{CS}(\zeta=0.6) - b^{CS}(\zeta=0)}{0.6} = \frac{1.26 - (-0.36)}{0.6} = 2.70$, giving the standard error for ζ as $SE(\zeta) = 1.38/2.70 = 0.51$.

For the 2000s period, the empirical Newey-West standard error with 4 lags equals $SE(\hat{b}^{SE}) = 1.18$. The numerical derivative is computed as $\frac{db^{CS}}{d\zeta} = \frac{b^{CS}(\zeta=0.6) - b^{CS}(\zeta=0)}{0.6} = \frac{-0.39 - (-0.31)}{0.6} = 0.44$, giving the standard error for ζ as $SE(\zeta) = 1.18/0.44 = 2.67$.

G Extension with richer price-wage dynamics

In reality, wage and price inflation are not as closely linked as in the model. This Section explores robustness to relaxing relation (23) in the main paper. Assume that there is an additional shock $v_{p,t+1}$ driving a wedge between price and wage dynamics, so equation (23) in the main paper is modified to

$$\pi_t \;\; = \;\; \pi_t^w - (1-\phi)x_{t-1} + v_{p,t}. \tag{A170}$$

The shock $v_{p,t}$ is assumed to be uncorrelated with all other shocks, iid, with standard deviation σ_p.

G.1 Macroeconomic solution with richer price-wage dynamics

The macroeconomic solution can now be described in terms of the same state vector as before, $Y_t = [x_t, \pi_t^w, i_t]'$, and the modified shock vector

$$v_t \;\; = \;\; \left[v_{x,t}, v_{\pi,t}, v_{i,t} + (1-\rho^i)\gamma^\pi v_{p,t}\right]'. \tag{A171}$$

Because the shock $v_{p,t}$ is iid it is unpredictable, and the macroeconomic state vector dynamics are described by

$$0 \;\; = \;\; F E_t Y_{t+1} + G Y_t + H Y_{t-1} + M v_t, \tag{A172}$$

where the matrices F, G, H, an M are as given by equation (A65) as before. The only difference for the macroeconomic dynamics is that the variance-covariance matrix of shocks needs to be replaced by

$$\Sigma_v \;\; = \;\; diag([\sigma_x^2, \sigma_\pi^2, \sigma_i^2 + (1-\rho^i)^2 (\gamma^\pi)^2 \sigma_p^2]). \tag{A173}$$

G.2 Asset pricing solutions with richer price-wage dynamics

The value function iterations for real bonds and consumption claims are unchanged. The solution for nominal bonds requires a minor modification. Using the properties of multivariate normals, the conditional distribution of $v_{p,t+1}$ conditional on $\epsilon_{t+1} = A\Sigma v_{t+1}$ is given by

$$
\begin{aligned}
v_{p,t+1}\,|\epsilon_{t+1} &\sim N\left(vec_p\epsilon_{t+1}, \left(\sigma_p^{\perp}\right)^2\right), && \text{(A174)}\\
vec_p &= Cov(v_{p,t+1}, \epsilon_{t+1})' = (1-\rho^i)\gamma^{\pi}\sigma_p^2\left(A\Sigma e_3'\right)', && \text{(A175)}\\
\left(\sigma_p^{\perp}\right)^2 &= \sigma_p^2 - vec_p vec_p'. && \text{(A176)}
\end{aligned}
$$

We can then write $v_{p,t+1}$ as the sum of two independent components

$$
v_{p,t+1} = vec_p\epsilon_{t+1} + \epsilon_{t+1}^{\perp}, \tag{A177}
$$

where $\epsilon_{t+1}^{\perp}$ is independent of ϵ_{t+1} with standard deviation $\sigma_p^{\perp}$.

We integrate analytically over $\epsilon_{t+1}^{\perp}$ to obtain the nominal bond price iteration. With price inflation given by $\pi_{t+1} = Y_{2,t+1} - (1-\phi)Y_{1,t} + v_{p,t+1}$ the n-period nominal bond price can be written as:

$$
\begin{aligned}
B_n^{\$}(\tilde{Z}_t, \hat{s}_t, x_{t-1}) &= \mathbb{E}_t\left[\exp\left(m_{t+1} - \xi_t - v_{p,t+1} - Y_{2,t+1} + (1-\phi)Y_{1,t} + b_{n-1}^{\$}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, B^{\$}x_t)\right)\right]\\
&= \mathbb{E}_t\left[\exp\left(m_{t+1} - \xi_t - vec_p\epsilon_{t+1} + \frac{1}{2}\left(\sigma_p^{\perp}\right)^2 - Y_{2,t+1} + (1-\phi)Y_{1,t}\right.\right.\\
&\quad \left.\left. + b_{n-1}^{\$}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, B^{\$}x_t)\right)\right]\\
&= \mathbb{E}_t\left[\exp\left(-\bar{r} + \frac{1}{2}\left(\sigma_p^{\perp}\right)^2 - e_3A^{-1}\tilde{Z}_t - \frac{\gamma}{2}(1-\theta_0)(1-2\hat{s}_t)\right.\right. \qquad \text{(A178)}\\
&\quad \left.\left. -\gamma(1+\lambda(\hat{s}_t))\sigma_c\epsilon_{1,t+1} - \left(e_2A^{-1} + vec_p\right)\epsilon_{t+1} + b_{n-1}^{\$}(\tilde{Z}_{t+1}, \hat{s}_{t+1}, x_t)\right)\right].
\end{aligned}
$$

The two-period nominal log bond price equals

$$
\begin{aligned}
b_2^{\$} &= -e_3[I+B]A^{-1}\tilde{Z}_t + \frac{1}{2}v_{\$}\Sigma_v v_{\$}\prime + \frac{1}{2}\left(\sigma_p^{\perp}\right)^2\\
&\quad + \gamma\left(\lambda(\hat{s}_t) + 1\right)\Sigma_M\Sigma_v v_{\$}' - 2\bar{r}. && \text{(A179)}
\end{aligned}
$$

where now the vector $v_{\$}$ reflects the loading of the additional inflation shock

$$
v_{\$} = \left(e_3 + e_2\right)\Sigma + vec_pA\Sigma. \tag{A180}
$$

Similarly, we now analytically derive the two-period risk-neutral nominal bond price, incorporating the assumption that $\pi_{t+1} = Y_{2,t+1} - (1-\phi)Y_{1,t} + v_{p,t+1}$:

$$P_{2,t}^{\$,rn} = \exp(-\xi_t)\,\mathbb{E}_t\left[M_{t+1}^{rn}P_{1,t+1}^{s,rn}\exp(-Y_{2,t+1}+(1-\phi)Y_{1,t})\right] \tag{A181}$$

$$= \exp(-r_t)\,\mathbb{E}_t\left[\exp(-Y_{3,t+1}-v_{p,t+1}-Y_{2,t+1}+(1-\phi)Y_{1,t}-\bar{r})\right]. \tag{A182}$$

Thus, the risk-neutral nominal bond price for an n-period bond is given by, where $v_{p,t+1}$ follows equation (A177):

$$\begin{aligned}
B_n^{\$,rn}\left(\tilde{Z}_t,\hat{s}_t,x_{t-1}\right) &= \mathbb{E}_t\left[\exp\left(-\bar{r}-e_3A^{-1}\tilde{Z}_t-e_2A^{-1}\epsilon_{t+1}-v_{p,t+1}+b_{n-1}^{\$,rn}\left(\bar{Z}_{t+1},\hat{s}_{t+1},x_t\right)\right)\right] \\
&= \mathbb{E}_t\left[\exp\left(-\bar{r}-e_3A^{-1}\tilde{Z}_t-e_2A^{-1}\epsilon_{t+1}-vec_p\epsilon_{t+1}+\frac{1}{2}\left(\sigma_p^{\perp}\right)^2\right.\right. \\
&\quad \left.\left.+b_{n-1}^{\$,rn}\left(\bar{Z}_{t+1},\hat{s}_{t+1},x_t\right)\right)\right] \\
&= \mathbb{E}_t\left[\exp\left(-\bar{r}-e_3A^{-1}\tilde{Z}_t+\frac{1}{2}\left(\sigma_p^{\perp}\right)^2-\left(e_2A^{-1}+vec_p\right)\epsilon_{t+1}\right.\right. \\
&\quad \left.\left.+b_{n-1}^{\$,rn}\left(\bar{Z}_{t+1},\hat{s}_{t+1},x_t\right)\right)\right].
\end{aligned} \tag{A183}$$

The risk-neutral log price of the two-period nominal bond is expressed as:

$$b_2^{\$,rn} = -e_3[I+B]A^{-1}\tilde{Z}_t+\frac{1}{2}v_{\$}\Sigma_v v_{\$}'+\frac{1}{2}\left(\sigma_p^{\perp}\right)^2-2\bar{r}. \tag{A184}$$

where $v_{\$}$ is defined by equation (A180).

G.3 Results with richer price-wage dynamics

Overall, I find that asset pricing moments are robust and indeed almost indistinguishable from the baseline even when the volatility of this new shock is set to a very high value. For this comparison, I set the volatility of the price-wage dynamics shock in equation (A170) to $\sigma_p = 0.55$ and leave all other parameters unchanged as in Table 1 in the main paper. This magnitude for σ_p is comparable to the supply shock volatility and equal to the volatility of monetary policy shocks in the 1980s calibration, so this should be regarded as plausible upper bound.

Table A5 compares the asset pricing moments for the 1980s and 2000s calibrations with $\sigma_p = 0$ (the baseline in the main paper) and $\sigma_p = 0.55$. For ease of comparison the model moments from Table 2 in the main paper are replicated in columns labeled "Model Baseline". Table A5 shows that all macro and asset pricing moments for the two calibrations are qualitatively and even quantitatively unchanged. Further, Figure A7 re-does the main model counterfactuals (Figure 4 in the main paper) while setting $\sigma_p = 0.55$. Figure A7 is visually indistinguishable from Figure 4 in the main paper, showing that the central model counterfactuals are insensitive to assuming richer price-wage dynamics.

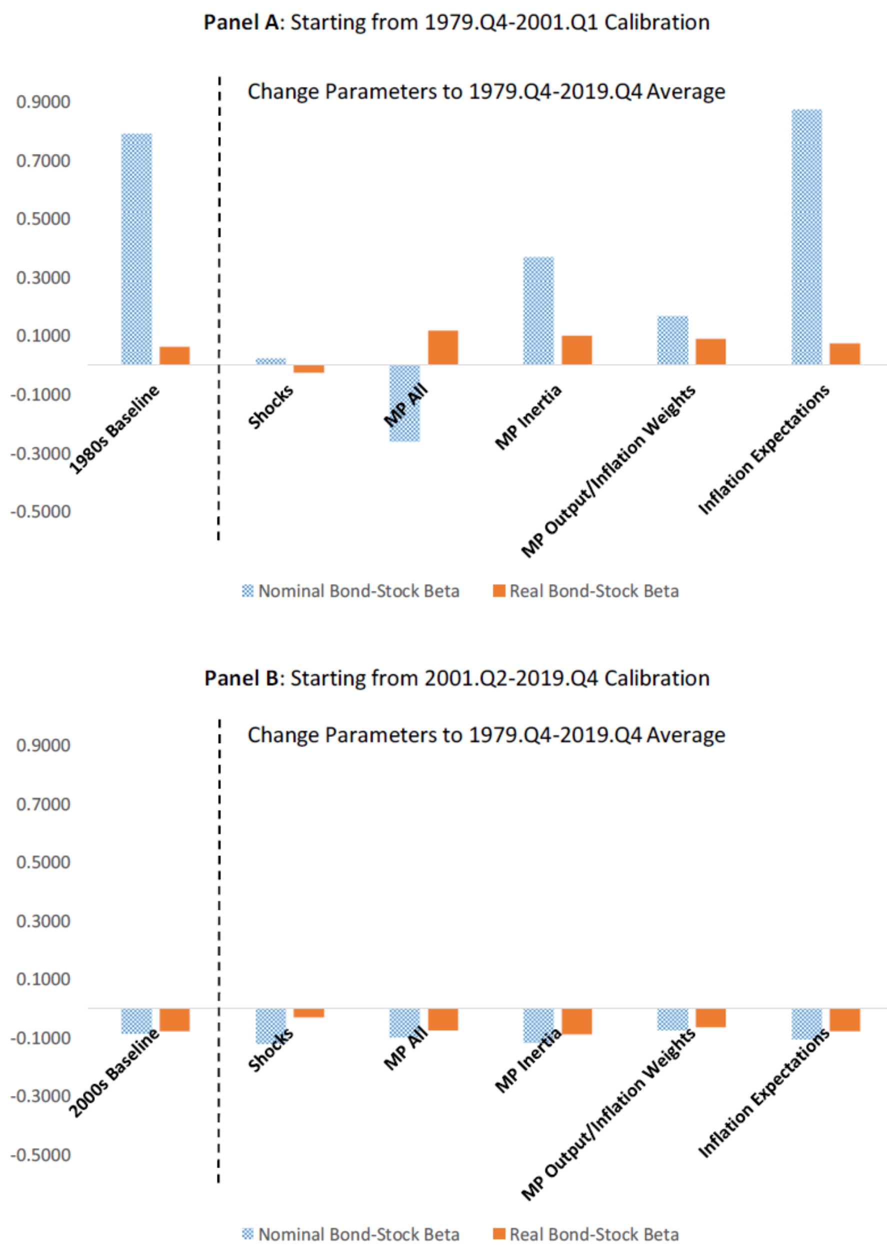
Intuitively, this new shock enters consumption dynamics because the monetary policy rule depends on price inflation in the model. Nominal bond prices do depend on this new shock, but the macroeconomic impact is transitory and hence the impact on the risk properties of nominal bonds is quantitatively small. To keep the exposition and model in the main paper simple, the main paper hence abstracts from the richer price-wage dynamics.

Table A5: Asset Prices and Macro Volatilities with Richer Price-Wage Dynamics

	1979.Q4-2001.Q1		2001.Q2-2019.Q4	
	Model Baseline	Model $\sigma_p = 0.55$	Model Baseline	Model $\sigma_p = 0.55$
Asset Prices: Stocks				
Equity Premium	7.33	7.33	9.15	9.16
Equity Vol	14.95	14.95	19.29	19.29
Equity Sharpe Ratio	0.49	0.49	0.47	0.47
AR(1) pd	0.96	0.96	0.93	0.93
1 YR Excess Returns on pd	-0.38	-0.38	-0.38	-0.38
1 YR Excess Returns on pd (R^2)	0.06	0.06	0.14	0.14
Asset Prices: Bonds				
Expected Bond Exc. Return	4.26	4.33	-0.84	-0.83
Return Vol	15.82	15.96	2.12	2.11
Nominal Bond-Stock Beta	0.86	0.87	-0.09	-0.09
Real Bond-Stock Beta	0.05	0.05	-0.08	-0.08
1 YR Excess Return on Yield Spread*	1.26	1.34	-0.31	-0.29
1 YR Excess Return on Yield Spread (R^2)	0.01	0.01	0.01	0.01
Macroeconomic Volatilities				
Std. Annual Cons. Growth*	0.76	0.76	1.59	1.60
Std. Annual Change Fed Funds Rate*	1.64	1.64	0.65	0.69
Std. Annual Change 10-Year Subj. Infl. Forecast*	0.62	0.62	0.12	0.12

This table replicates the model moments from Table 2 in the main paper in columns labeled "Model Baseline". Columns labeled "Model $\sigma_p = 0.55$ show model moments with $\sigma_p = 0.55$ for the shock to the price-wage dynamics (A170).

Figure A7: Counterfactual Bond-Stock Betas with Richer Price-Wage Dynamics



This figure is analogous to Figure 4 in the main paper, except that it sets $\sigma_p = 0.55$ for the shock to the price-wage dynamics (A170).

References

Beaudry, Paul and Franck Portier (2006) “Stock prices, news, and economic fluctuations,” *American Economic Review*, 96 (4), 1293–1307.

Bianchi, Francesco, Sydney C Ludvigson, and Sai Ma (2022) “Belief distortions and macroeconomic fluctuations,” *American Economic Review*, 112 (7), 2269–2315.

Bianchi, Javier and Guido Lorenzoni (2021) “The prudential use of capital controls and foreign currency reserves,” *Handbook of International Economics Vol. V.*

Bordalo, Pedro, Nicola Gennaioli, Rafael La Porta, and Andrei Shleifer (2020) *Expectations of fundamentals and stock market puzzles*: National Bureau of Economic Research.

Campbell, John Y and John H Cochrane (1999) “By force of habit: A consumption-based explanation of aggregate stock market behavior,” *Journal of Political Economy*, 107, 205–251.

Campbell, John Y and Sydney Ludvigson (2001) “Elasticities of Substitution in Real Business Cycle Models with Home Production,” *Journal of Money, Credit and Banking*, 33 (4), 847–875.

Campbell, John Y, Carolin Pflueger, and Luis M Viceira (2020) “Macroeconomic drivers of bond and equity risks,” *Journal of Political Economy*, 128 (8), 3148–3185.

Christiano, Lawrence H, Martin Eichenbaum, and Charles L Evans (2005) “Nominal rigidities and the dynamic effects of a shock to monetary policy,” *Journal of Political Economy*, 113 (1), 1–45.

Coibion, Olivier and Yuriy Gorodnichenko (2015) “Information rigidity and the expectations formation process: A simple framework and new facts,” *American Economic Review*, 105 (8), 2644–2678.

Davis, J Scott (2012) “Inflation expectations have become more anchored over time,” *Economic Letter*, 7, 1–4.

Greenwood, Jeremy, Zvi Hercowitz, and Gregory W Huffman (1988) “Investment, capacity utilization, and the real business cycle,” *American Economic Review*, 402–417.

Kekre, Rohan and Moritz Lenel (2022) “Monetary policy, redistribution, and risk premia,” *Econometrica*, 90 (5), 2249–2282.

Krishnamurthy, Arvind and Annette Vissing-Jorgensen (2012) “The aggregate demand for Treasury debt,” *Journal of Political Economy*, 120 (2), 233–267.

Lucas, Robert E. Jr. (1988) “On the Mechanics of Economic Development,” *Journal of Monetary Economics*, 22, 3–42.

Maćkowiak, Bartosz and Mirko Wiederholt (2015) “Business cycle dynamics under rational inattention,” *Review of Economic Studies*, 82 (4), 1502–1532.

Rotemberg, Julio J (1982) "Monopolistic price adjustment and aggregate output," *Review of Economic Studies*, 49 (4), 517–531.

Rotemberg, Julio J and Michael Woodford (1997) "An Optimization-Based Econometric Framework for the Evaluation of Monetary Policy," *NBER Macroeconomics Annual*, 12, 297–346.

Smets, Frank and Rafael Wouters (2007) "Shocks and frictions in US business cycles: A Bayesian DSGE approach," *American Economic Review*, 97, 586–606.

Stock, James H and Mark W Watson (2007) "Why has US inflation become harder to forecast?" *Journal of Money, Credit and Banking*, 39, 3–33.

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